EGIN

A cross-platform mobile and web application for managing a land plot

Probation Period 1 · Project Report

Group: SE-2414

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Submission version: prob-period-1
Eleven report sections · editable diagram sources · supporting evidence

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Submission version and supporting materials

This PDF combines Sections 1–10 and References as Section 11 from `docs/DIP`. The report, editable sources and evidence are packaged in the repository version tagged prob-period-1.

The application source baseline is b0b0f46. Historical source inspection and verification results recorded in the sections retain their original 8cc0ca3 attribution. The submission does not convert historical or incomplete checks into new successful runs.

Material	Accessible link and purpose
Frozen repository version	https://github.com/Adilzhan17/egin/tree/prob-period-1 Source code, report sections and evidence at prob-period-1
Live web prototype	https://egin.perricheno.com Public web application; a live service may change after submission
Staging prototype	https://dev-egin.perricheno.com Web staging environment
Developer API portal	https://api-egin.perricheno.com Public API documentation and developer interface
Editable diagram sources	https://github.com/Adilzhan17/egin/tree/prob-period-1/docs/DIP/diagrams PlantUML sources and Mermaid navigation source
Interface evidence	https://github.com/Adilzhan17/egin/tree/prob-period-1/docs/DIP/evidence/ux Screenshots, reproduction script and capture metadata
Verification evidence	https://github.com/Adilzhan17/egin/tree/prob-period-1/docs/DIP/evidence/verification Retained execution record; see Section 9 for qualifications
Editable report sources	https://github.com/Adilzhan17/egin/tree/prob-period-1/docs/DIP All eleven Markdown documents and the reproducible PDF builder

> Delivery state: the web/PWA implementation is available; Capacitor support is configured. A tested store-ready Android/iOS release, pending integrations and completed outcome evaluation are separate milestones. Live prototype links are convenience access points; the tagged repository and capture metadata define the report's evidence version.

1. Project Proposal

Editable section source · PART-1-PROPOSAL.md

Parameter	Value
Project title	EGIN — A cross-platform mobile and web application (Capacitor & PWA) for managing a land plot
Team and roles	**Rustam Seitzhan** — Full-Stack Developer *(Offline Synchronisation Engine, WebAuthn & Capacitor Passkeys, Cross-Device QR Sign-in, Security, CI/CD & Blue/Green Deployment)* **Adilzhan Amangeldi** — Full-Stack Developer *(EGKN Cadastral GIS Adapter & Reprojection, 3D Crop Scene & Weather Module, 1C OData Connector, External API & Developer Portal)*
Supervisor	Shuteyeva Gulnur Serikovna
Repository	`https://github.com/Adilzhan17/egin`

1.1 Problem statement

Owners and managers of small farms in Kazakhstan have to assemble a picture of their plot from scattered sources: boundaries from the public EGKN cadastral map, weather from separate services, and field records kept apart from the accounting system. Cloud-only systems require a stable connection, which is often weak or unavailable in rural areas.

Published studies support this context: Kazakh research names weak and unstable mobile coverage in rural areas, a shortage of skills and local specialists, and the absence of a single digital ecosystem among the main obstacles to agricultural digitalisation [2]; an FAO report notes low awareness of digital technologies among medium and small farmers [3]; and a systematic review focused on smallholders identifies infrastructure, cost and digital literacy as the primary adoption barriers [1].

1.2 Target users

- **Primary:** owners and managers of small crop farms and peasant farms (KH), and agronomists or field workers who keep plot-level records in the field on Android and iOS smartphones.
- **Secondary:** farm accountants who maintain records in 1C; developers of external integrations using the public `/v1` API and the developer portal.

1.3 Aim

To develop a cross-platform mobile and web application (built on a single codebase using Capacitor for Android and iOS alongside an installable PWA) that brings together on one screen the plot boundaries (by cadastral number), a local weather forecast, field records with photos and audio, a sensor registry and reports, while continuing to work offline and synchronising data between devices safely.

1.4 Objectives

1. Study user needs, existing farm management systems and technology alternatives for cross-platform mobile delivery.
2. Formulate functional and non-functional requirements with measurable acceptance criteria.
3. Design the system architecture: system context diagram, use case diagram, component diagram (including the Capacitor cross-platform bridge), deployment diagram and key sequence diagrams.
4. Implement the offline store and synchronisation engine with monotonic sequence cursors, record versions, tombstones for deletions and explicit conflict resolution.
5. Implement passwordless sign-in and registration via WebAuthn and Capacitor passkey plugins (`@capgo/capacitor-passkey`) with a one-time recovery code and cross-device QR authentication.
6. Implement plot creation: search by cadastral number via the public EGKN map with UTM-to-WGS84 reprojection, GeoJSON import and export, and geolocation as a fallback.
7. Implement a weather module based on the coordinates of the selected plot, an event centre and report export (HTML, CSV, JSON) using Web Share and Capacitor Filesystem/Share APIs.
8. Implement a sensor registry (QR, manual entry) and prepare the contract for telemetry integration.
9. Implement read-only import from 1C via standard OData and an external `/v1` API with scoped keys and a developer portal.
10. Develop and carry out a verification plan (automated server and browser tests in Chromium and WebKit, plus checks on Android and iOS devices).

1.5 Expected outcomes

- A cross-platform application built from a single TypeScript/React codebase (`apps/mobile`): deployed on the web as an installable PWA (production at `https://egin.perricheno.com` and staging at `https://dev-egin.perricheno.com`) and packaged via **Capacitor** for **Android and iOS** (using `CapacitorHttp`, `Filesystem`, `Share` and `@capgo/capacitor-passkey`).
- A complete documentation package: proposal, literature review, system and technology comparisons, requirements specification, architectural decisions and diagrams, and verification plan.
- An automated test suite (Node.js server tests and Playwright browser scenarios in Chromium and WebKit) integrated into GitHub Actions CI.

1.6 Project scope

In scope (core v0.3 implemented; cross-platform Capacitor delivery configured):

- Cross-platform mobile shell (Capacitor for Android and iOS) and installable PWA sharing one UI and offline engine.
- 3D crop scene on the home screen, a dedicated weather tab (Open-Meteo), agricultural news (ElDala.kz feed) and an event centre.
- Cadastral map of plots: search by 11–12-digit cadastral number, selection by tap, Polygon contours with inner rings, GeoJSON import and export, and offline storage of boundaries and public parcel snapshots.
- Offline-first storage and synchronisation between devices, explicit conflict resolution, and background sending via Background Sync where supported.

- User profile, WebAuthn / Capacitor passkey registration and login, QR sign-in on a second device, one-time recovery code rotation, and push notifications.
- Sensor registry and QR format; offline report export (HTML, CSV, JSON); read-only import from 1C over HTTPS OData; external `/v1` API with hashed scoped keys and a developer portal (`https://api-egin.perricheno.com`).
- Zero-downtime blue/green deployment behind a persistent Nginx gateway (`deploy/manage.py`), with automated online SQLite backups.

Out of scope for the current stage:

- Separate platform-specific codebases written in pure Kotlin or Swift (all mobile targets are delivered cross-platform through Capacitor).
- Receiving live sensor telemetry and automatic alerts based on it.
- Growth-phase analytics and soil measurements (the 3D scene is explicitly labelled as an example until analytics is connected).
- eGov sign-in and contractual Smart Bridge `LandCoordinates` integration (explicitly marked "В разработке").
- Live data exchange with AgroSignal and AgroStream (only owner-scoped configuration preparation is included).
- Multiple disjoint polygons within a single plot record.
- Automatic Capacitor APK/AAB/IPA packaging on every CI push (mobile packaging is triggered only on an explicit release command).

1.7 Engineering complexity

Component	Source of engineering complexity
Offline synchronisation	Monotonic sequence cursor (`seq`), per-record `version`, tombstones (`deleted=1`), payload-equality idempotent retries, explicit conflict resolution, and a single engine (`sync-engine.js`) shared by the UI and the service worker
Passkey & Cross-Platform Auth	Server-side verification of challenge, RP ID, origin, cryptographic signature and user verification (`@simplewebauthn/server`); cross-platform passkey bridge (`@capgo/capacitor-passkey`) with Android Digital Asset Links and iOS Associated Domains; browser-bound QR login; SHA-256 hashed one-time recovery code rotation
Cadastral map adapter	Reprojection from regional UTM zones (`EPSG:32639`–`32645`) to `WGS84` via `proj4`, inner ring area subtraction, rate limits (24/60/240 req/min), TLS verification with a bundled Sectigo intermediate CA, and strict stripping of personal data (names, IIN/BIN, contacts)
1C connector	Parsing of OData XML `$metadata` and namespace aliases, bounded paged reading (≤ 1,000 rows, ≤ 8 MiB, ≤ 90 s), SSRF protection (public IPv4 over HTTPS/443 only, pinned DNS, no redirects), AES-256-GCM password encryption, and idempotent owner-scoped import that preserves manual edits

Component	Source of engineering complexity
External API and portal	Keys with granular read/write scopes and 7/30/90/365-day expiry, only SHA-256 secret hashes stored, 60 req/min shared rate limits, version-aware updates via `If-Match`, and separate portal origin/session isolation
Deployment and tests	Blue/green container switching behind a permanent Nginx gateway, instant rollback, online SQLite backups; browser tests using a virtual WebAuthn authenticator across Chromium and WebKit

1.8 Realism of the scope and main risks

- **Core readiness:** the core application, server modules and Capacitor cross-platform configuration are already implemented in version 0.3, while full store-ready Android/iOS packaging is scheduled only for explicit release milestones.
- **Public cadastral interface:** the EGKN adapter uses the public map endpoints rather than a contractual Smart Bridge integration, so the upstream interface may change. *Mitigation:* GeoJSON import/export and device geolocation are implemented as full fallback paths, and contractual access to `LandCoordinates` is defined as the target commercial state.
- **Weather API licensing:** Open-Meteo is free for non-commercial use under CC BY 4.0; commercial operation requires transitioning to a paid plan.
- **iOS Capacitor passkeys:** iOS Associated Domains for the Capacitor passkey plugin require an Apple Developer Team ID (`apple-app-site-association`), whereas web passkeys in Safari/PWA work immediately on the domain without it.

2. Evidence of the Relevance

Editable section source · PART-2-EVIDENCE-OF-RELEVANCE.md

2.1 Stakeholder engagement and relevance to EGIN

During the initial exploration of EGIN, the project team held informal discussions with approximately **10–15 potential users: farmers and agronomists in Kazakhstan**. According to the team's account, the participants included owners or managers of substantial agricultural land holdings, with areas described in hectares. The discussions concerned the project idea, its potential usefulness and problems in the agricultural sector. Exact farm sizes, participant-level responses, meeting dates and interview transcripts have not yet been included in the evidence package.

The team also **met with Zhiguli Dairabayev, identified by the team as a representative of the Auyl party**, and reports that he expressed support for promoting the project. His official parliamentary profile provides background on his agricultural policy involvement [19]. **This link identifies the stakeholder; it is not a published record of the team's meeting.** A dated meeting note, photograph with context, correspondence or an event publication is still needed to document that particular encounter.

These activities establish contact with people connected to the intended application domain and provide a reasonable basis for further requirements discovery. EGIN's implemented plot management, cadastral lookup, coordinate-based weather, local data storage, event centre and report export provide concrete scenarios to discuss with those stakeholders. The project therefore has an inspectable prototype around which subsequent validation can be organised, rather than only an abstract product proposal.

However, general interest in the idea does not establish that any particular feature is required, that users will adopt the product, or that the system improves farm outcomes. No participant quotations, feature rankings, willingness-to-pay figures or measured productivity gains are claimed here. **Support for project promotion does not constitute an official partnership, a procurement commitment, a signed client agreement or acceptance of the delivered software.**

The current evidence is exploratory. The proposal primarily targets small farms, whereas some reported discussions involved managers of larger holdings. Follow-up validation should therefore distinguish farm size, role, device access and existing work practices, rather than treating these groups as interchangeable.

2.2 Evidence register and strength of claims

The evidence review concerns repository snapshot `8cc0ca3f73052b030dcddb990b8a5460be9ddcdf`, reviewed on **23 October 2026**, together with the project owner's account supplied during documentation preparation. That date is the review date, not the date of the interviews or stakeholder meeting.

ID	Evidence	What it supports	Current status and limitation
E2-01	The team's account of discussions with approximately 10–15 potential users: farmers and agronomists in Kazakhstan	Initial contact with potential users and discussion of the agricultural problem domain	**Team-reported.** Individual records, dates, consent and attributable findings are not yet attached; this is not a verified count of structured interviews.
E2-02	The team's account of its meeting with Zhiguli Dairabayev and his support for promotion	Stakeholder engagement and interest in communicating the project	**Team-reported.** The official profile [19] supplies background only. Meeting-specific documentation and the precise scope of support remain to be attached. No formal agreement is asserted.
E2-03	The literature and regional context reviewed in Part 3, with sources in References	A secondary-source rationale for investigating rural connectivity, usability and fragmented agricultural information	**Documented literature rationale.** Sector-level findings are not observations from these particular participants, nor proof of EGIN's effectiveness.
E2-04	The application source, test scenarios and prototype inventory in Part 8 and Part 9	The existence of software mechanisms that can be evaluated against the proposed needs	**Inspectable implementation evidence.** Code and screenshots establish capability or presentation, not user demand or successful field adoption.
E2-05	Owner-defined project constraints in AGENTS.md and the scope in Part 1	Phone-first design, the prominent crop view, clear eGov limitations, environment isolation, controlled deployment and developer-portal boundaries	**Recorded project direction.** These constraints come from the project owner and design scope; they are not quotations from farmers or a signed external client contract.

Only E2-01 and E2-02 concern direct engagement reported by the team. E2-03 provides secondary context, E2-04 provides technical evidence and E2-05 records product decisions. These categories must remain separate when presenting requirements to the committee.

2.3 Requirements hypotheses to validate

The following hypotheses connect the project to testable user needs. They are **proposed interpretations**, not reconstructed interview results. Requirement identifiers refer to Part 6.

ID	Hypothesis	Relevant implementation and requirement	Evidence needed to accept or revise it
H1	Access to saved plot information and reliable local changes are useful when a field worker temporarily loses connectivity.	Local field and sensor data in IndexedDB, synchronisation queue and explicit conflicts; FR-FLD-6, FR-SYNC-1–8, NFR-REL-1. Manual diary entry is not an active user flow in the current event-centre UI.	Observe a real task under the participant's normal connectivity conditions; record which information must remain available, and test recovery after reconnection.
H2	Finding a plot by an existing cadastral identifier can reduce the effort of establishing its boundary compared with a user's current manual process.	EGKN search, preview and confirmation, with GeoJSON/geolocation fallbacks; FR-FLD-1–4, NFR-USE-4.	A paired task with the same plot and verified identifier; record completion time, boundary correctness, failures and participant comments.
H3	Passwordless access with an explicit recovery path can be understandable and usable for the intended users.	WebAuthn registration, recovery and cross-device QR; FR-AUTH-1–4.	A first-use and recovery walkthrough on real participant devices; record assistance, errors and successful completion. Security tests alone do not establish usability.
H4	A read-only connection to an existing 1C system may reduce duplicate entry for farms that actually use compatible OData publication.	The 1C connector and owner-scoped import; FR-INT-1–2.	Confirm which participants use 1C, obtain an authorised test instance and compare a defined import task with their current process. No client deployment is claimed.
H5	Presenting a forecast for the selected plot can make locating relevant weather information easier than switching between separate tools.	Plot-coordinate weather and freshness indicators; FR-WTH-1–3.	Compare retrieval tasks using the same location, forecast source and observation time; assess location selection and freshness comprehension, not forecast accuracy or crop yield.

The prominent 3D crop view is an owner-defined design choice (E2-05). It should be evaluated for comprehension and usability; it is not presented as a farmer-validated requirement or a measurement-driven digital twin. Likewise, the existence of chat, marketplace and trade modules does not demonstrate demand for those extensions: their stakeholder value requires separate evidence.

2.4 Interview record structure

For the assessment, prepare **3–5 traceable discussion records** selected from the reported engagement, or collect follow-up records where earlier notes cannot be recovered. Do not convert the estimated 10–15 participants into completed structured interviews retrospectively. The slots below are a collection template, not five already documented participants.

2. Evidence of the Relevance · Table 2.3

Participant	Role	Agricultural Context	Discussed Problem	Feedback	Requirement Implication
P01 — record pending	To be recorded	Farm scale, region, crops and current tools: to be recorded	To be taken from a dated discussion note	Verified paraphrase or consented quotation pending	Link a specific observation to an FR/NFR or hypothesis after review
P02 — record pending	To be recorded	To be recorded; do not infer from another participant	Source record pending	Source record pending	Pending evidence review
P03 — record pending	To be recorded	To be recorded; do not infer from another participant	Source record pending	Source record pending	Pending evidence review
P04 — optional additional record	To be recorded	To be recorded	Source record pending	Source record pending	Pending evidence review
P05 — optional additional record	To be recorded	To be recorded	Source record pending	Source record pending	Pending evidence review

Each completed record should additionally contain its date, discussion format, interviewer, participant permission, supporting-file identifier and the distinction between a direct quotation and a team interpretation. Use pseudonyms in the report where appropriate. Keep identifiable source material in an agreed restricted location and attach only material approved for the defence.

Suggested follow-up questions are: What plot-related task was last performed? Which tools were used? Where was information copied or searched for twice? What happened when connectivity was lost? Which proposed EGIN task would be useful, unnecessary or confusing? What would prevent adoption? Ask these questions before presenting a preferred feature, and record negative as well as positive feedback.

2.5 Supporting materials to attach

Material	Purpose	Current availability
Dated notes for at least three substantive user discussions, ideally five	Establish who was consulted, what was discussed and how a requirement follows from the discussion	Not supplied in the current repository evidence package
An anonymised participant/context register	Distinguish farmers, agronomists and managers, and separate small-farm needs from larger-operation needs	Template provided in Section 2.4; individual records pending
A contextualised photograph, correspondence or dated note of the Dairabayev meeting	Document that the reported meeting took place	Meeting reported by the team; meeting-specific attachment pending
Recording excerpts or a participant-approved summary, where available and permitted	Support the substance of a discussion rather than merely the presence of attendees	Not supplied; do not invent transcripts or quotations
A requirement decision log linking evidence IDs to FR/NFR IDs	Show which decisions changed because of stakeholder input and which remain design hypotheses	Initial mapping in Sections 2.2–2.3; participant-level validation pending
An actual client letter or agreement, only if one is later obtained	Establish a defined client relationship and its agreed scope	No signed client agreement is claimed or required as a substitute for the reported meeting

A photograph can support attendance but cannot by itself establish agreement on requirements. A public biography supports identity but cannot establish that a private meeting occurred. Attach these materials with accurate captions and the appropriate claim scope.

2.6 Implications for the next evaluation stage

The available account supports continued investigation of EGIN with agricultural users, while the implementation provides tasks that can be demonstrated and measured. The next step is to make the existing engagement traceable, select representative tasks and execute the protocol in Part 9. Requirements should then be retained, revised or rejected on the basis of recorded observations.

Criterion 2 should therefore be presented as **stakeholder engagement reported; supporting records and feature-level validation pending**. The report does not claim that promotional support or an estimated participant count alone satisfies the evidence standard.

3. Literature Review

Editable section source · PART-3-LITERATURE-REVIEW.md

3.1 Selection method

The literature search covered the six technical domains that govern EGIN's design decisions: barriers to the adoption of digital agriculture by smallholders, the regional context in Kazakhstan, IoT sensors and digital twins in crop farming, local-first offline synchronisation and PWAs, WebAuthn passkey usability and security, and open weather data sources.

Selection criteria: peer-reviewed publications from 2021–2026, supplemented by two foundational computer-science papers on local-first software and Conflict-free Replicated Data Types (CRDTs), one international organisation report (FAO) on Kazakhstan, and the official reference for the weather API used in the system. In total, the review analyses **13 sources**, including **11 scientific publications** (exceeding the minimum of 8) and **11 sources published within the last five years**.

3.2 Critical review

Adoption barriers and the Kazakh context

The systematic review by Gumbi et al. demonstrates that digital agriculture systems remain much less accessible to smallholder farms than to large agro-enterprises: the primary barriers are rural connectivity infrastructure, subscription costs and digital literacy, compounded by a weak perceived value of complex platforms [1]. For Kazakhstan, Abdrazakova et al. reach consistent conclusions, highlighting weak and unstable mobile coverage in rural districts, a shortage of local IT/agronomy specialists, and the absence of a unified digital ecosystem [2]. The FAO country report likewise records low awareness of digital tools among small and medium Kazakh farmers and a weakly developed extension and advisory system [3].

A limitation of these studies is that they analyse the agricultural sector at a macro level rather than evaluating specific mobile interfaces. For EGIN, their findings translate into three strict architectural constraints: **offline-first operation**, a **low entry barrier** (phone-first UI, free web access without mandatory registration for viewing the home screen and weather), and **reuse of existing structured data** (state cadastre and 1C accounting) instead of forcing manual entry from scratch.

Sensors and digital twins

Ndjuluwa et al. systematise IoT sensors, communication channels and network protocols in crop farming, covering soil moisture, temperature monitoring and yield estimation [4]. Recent reviews of agricultural digital twins classify crop, weather and soil models while emphasising major practical barriers: the requirement for continuous high-quality sensor streams, privacy concerns, complex multi-source integration [5], high data-management overhead [6], and the early maturity stage of crop-monitoring twins [7].

The critical engineering conclusion for EGIN is that **a 3D visualisation must not be presented as a digital twin unless backed by verified field measurements**. Accordingly, EGIN explicitly labels its 3D crop scene as an illustrative example until analytics is connected, and the sensor registry records device metadata without imitating live telemetry.

Offline-first, synchronisation and PWA

Kleppmann et al. formulate the *local-first* software principles, placing the primary copy of user data in local storage on the client device and treating the server as a secondary synchronisation peer [8]. While Conflict-free Replicated Data Types (CRDTs) guarantee strong eventual consistency without central coordination [9], Kleppmann et al. note unresolved user-interface challenges when exposing automatic merges to end users [8], and CRDT metadata increases storage and implementation complexity.

Because plot records in a small farm are edited by one or a few people rather than concurrently by dozens of collaborators, EGIN adopts a simpler, auditable model: a monotonic server sequence cursor (`seq`), per-record integer versions, tombstones for deletions, and an **explicit user choice when a version conflict occurs**. The viability of delivering offline-capable tools to smallholder farmers via a Progressive Web App (PWA) is further supported by recent work targeting low-connectivity rural environments [12].

Passkey authentication

Lassak et al. investigate FIDO2/WebAuthn deployment barriers and identify account recovery, cross-device friction, platform inconsistencies and user mental models as the dominant challenges [10]. In a large-scale measurement study of 872 passkey-enabled websites, Jannett et al. uncover widespread inconsistencies in key lifecycle management, continued support for deprecated algorithms, and 15 concrete attack vectors—including unauthorized passkey deletion and account lockout [11].

These two security studies directly define EGIN's authentication requirements and test cases: issuing a **one-time recovery code with automatic rotation** on use, **forbidding deletion of the last remaining passkey**, binding cross-device QR sign-in to the initiating browser cookie and a verified passkey assertion, and testing against key-deletion lockout scenarios.

Weather data

Open-Meteo aggregates numerical weather prediction models from national meteorological services behind an open API licensed under CC BY 4.0 (attribution required, free for non-commercial use) [13]. Its limitation is that gridded forecast models do not replace in-field microclimate stations; therefore, EGIN presents Open-Meteo data strictly as a coordinate-based forecast and marks cached forecasts as stale when viewed offline.

3.3 Review conclusions

1. The requirement for offline-first operation and minimal onboarding friction is directly supported by both global smallholder reviews [1] and Kazakhstan-specific studies [2], [3].
2. Literature on agricultural digital twins [5]–[7] justifies strict interface honesty: separating the 3D visual scene and sensor registry from unimplemented telemetry and growth-phase computation.
3. Foundational local-first and CRDT works [8], [9] and recent USENIX Security studies on passkeys [10], [11] provide concrete architectural trade-offs and security verification scenarios for EGIN's sync engine and WebAuthn implementation.

3.4 Structured analysis table of publications

3. Literature Review · Table 3.1

#	Source	Type and year	Key findings	Limitations	Use in EGIN
1	Gumbi N., Gumbi L., Twinomurinzi H. *Towards Sustainable Digital Agriculture for Smallholder Farmers: A Systematic Literature Review.* Sustainability, 2023, 15(16), 12530	Scientific, 2023, systematic review	Primary barriers: rural infrastructure, cost, digital literacy; low early adoption; unclear perceived value of complex systems	Covers smallholders globally rather than Kazakhstan; does not evaluate specific mobile UIs	Rationale for offline-first architecture, phone-first layout and free web entry
2	Abdrazakova A., Majerová I., Abdildin N. *Digital Agriculture: Challenges and Prospects in Kazakhstan.* Silesian University in Opava, 2022, pp. 1–15	Scientific (conference paper), 2022	Weak rural mobile coverage, shortage of local specialists, and absence of a unified digital ecosystem in Kazakhstan	Relies on sector-level statistics and macroeconomic surveys rather than field-level usability studies	Problem statement context for Kazakhstan; justification for offline boundary & record storage

#	Source	Type and year	Key findings	Limitations	Use in EGIN
3	FAO. *Promotion and Exchange of Good Practices on the Use of Digital Technologies in Agriculture: Republic of Kazakhstan.* April 2022	International organisation report, 2022	Low awareness of digital tools among small and medium farmers; weakly developed advisory services	Situational policy report, not a peer-reviewed experimental study	Confirms the need for simple onboarding and reuse of familiar identifiers (cadastral number, 1C)
4	Ndjuluwa L.N.P., Adebisi J.A., Dayoub M. *Internet of Things for Crop Farming: A Review of Technologies and Applications.* Commodities, 2023, 2(4), 367–381	Scientific, 2023, review	Systematises agricultural sensors, communication channels and protocols for soil monitoring and yield estimation	Technical survey without UX evaluation or smallholder cost analysis	Defines the sensor registry domain model (`moisture`, `temperature`, `weather`) and future telemetry contract
5	Peladarinos N. et al. *Enhancing Smart Agriculture by Implementing Digital Twins: A Comprehensive Review.* Sensors, 2023, 23(16), 7128	Scientific, 2023, review	Classifies crop, weather and soil twins; highlights data quality, privacy and multi-source integration barriers	Broad survey; does not evaluate lightweight mobile deployments on small farms	Justification for not presenting the 3D scene as a digital twin without live measurements
6	Escribà-Gelonch M. et al. *Digital Twins in Agriculture: Orchestration and Applications.* J. Agric. Food Chem., 2024, 72(19), 10737–10752	Scientific, 2024, review	Lifecycle analysis of agricultural digital twins; high complexity of orchestrating heterogeneous data streams	Focuses on large-scale production infrastructure rather than mobile field notebooks	Defines scope boundaries and data-quality prerequisites for future analytics
7	Melesse T.Y. *Digital twin-based applications in crop monitoring.* Heliyon, 2025, 11(2), e42137	Scientific, 2025, review	Compares physics-based, agent-based, data-driven and hybrid crop models; notes early stage of practical adoption	Review article without new field experiments	Confirms that growth phases must not be computed from synthetic or missing inputs
8	Kleppmann M., Wiggins A., van Hardenberg P., McGranaghan M. *Local-first software: you own your data, in spite of the cloud.* Onward! 2019, ACM, pp. 154–178	Scientific (foundational), 2019	Formulates local-first principles (primary local copy, offline operation, multi-device sync) and notes UI challenges of automatic merge models	Oriented toward collaborative document editing; prototype-based essay	Core architectural pattern: IndexedDB as the primary working store with background sync
9	Shapiro M., Preguiça N., Baquero C., Zawirski M. *Conflict-free Replicated Data Types.* SSS 2011, LNCS 6976, pp. 386–400	Scientific (foundational), 2011	Formalises conditions for replica convergence without coordination (strong eventual consistency)	Theoretical distributed-systems paper; does not address mobile UX or storage overhead	Evaluated as an alternative in Section 5.3; rejected in favour of explicit version conflict resolution

#	Source	Type and year	Key findings	Limitations	Use in EGIN
10	Lassak L. et al. *Why Aren't We Using Passkeys?* USENIX Security Symposium, 2024	Scientific, 2024, empirical study	Identifies account recovery, cross-device friction and technical edge cases as major FIDO2 adoption barriers	Interviews conducted with organisation security specialists rather than end users	Drives requirements for one-time recovery code rotation and cross-device QR login
11	Jannett L. et al. *The State of Passkeys: Studying the Adoption and Security of Passkeys on the Web.* USENIX Security Symposium, 2026	Scientific, 2026, measurement study	Analyses 872 passkey sites; finds inconsistent key management, deprecated algorithms and 15 attack types including key-deletion lockout	Evaluates web relying parties in general rather than agricultural PWAs	Defines security rules (forbidding last-key deletion, strict RP/origin checks) and test cases
12	Tantaroudas N., Karachalios I., McCracken A. *Retrieval-grounded multilingual LLM assistance for island smallholder farmers.* Open Research Europe, 2026	Scientific (version 1, open peer review), 2026	Demonstrates a lightweight PWA architecture tailored to smallholder farmers with low digital literacy and poor connectivity	Focuses on an LLM advisory chatbot rather than cadastral plot record-keeping	Supports the choice of a PWA delivery model for rural smallholders
13	Zippenfenig P. *Open-Meteo.com Weather API.* Zenodo, 2023, doi: 10.5281/zenodo.7970649	Software with official citation reference, 2023	Aggregates open national weather service models; CC BY 4.0 licence; free for non-commercial use	Gridded forecast model rather than in-field station telemetry; commercial use requires a paid plan	Weather data provider for plot coordinates (10-day and hourly forecasts)

4. Existing System Comparison

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Four established agricultural management systems available to farms in Kazakhstan and documented in open sources are compared against EGIN v0.3 across **eight consistent criteria**: **Cropwise Operations** (Syngenta Digital) [14], **EOSDA Crop Monitoring** [15], **AGRIVI** [16], and **AgroSignal.Expert** [17].

4. Existing System Comparison · Table 4.1

Criterion	EGIN v0.3 [18]	Cropwise Operations [14]	EOSDA Crop Monitoring [15]	AGRIVI [16]	AgroSignal.Expert [17]
1. Offline operation	**Yes:** IndexedDB store, sync queue, saved plot boundaries and cached weather work offline; basemap tiles require network	**Yes:** mobile field data viewing and scouting after initial synchronisation	**Yes:** offline and online modes in the mobile scouting app	**Yes:** mobile app supports offline field record entry with subsequent sync	**Yes:** offline field inspections in the "AgroSignal.Scouting" Android app with later upload
2. Weather for a specific plot	**Yes:** Open-Meteo by selected plot coordinates (current, hourly, 10-day forecast, 24 h cache)	**Yes:** localised weather forecast per field	**Yes:** 14-day field forecast, historical archive, weather risk alerts	**Yes:** field-level weather forecasts and condition alerts	**Yes:** weather module and integration with local weather stations
3. Satellite monitoring and vegetation indices	**No** (out of scope: 3D scene is an illustrative example; growth phase is not computed)	**Yes:** satellite vegetation monitoring (NDVI), field zoning and scouting maps	**Yes:** NDVI, NDRE, MSAVI indices, zoning and vegetation maps	**Yes:** satellite imagery and NDVI vegetation health monitoring	**Yes:** NDVI vegetation maps in the web platform
4. Sensors and IoT	**Registry only:** QR and manual sensor registration linked to a plot; live telemetry reception is out of scope	**Yes:** integration with GPS machinery telematics and weather stations	**Partial:** soil moisture analytics (satellite-derived) and partner station integrations	**Yes:** integration with third-party IoT weather stations and soil sensors	**Yes:** telemetry from weather stations, soil sensors and machinery fuel/GPS sensors

Criterion	EGIN v0.3 [18]	Cropwise Operations [14]	EOSDA Crop Monitoring [15]	AGRIVI [16]	AgroSignal.Expert [17]
5. Plot boundaries and KZ cadastre	**Yes:** built-in search and tap-identification via the public Kazakhstan EGKN map (UTM → WGS84), plus GeoJSON import/export	**No built-in KZ EGKN lookup:** manual drawing or Shapefile/KML/GeoJSON import	**No built-in KZ EGKN lookup:** manual drawing or Shapefile/KML/GeoJSON import	**No built-in KZ EGKN lookup:** manual polygon mapping or file import	**No built-in KZ EGKN lookup:** KML/Shapefile import or manual contouring
6. Accounting and external integrations	**1C via standard OData** (read-only, encrypted credentials) + public `/v1` **REST API** with scoped keys and developer portal	Machinery GPS telematics, ERP connectors for enterprise clients	Custom enterprise API (on paid plans)	Agworld, ERP and accounting integrations on enterprise tiers	1C integration module, telematics and partner API
7. Access and pricing model	**Free web access** (no contract required to start; commercial tier undefined)	**Contract-gated:** mobile app available only to registered Cropwise customers via sales	**Freemium / Per-hectare subscription:** limited free demo field, paid plans by area/users	**Commercial subscription:** quote-based enterprise and farm plans	**Commercial SaaS:** contract-based deployment via sales
8. Client platform	**Cross-platform (Capacitor for Android & iOS + Web PWA from a single codebase)**	Web portal, separate iOS and Android apps	Web portal, iOS and Android scouting apps	Web portal, iOS and Android apps	Web portal; Android-only app for field scouting

Conclusions from the comparison

1. **Where existing platforms are stronger:** mature commercial systems (Cropwise, EOSDA, AGRIVI, AgroSignal) provide satellite vegetation indices (NDVI) and live machinery/sensor telemetry. EGIN deliberately excludes satellite analytics and live telemetry from its current scope rather than attempting to duplicate heavyweight agronomic platforms.
2. **EGIN's specific niche:** none of the global platforms provide built-in plot lookup by a **Kazakhstan 11–12-digit cadastral number** against the public EGKN map; they require manual polygon drawing or uploading pre-prepared GIS files (Shapefile/GeoJSON). EGIN combines instant Kazakh cadastral boundary lookup, coordinate weather, offline field records with media, read-only 1C OData import, and passwordless passkey sign-in in a single cross-platform application (Capacitor for Android/iOS + Web PWA).

3. **Low entry barrier:** competing full-featured systems require sales onboarding or per-hectare subscriptions (Cropwise, AGRIVI, AgroSignal), whereas EGIN opens immediately in a mobile browser as a PWA and packages into Android/iOS apps via Capacitor without maintaining separate platform-specific codebases [1]–[3].

5. Technology Comparison

Editable section source · PART-5-TECHNOLOGY-COMPARISON.md

Six key architectural decisions were evaluated against the project requirements, with three alternatives considered for each decision.

5.1 Client platform (Cross-platform strategy)

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. Cross-platform Web & Mobile via Capacitor + PWA *(chosen)*	Single TypeScript/React codebase (`apps/mobile`) for **Android, iOS and Web**; Capacitor packages the local `dist` bundle into Android/iOS apps and bridges device APIs (`CapacitorHttp`, `Filesystem`, `Share`, `@capgo/capacitor-passkey`, camera, microphone, geolocation); simultaneous zero-install web access as a PWA with Service Worker offline caching	Background Sync API is supported in Chromium (on WebKit/Safari, queued changes sync when the app is opened); Web Push on iOS requires Home Screen PWA installation (iOS 16.4+); iOS Associated Domains for the Capacitor passkey plugin require an Apple Developer Team ID	**Chosen:** delivers cross-platform Android, iOS and Web apps from one codebase
B. Platform-specific native development (Kotlin for Android + Swift for iOS)	Direct low-level OS API access without a bridge layer	Requires writing and maintaining two separate mobile codebases plus a third web application for browser access; triples development and testing effort for a two-person team	**Rejected:** EGIN is designed as a cross-platform project rather than separate native apps
C. Widget-based cross-platform frameworks (Flutter / React Native)	Single codebase for Android and iOS stores	Weak fit for a web-first offline PWA with heavy WebGL (Three.js 3D crop scene) and DOM/canvas mapping (Leaflet); would require separate web maintenance or rewriting the existing v0.3 core	**Rejected:** Capacitor provides superior web/mobile code reuse for EGIN's stack

5.2 Server platform and data store

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. Node.js 22 (Express) + SQLite in WAL mode *(chosen)*	Zero-dependency single-node operation (`node:sqlite`); atomic `BEGIN IMMEDIATE` transactions; fast local reads; simple online file backups (`/data/backups/`) shared safely across blue/green slots	Single-writer concurrency ceiling; per-profile quotas enforced (10,000 records and 128 MB of binary assets, ≤ 8 MB per asset)	**Chosen** for the core diploma scope

Alternative	Advantages for EGIN	Limitations and risks	Verdict
B. Node.js + PostgreSQL	High write concurrency, horizontal scaling, rich relational indexing	Requires operating and backing up a separate DBMS container; higher memory footprint on a single host (used only in out-of-scope platform/trade modules)	**Deferred:** candidate for future multi-tenant scaling
C. Backend-as-a-Service (Firebase / Firestore)	Built-in client offline cache and rapid initial prototyping [8]	Vendor lock-in; risk of violating Kazakhstan's data residency requirements when hosted in foreign cloud regions; unpredictable per-read/write costs	**Rejected**

5.3 Offline storage and synchronisation model

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. IndexedDB + monotonic `seq` cursor, integer `version`, tombstones, and explicit user conflict resolution *(chosen)*	Guarantees zero silent data loss (`NFR-REL-1`); deterministic server validation; deletions propagate reliably as `deleted=1` tombstones; payload-equality checks make retries idempotent; one engine (`sync-engine.js`) serves both UI and Service Worker across Web and Capacitor	Requires the user to choose a version when a concurrent offline edit occurs on two devices	**Chosen:** optimal for small-farm plots edited by 1–3 users
B. CRDTs (Automerge / Yjs)	Automatic conflict-free replica convergence without central coordination [8], [9]	High metadata and memory overhead; complex schema migrations; unresolved UX ambiguity when automatic merges alter structured field records [8]	**Rejected** for current scope; candidate only if real-time multi-user editing is needed
C. Last-Write-Wins (server timestamp priority)	Trivial server implementation	Silently overwrites and loses field observations recorded offline when two devices sync out of order	**Rejected:** violates data-integrity requirements

5.4 Authentication

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. Passkeys (WebAuthn + `@capgo/capacitor-passkey`) + one-time recovery code rotation + cross-device QR login *(chosen)*	Phishing-resistant public-key cryptography across web browsers and Capacitor mobile shells; biometric data never leaves the device; QR flow allows signing in on a second device; forbidding last-key deletion prevents self-lockout	Account recovery and user unfamiliarity noted in literature [10], [11] — mitigated by issuing a rotating one-time recovery code at registration	**Chosen**
B. Email / phone + password	Familiar mental model for traditional web users	Vulnerable to phishing and password reuse; requires external SMTP/SMS delivery infrastructure and secret storage	**Rejected**
C. eGov QR / EDS or SMS OTP	High familiarity in Kazakhstan	Official eGov integration requires Smart Bridge approval (currently shown as "В разработке"); commercial SMS gateways incur per-login operational costs	**Deferred** until official eGov integration is approved

5.5 Plot boundaries and map engine

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. Leaflet + OpenStreetMap / KZ aerial basemaps + server-side adapter for the public EGKN map *(chosen)*	Works without proprietary map API keys in both browser and Capacitor WebView; retrieves exact parcel contours by 11–12-digit cadastral number; reprojects UTM (`EPSG:32639`–`32645`) to `WGS84`; strips PII; saves boundaries offline	Relies on the public EGKN web map interface, which may change; requires bundling the Sectigo intermediate CA certificate and enforcing strict rate limits	**Chosen** (with GeoJSON and geolocation fallbacks)
B. Contractual Smart Bridge `LandCoordinates` service	Official government SLA and legally fixed API contract	Requires formal legal onboarding and approval (not yet arranged)	**Target commercial state**
C. Manual polygon drawing and GeoJSON upload only	Zero external service dependencies	Drawing boundaries on a small phone screen is slow and error-prone; farmers rarely have ready-made GeoJSON files	**Retained as fallback** alongside Alternative A

5.6 Weather data source

Alternative	Advantages for EGIN	Limitations and risks	Verdict
A. Open-Meteo API *(chosen)*	Free for non-commercial use under CC BY 4.0 [13]; no API key required for development; coordinate-level resolution with current, hourly and 10-day metrics	Commercial deployment requires purchasing a commercial API plan; gridded forecasts do not replace in-field weather stations	**Chosen**
B. Commercial global weather APIs (OpenWeather / Tomorrow.io / Google Weather)	Contractual SLA for commercial workloads	Paid tiers required for full 10-day hourly resolution; API key management overhead	**Backup option** for commercial scaling
C. National meteorological service (Kazhydromet)	Official observations from domestic synoptic stations	Lacks an open, self-service REST API offering free coordinate-interpolated 10-day hourly JSON forecasts	**Rejected**

6. Requirements Specification

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6.1 Conventions

Identifiers. `FR-<area>-<n>` for functional requirements and `NFR-<area>-<n>` for non-functional requirements. IDs are stable. A requirement is never renumbered, only marked *Deprecated*.

Priority (MoSCoW).

Priority	Meaning for the diploma
M — Must	Mandatory for the final defence. The project aim is not met without it.
S — Should	Important, planned for the final version. It can be descoped only with a written justification.
C — Could	Desirable if time allows. It does not affect achievement of the aim.
W — Won't (this stage)	Explicitly out of scope (see 1.6). It is listed so that the boundary can be verified.

Verification method.

Code	Method
T	Automated test, run in CI by `pnpm test` (server, unit) or by a Playwright scenario in Chromium and WebKit.
D	Demonstration on a smartphone (Capacitor Android/iOS build or mobile browser / installed PWA) on the staging environment `dev-egin.perricheno.com`.
I	Inspection of code (`server.mjs`, `prepare-native.mjs`, etc.), configuration or database schema.

Code	Method
A	Analysis or measurement (Lighthouse, timing, log review).

Status describes implementation identified in the source, not automatic acceptance of every criterion: *Implemented*, *Partial*, *Planned*, or *Out of scope*. The documentation review uses commit `8cc0ca3f73052b030dcddb990b8a5460be9ddcdf` (23 October 2026). No submission tag existed at that historical inspection. The later `prob-period-1` tag packages this report and preserves the original review baseline. Executed checks, environment limitations and unmeasured targets are recorded separately in Part 9. Capacitor configuration does not by itself establish a tested Android/iOS release.

Traceability. The *Source* column links each requirement to literature (3.x, reference `[n]`), owner-defined project constraints (Section 2.2, E2-05), a hypothesis (Section 2.3) or a technology decision (5.x). Owner constraints are not a signed external client agreement; the team-reported discussions are distinguished from feature-level findings in Part 2.

Rule for "Must". Every **M** requirement has at least one acceptance criterion that is binary (pass/fail) and a concrete verification method (`T`, `D`, `A` or `I` on exact code lines). This satisfies the committee criterion *"whether the fulfilment of each mandatory requirement can be verified"*.

6.2 Actors

Actor	Description
Farmer	Owner or manager of a small farm. Main user of plots, weather, records and reports across Android, iOS (Capacitor) and Web PWA.
Field worker	Agronomist or brigadier. Creates records with photo and audio in the field, often without a network.
Accountant	Maintains the farm's 1C infobase and provides read-only OData access.
External developer	Integrates with EGIN through the `/v1` API and the developer portal.
Operator	Project team. Deploys, monitors and restores the service.
External systems	EGKN public cadastral map, Open-Meteo, ElDala.kz, 1C OData, Web Push services, device authenticator (WebAuthn / Capacitor Passkey).

6.3 Functional requirements

6.3.1 Authentication and account (AUTH)

Platform qualification. Web passkey support is implemented. The checked-in `native-config.json` has `iosPasskeyReady: false`; the native iOS passkey path therefore remains unavailable until the required association/configuration is supplied. A web test result does not accept the full native requirement.

6. Requirements Specification · Table 6.4

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-AUTH-1	The system shall let a user register and sign in with a passkey (WebAuthn in browsers and `@capgo/capacitor-passkey` in Capacitor Android/iOS shells), without a password.	M	(1) Registration creates a profile and one passkey. (2) The server rejects an assertion with a wrong challenge, RP ID, origin or signature, or without user verification (each case returns 4xx and creates no session). (3) Biometric data is never sent to the server.	T (`api.test.mjs`, `auth.cjs` with a virtual authenticator), D	5.4, [10], 2.2 E2-05	Partial — web implemented; native iOS passkey configuration pending
FR-AUTH-2	At registration the system shall issue a one-time recovery code and let the user restore access with it.	M	(1) The code is shown once, and only its SHA-256 hash is stored. (2) Successful recovery invalidates the old code and issues a new one. (3) Reusing a code is rejected.	T (`api.test.mjs`, `auth.cjs`)	[10] (recovery problem), H3 (2.3)	Implemented
FR-AUTH-3	A signed-in user shall be able to add a second passkey and delete passkeys, except the last one.	M	Deleting the last passkey is refused with 409 and an explanatory message. After deletion, the removed key cannot sign in.	T (`auth.cjs`)	[10], [11]	Implemented
FR-AUTH-4	The system shall support sign-in on another device by QR: the new device shows a QR, and the signed-in phone confirms it with a 6-digit code match and a passkey.	S	(1) The QR expires after 2 minutes. (2) The session is issued once, only to the browser that created the QR. (3) Approval without a fresh passkey assertion is impossible. (4) A foreign-origin QR is rejected.	T (`qr-auth.test.mjs`, `auth.cjs`, `qr-scanner.cjs`)	5.4	Implemented

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-AUTH-5	The system shall show eGov sign-in as "В разработке" and shall never present the EGIN QR as an eGov QR.	M	The UI contains no active eGov login control, and no EGIN QR screen mentions eGov as its issuer.	I, D	2.2 E2-05, 5.4	Implemented
FR-AUTH-6	The home screen, weather and news shall be available without signing in. A guest may keep local plot data and view locally available records; server synchronisation and private server data require authentication.	S	An anonymous user opens `/` and can save a plot locally. `/api/sync` without a session returns 401. Private `/v1` resources require a valid scoped API key rather than a browser session.	T (`browser.cjs`, `api.test.mjs`, `developer.test.mjs`), I (`App.tsx`, `FieldsPage.tsx`)	[1] (low entry barrier)	Implemented

6.3.2 Plots and cadastre (FLD)

6. Requirements Specification · Table 6.5

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-FLD-1	The user shall be able to find a plot by its 11–12-digit cadastral number and see its boundary before saving.	M	(1) Spaces, hyphens and colons in the number are normalised. (2) For a valid existing number the contour is shown on the map in WGS84, reprojected from district UTM EPSG:32639–32645. (3) Nothing is saved until the user confirms "Add". (4) An invalid or unknown number gives a clear error, not an empty plot.	T (`cadastre.test.mjs`, `land.test.mjs`, `map-picker.cjs`)	1.7, H2 (2.3)	Implemented
FR-FLD-2	The user shall be able to select a plot by tapping the map at zoom ≥ 17.	S	A tap inside a contour selects it, and a tap inside an inner ring (hole) does not. Several exact matches are offered for choice.	T (`cadastre.test.mjs`, `map-picker.cjs`)	1.7	Implemented
FR-FLD-3	The system shall store a public snapshot of plot details (address, purpose, category, registered area, cadastral value, source URL and fetch time) and shall not store personal data.	M	(1) Personal names, IIN/BIN of individuals and contacts are absent from the API response and the database. (2) Unknown fields are rejected before the transaction. (3) Missing data is shown as "not provided", never as zero or "no owner".	T (`cadastre-details.test.mjs`), I	Legal/ethical constraint, 1.8	Implemented

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-FLD-4	The user shall be able to import and export plot boundaries as GeoJSON.	M	Import accepts files ≤ 2 MiB, ≤ 200 features and ≤ 2,000 vertices in total. It rejects self-intersecting or touching rings with a message. Export, then re-import, gives the same geometry, including holes.	T (`land.test.ts`, `workspace.cjs`)	1.8 (fallback if EGKN changes)	Implemented
FR-FLD-5	If no boundary is available, the user shall be able to create a plot from current geolocation as a point.	S	The point plot is saved offline and used for weather. The UI marks it as "without boundaries".	T (`browser.cjs`)	1.8	Implemented
FR-FLD-6	Saved plot boundaries shall be available offline.	M	After a successful load, switching the device to airplane mode and reloading still shows the list of plots and their contours (the basemap may be missing).	T (`browser.cjs`, offline reload), D	[2], [8]	Implemented
FR-FLD-7	The system shall show both the computed geometric area and the registered area, labelled separately.	C	Both values are visible on the plot passport and are never merged.	T (`parcel-passport.cjs`)	Data honesty	Implemented
FR-FLD-8	A plot consisting of several separate polygons.	W	—	—	1.6	Out of scope

6.3.3 Offline storage and synchronisation (SYNC)

6. Requirements Specification · Table 6.6

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-SYNC-1	All create, edit and delete operations shall work without a network and shall be queued locally (IndexedDB).	M	In offline mode the user creates a plot and a record with a photo, edits it and deletes another record. After reload, all changes are still present and shown as "waiting for sync".	T (`workspace.cjs`, `sync-ui.cjs`)	[2], [8], 5.3	Partial
FR-SYNC-2	The queue shall be sent automatically when the network returns, when the app opens and every 60 s, and manually on request.	M	After the connection is restored, the queue empties without user action. The pending counter reaches 0 and the server holds the records.	T (`workspace.cjs`, `sync-batches.test.ts`)	5.3	Implemented

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-SYNC-3	Each record shall carry a version. A concurrent change on two devices shall produce an explicit conflict that the user resolves. Silent overwriting is not allowed.	M	Two browser contexts edit the same record offline and then sync. The second sync shows a conflict with both versions. After the user's choice, both devices converge to the chosen version.	T (`workspace.cjs`, two browsers; `api.test.mjs`; `conflict-summary.test.ts`)	[8], [9], 5.3	Implemented
FR-SYNC-4	Deletions shall be synchronised as tombstones, so a deleted record does not reappear.	M	A record deleted on device A disappears on device B after sync, and does not come back after B syncs its older copy.	T (`api.test.mjs`, `workspace.cjs`)	5.3	Implemented
FR-SYNC-5	Repeating a sync request with the same payload shall be idempotent.	M	Sending the same batch twice creates no duplicates and does not increase versions.	T (`api.test.mjs`)	1.7	Implemented
FR-SYNC-6	Background sending shall use Background Sync where the browser supports it. Elsewhere (including WebKit and Capacitor shells), sending continues when the app is opened.	S	In Chromium a queued change is sent by the service worker after the tab is closed. In WebKit/Capacitor it is sent on the next opening.	T (`sync-ui.cjs`), D	5.3	Implemented
FR-SYNC-7	Attachments (photo, audio) shall be downloaded on a second device after sync.	S	A photo attached on device A opens on device B after sync, and is still available offline after that.	T (`workspace.cjs`)	—	Implemented
FR-SYNC-8	Data of different profiles on one device shall be isolated.	M	After switching profiles, no records, queue entries or cached attachments of the previous profile are visible or sent.	T (`api.test.mjs` isolation, `workspace.test.ts`)	Security	Implemented

6.3.4 Weather (WTH)

6. Requirements Specification · Table 6.7

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-WTH-1	The system shall show current weather, an hourly forecast and a 10-day forecast for the coordinates of the selected plot.	M	Changing the selected plot changes the request coordinates. Data of another point is never shown for the plot.	T (`open-meteo.test.ts`, `home-weather.cjs`, `weather.test.ts`)	H5 (2.3), 5.6	Implemented
FR-WTH-2	The last forecast shall be cached per coordinates for up to 24 h and marked as stale when it is outdated or shown offline.	M	Offline, the cached forecast is shown with its timestamp and a "stale" label. It is never presented as a current warning.	T (`home-weather.cjs`, `weather-presentation.test.ts`)	[2]	Implemented
FR-WTH-3	Missing indicators (for example heat index or thunderstorm probability) shall be shown as "no data", never as invented values.	S	With a fixture that lacks a field, the card shows "no data".	T (`weather-presentation.test.ts`)	Data honesty	Implemented

6.3.5 Field records, events and reports (REC, EVT, RPT)

Current UI boundary. `App.tsx` routes both `/events` and `/journal` to `EventsPage`, which displays existing records without a manual diary composer. The underlying record/attachment store and sync mechanisms remain, and `workspace.cjs` seeds legacy records directly for its tests. FR-REC-1 and the manual-record portion of FR-SYNC-1 are therefore **Partial** as end-user requirements. Seeded tests of FR-SYNC-3/7 establish data mechanics, not a current user-facing diary editor. Restoring an editor or revising that acceptance scope requires an explicit product decision; see Part 8.

6. Requirements Specification · Table 6.8

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-REC-1	The user shall be able to attach photos and audio to a plot record, offline as well.	M	A photo (`image/jpeg|png|webp`) or audio clip (`audio/webm|ogg|mp4|mpeg|wav`) is saved offline and uploaded via `PUT /api/assets/:id`. Unsupported MIME types are rejected with 400 (`api.test.mjs`), and payloads > 8 MB are rejected by `express.raw({limit:'8mb'})` in `server.mjs:163`.	T (`api.test.mjs`, `browser.cjs`, `workspace.cjs`), I (`server.mjs:162-168`)	[2]	Partial

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-EVT-1	The event centre shall show the weather state of the selected plot, boundary imports, sync conflicts, 1C exchange history and sensor registry state, with filters by source, plot and severity.	S	Each event type appears after its trigger. Filters reduce the list correctly. The action button opens the related screen (forecast, conflict, connection).	T (`events.cjs`, `events.test.ts`)	Single screen, [2]	Implemented
FR-RPT-1	The user shall be able to export a report for a selected plot and period as HTML (with photos and audio), CSV and JSON.	M	(1) All three formats are produced offline from the local copy. (2) CSV cells that start with `= + - @` are neutralised. (3) HTML escapes user text. (4) The files are shared through Web Share / Capacitor Filesystem & Share, or downloaded where sharing is unsupported.	T (`reports.cjs`, `reports.test.ts`)	1.4	Implemented

6.3.6 Sensors (SNS)

6. Requirements Specification · Table 6.9

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-SNS-1	The user shall be able to register a sensor by scanning a QR code (camera or image) or by typing its ID, and link it to a plot.	S	Only the `egin://sensor?...` and `{"kind":"egin-sensor"}` formats are accepted. An ID outside `[A-Za-z0-9_-]{3,64}` is rejected. URLs in the QR are never opened automatically.	T (`qr-scanner.cjs`, `qr-recognition.cjs`, `workspace.cjs`)	1.6	Implemented
FR-SNS-2	The UI shall not present a registered sensor as connected or as delivering data.	M	No sensor screen shows readings or an "online" state while there is no telemetry.	I, T (`readiness.cjs`)	[5]–[7]	Implemented
FR-SNS-3	Receiving sensor telemetry and automatic alerts.	W	—	—	1.6	Out of scope

6.3.7 Home screen and 3D (HOME)

6. Requirements Specification · Table 6.10

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-HOME-1	The home screen shall open with a 3D crop scene, followed by a summary of the selected plot (boundaries and EGKN snapshot, forecast freshness, linked sensors, next step).	M	At 360×800 and 390×844 the 3D canvas is the first element and is fully visible above the fold. The summary reflects the selected plot.	T (`browser.cjs`, `home-weather.cjs`)	2.2 E2-05	Implemented
FR-HOME-2	The 3D scene shall reflect the real weather of the plot (time of day, rain, snow, clouds, wind). Manual preview shall not change the real forecast or the crop phase.	S	With a weather fixture of "rain, night", the scene shows rain and a night palette.	T (`home-weather.cjs`)	—	Implemented
FR-HOME-3	The 3D scene shall not be presented as a digital twin and shall not show a crop phase computed from invented data.	M	The UI labels the scene as an example until analytics is connected.	I, D	[5]–[7]	Implemented
FR-HOME-4	The home screen shall show an auto-updated agricultural news feed (ElDala.kz) and keep the last feed offline.	C	If the source fails, the last feed is returned with `stale=true`. Without a cache the server returns 503 and the UI shows an error state.	T (`news.test.mjs`, `news.cjs`)	—	Implemented

6.3.8 Integrations and external API (INT, API)

6. Requirements Specification · Table 6.11

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-INT-1	The user shall be able to connect a 1C infobase by its standard OData URL and credentials. The connection is checked by reading `$metadata` before it is saved.	S	Wrong credentials or an invalid URL are rejected before saving. The password is never returned by any API.	T (`one-c.test.mjs`, `one-c-transport.test.mjs`, `services.cjs`)	2.2 E2-05, H4	Implemented

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-INT-2	1C import shall be read-only and idempotent, and shall preserve manual EGIN edits.	S	(1) Repeating an import creates no duplicates. (2) A record edited in EGIN is skipped and logged as a conflict. (3) A row deleted in 1C does not delete EGIN data. (4) No write requests are sent to 1C.	T (`one-c-flow.test.mjs`)	2.2 E2-05	Implemented
FR-API-1	External developers shall access profile, fields, journal, sensor registry, attachments, weather and news through the `/v1` API with an open key and a secret key.	S	(1) A request without a key, with a revoked key or with an expired key returns 401. (2) A key without the required scope returns 403. (3) Updates without a correct `If-Match` return 412.	T (`developer.test.mjs`, `developers.cjs`)	2.2 E2-05	Implemented
FR-API-2	The developer portal shall issue, rotate and revoke keys with chosen scopes and an expiry of 7, 30, 90 or 365 days. The secret is shown only once.	S	After rotation, the old pair is rejected immediately. The secret is not retrievable after the creation dialog is closed.	T (`developers.cjs`)	2.2 E2-05	Implemented
FR-API-3	The public API reference (`/openapi.json`) shall match the installed routes.	S	The route-inventory test fails if a route is missing from `catalog.json`, or listed there but not installed.	T (`developer.test.mjs`)	AGENTS.md	Implemented
FR-INT-3	Contractual cadastral access via Land-Coordinates (Smart Bridge), eGov sign-in, AgroSignal and AgroStream data exchange.	W	Shown as "В разработке".	I	1.6	Out of scope

6.3.9 Notifications (PUSH)

6. Requirements Specification · Table 6.12

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source	Status
FR-PUSH-1	The user shall be able to subscribe the current device to Web Push (`/api/push/subscribe`), send a test notification (`/api/push/test`) and unsubscribe (`/api/push/unsubscribe`).	C	Only HTTPS endpoints on allowlisted push domains (`fcm.googleapis.com`, `web.push.apple.com`, `updates.push.services.mozilla.com`) are accepted (≤ 20 subscriptions per user, `server.mjs:171-174`). Test push arrives on Android and installed iPhone PWA.	I (`server.mjs:170-174`), D	—	Implemented

6.4 Non-functional requirements

6.4.1 Usability and accessibility (USE)

6. Requirements Specification · Table 6.13

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-USE-1	Phone-first layout for 320–480 px with safe-area insets across Web PWA and Capacitor shells.	M	At 320, 360, 390 and 430 px widths there is no horizontal scrolling. Content does not go under the notch or the home indicator (`env(safe-area-inset-*)`).	T (Playwright viewport checks in `browser.cjs`), D	[1], 2.2 E2-05
NFR-USE-2	White and dark green palette. The active bottom-navigation tab changes icon and text colour, with no filled tab background.	M	Visual check on Chromium and WebKit screenshots. Text contrast ≥ 4.5:1 (WCAG 2.1 AA).	A (Lighthouse/axe), I	2.2 E2-05
NFR-USE-3	Touch targets of at least 44×44 CSS px.	S	An axe/Lighthouse "tap targets" audit passes on the main screens.	A	[1] (digital literacy)
NFR-USE-4	Adding a plot by cadastral number shall take less time than drawing its contour manually.	S	In a usability session (n ≥ 3), the median time by number is lower than the median time by drawing. Target: ≤ 60 s from opening "Plots" to a saved plot.	A (Section 9)	H2 (2.3)

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-USE-5	The interface language is Russian, and every user-facing error states what to do next.	S	No raw error codes or stack traces appear in the UI (checked on fixture failures).	T, I	[3]

6.4.2 Performance (PERF)

6. Requirements Specification · Table 6.14

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-PERF-1	Fast first load on a mobile network.	S	Lighthouse mobile profile (Slow 4G, 4× CPU) on the staging site: LCP ≤ 4.0 s, Performance score ≥ 70.	A (Section 9)	[2]
NFR-PERF-2	Repeat load from the service worker cache or local Capacitor bundle works offline.	M	A second load in airplane mode renders the shell and the saved plots.	T (`app-update.cjs`, `browser.cjs`)	[2], [12]
NFR-PERF-3	The 3D scene shall not drain the battery while idle.	S	With no interaction, or with the tab hidden or scrolled off-screen, the render loop stops (0 frames requested for 5 s).	T (`scene-performance.cjs`)	—
NFR-PERF-4	Server responses for cadastral search are bounded.	M	Time limit 20 s, response ≤ 5 MiB, ≤ 2,000 points. If a limit is exceeded, the server returns a controlled error.	T (`cadastre.test.mjs`)	1.7

6.4.3 Reliability and availability (REL)

6. Requirements Specification · Table 6.15

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-REL-1	No data loss in the offline-to-online transition.	M	In the sync test suite, 100% of queued operations reach the server or end in an explicit conflict, with 0 silent losses.	T (`workspace.cjs`, `api.test.mjs`)	[8]

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-REL-2	Zero-downtime deployment and fast rollback.	M	During `deploy/manage.py deploy production`, `/api/health` keeps answering 200. `rollback production` restores the previous slot in ≤ 60 s.	D, A (deploy log)	5.1
NFR-REL-3	Consistent backups.	M	Every deploy creates an online SQLite copy in `/data/backups/`, plus the 1C key. A restore drill on staging opens the copy without integrity errors (`PRAGMA integrity_check` = ok).	D, A	5.2
NFR-REL-4	Degradation when external sources fail.	S	If EGKN, Open-Meteo or ElDala is unavailable, the app stays usable. It shows cached data with a label and offers GeoJSON or geolocation as the plot fallback.	T (news/ weather fixtures), D	1.8
NFR-REL-5	Only backward-compatible database migrations.	M	Each migration is reviewed. The previous release runs against the migrated schema in staging before the production switch.	I, D	AGENTS.md

6.4.4 Security and privacy (SEC)

6. Requirements Specification · Table 6.16

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-SEC-1	Session cookies are HttpOnly, Secure and SameSite=Lax (Strict for QR). Sessions expire after 30 days.	M	`HttpOnly` and `SameSite=Lax` verified in `api.test.mjs:27`; 30-day TTL (`30 * 86400000` ms, `Max-Age=2592000`) and expiration filter (`expires > Date.now()`) verified in `server.mjs:73,86`.	T (`api.test.mjs`), I (`server.mjs:73,86`)	[10], [11]
NFR-SEC-2	Data ownership is enforced on every record and attachment.	M	A request for another user's record or attachment returns 403/404 in all `/api/*` and `/v1` routes.	T (`api.test.mjs`, `developer.test.mjs`)	OWASP API1
NFR-SEC-3	Secrets are never stored in plaintext or in Git.	M	API keys are stored only as SHA-256 hashes. 1C passwords are encrypted with AES-256-GCM. Recovery codes are hashed. A repository scan (`git grep`, gitleaks) finds no secrets.	T, I, A	5.4

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-SEC-4	Rate limits.	M	Cadastral search: 24 requests/min per profile. `/v1`: 60 requests/min per key. 1C: 12 calls/min per owner. Exceeding a limit returns 429.	T (`cadastre.test.mjs`, `developer.test.mjs`, `one-c.test.mjs`)	1.7
NFR-SEC-5	Outbound connections (1C, EGKN) are restricted.	M	1C: public IPv4 over HTTPS/443 only, TLS verified, no redirects. Requests to private or loopback addresses are refused.	T (`one-c-transport.test.mjs`)	SSRF protection
NFR-SEC-6	Separate environments.	M	Production and staging have separate data volumes and passkey RP origins. A production passkey cannot sign in to staging.	D, I	AGENTS.md
NFR-SEC-7	Personal data minimisation.	M	Cadastral snapshots contain no personal names or IINs of individuals. Request logs exclude headers, body and query strings.	T (`cadastre-details.test.mjs`), I	Law of the RK "On Personal Data"

6.4.5 Compatibility, maintainability and deployability (CMP, MNT)

6. Requirements Specification · Table 6.17

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-CMP-1	Cross-platform client support: Capacitor Android (API 24+) and iOS shells, Chrome/Edge on Android 10+, and Safari on iOS 16.4+ (browser and installed PWA).	M	All Playwright scenarios pass in Chromium and WebKit. Capacitor configuration (`prepare-native.mjs`, `native-origins.json`) packages the local `dist` bundle reproducibly.	T, I, D	5.1
NFR-CMP-2	The web app is installable as a PWA and offers updates without data loss.	S	The "Update now" prompt appears for a new version. After the update, the queue and saved records are intact.	T (`app-update.cjs`)	[12]

ID	Requirement	Pri.	Acceptance criteria / metric	Verif.	Source
NFR-MNT-1	Every change passes CI: typecheck, server tests and browser tests.	M	`main` is protected. A PR cannot be merged with failing checks.	I (`.github/workflows/checks.yml`), A (CI history)	Git requirements
NFR-MNT-2	One synchronisation engine is shared by the app and the service worker across Web and Capacitor.	S	`entities/workspace/sync-engine.js` is the only sync implementation (inspection).	I	5.3
NFR-MNT-3	Capacitor mobile package builds (APK/AAB/IPA) run only on an explicit release request and never automatically on push or PR.	M	Inspection of `.github/workflows/checks.yml`: the mobile packaging job requires `github.event_name == 'workflow_dispatch' && inputs.build_native`.	I (`.github/workflows/checks.yml`)	2.2 E2-05

6.4.6 Capacity limits (CAP)

6. Requirements Specification · Table 6.18

ID	Requirement	Pri.	Acceptance criteria	Verif.	Source
NFR-CAP-1	Per-profile storage quotas.	S	Up to 10,000 records (`server.mjs:155`) and 128 MB of binary assets (`server.mjs:166`) per profile, ≤ 8 MB per asset (`server.mjs:163`). Exceeding a quota returns HTTP 413 inside the transaction without partial writes.	I (`server.mjs:155,163,166`), T (`api.test.mjs`)	5.2

6.5 Traceability summary

Need / evidence	Requirements
Unstable rural connectivity [2], local-first [8]	FR-SYNC-1…8, FR-FLD-6, FR-WTH-2, NFR-REL-1, NFR-PERF-2
Cross-platform mobile & web delivery from one codebase [1], [12]	FR-AUTH-1, FR-RPT-1, NFR-CMP-1, NFR-CMP-2, NFR-MNT-2, NFR-MNT-3
Low digital literacy and entry barrier [1], [3]	FR-AUTH-1, FR-AUTH-6, NFR-USE-1…5

Need / evidence	Requirements
Reuse existing data (cadastre, 1C) instead of manual entry	FR-FLD-1…4, FR-INT-1…2
Passkey recovery and lockout protection [10], [11]	FR-AUTH-2, FR-AUTH-3, FR-AUTH-4
Do not imitate a digital twin without measurements [5]–[7]	FR-HOME-3, FR-SNS-2
Owner-defined project constraints (2.2, E2-05)	FR-AUTH-5, FR-HOME-1, FR-API-1…3, NFR-USE-1…2, NFR-SEC-6, NFR-MNT-3
Hypotheses H2 and H5 (2.3)	NFR-USE-4, FR-WTH-1 (evaluation planned in Section 9)

6.6 Summary by priority

Priority	FR	NFR	Total
Must	21	18	39
Should	15	9	24
Could	3	0	3
Won't (this stage)	3	0	3

The counts above classify requirements by priority; they are not a pass-rate summary. Not every **Must** requirement is accepted: manual diary creation is not exposed by the current UI, physical-device/native compatibility and operational measurements still need evidence, and `main` was not protected at review time (NFR-MNT-1). Part 9 records actual check outcomes and planned measurements; Part 10 records the corresponding work and risks. The `prob-period-1` submission tag does not establish a completed native release or field evaluation.

7. Architecture and Diagrams

Editable section source · PART-7-ARCHITECTURE.md

The architecture diagrams have editable PlantUML sources in `docs/DIP/diagrams/` and readable exports (SVG and PNG) in `docs/DIP/diagrams/export/`. To regenerate the SVG exports from the repository root, run `java -jar plantuml.jar -tsvg -o export docs/DIP/diagrams/*.puml`; repeat with `-tpng` for PNG. A public source file can also be POSTed to `https://kroki.io/plantuml/svg` or `https://kroki.io/plantuml/png`. The prototype navigation diagram has a separate editable Mermaid source, SVG export and PNG export.

Evidence boundary. These diagrams include intended native and manual-record flows. Capacitor support is configured, but an accepted native release is not established. Manual record creation remains partial because the current event-centre interface has no manual composer, as detailed in Part 6 and Part 8. Architecture diagrams describe structure and intended interactions; acceptance evidence and outstanding checks are recorded separately in Part 9.

#	Diagram	Source	Export
7.2	System context *(required)*	PlantUML source	SVG / PNG
7.3	Use case *(required)*	PlantUML source	SVG / PNG
7.4	Component diagram *(required)*	PlantUML source	SVG / PNG
7.5	Deployment *(supplementary)*	PlantUML source	SVG / PNG
7.6	Sequence 1: offline record, sync and conflict *(supplementary)*	PlantUML source	SVG / PNG
7.7	Sequence 2: plot lookup by cadastral number *(supplementary)*	PlantUML source	SVG / PNG

7.1 Architectural overview

EGIN is a **local-first, cross-platform client–server system**. The source of truth for the user's working copy is the phone (IndexedDB), and the server acts as a synchronisation and integration hub. This follows from the evidence in Section 2 and the literature in Section 3: rural connectivity is unstable [2], and the local-first approach keeps the application usable without a network [8].

The system has four deployable parts:

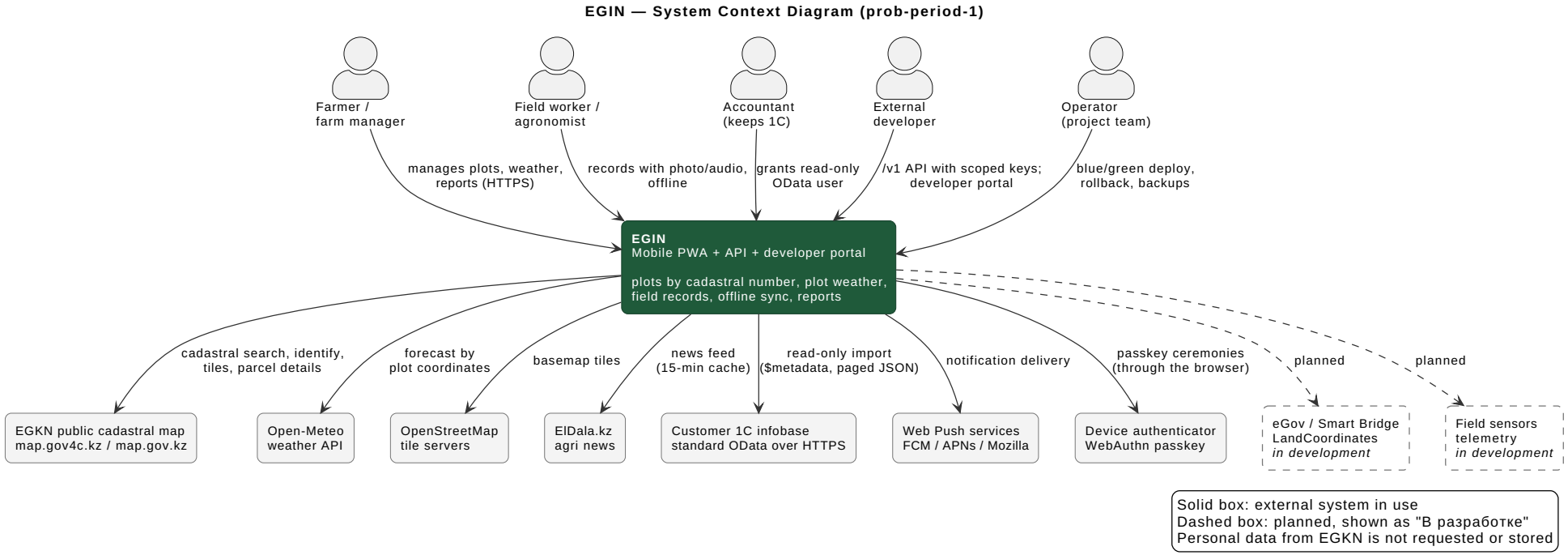
1. **Cross-platform client** (`apps/mobile`): a single React/TypeScript/Vite codebase delivered both as an installable **Web PWA** and as **Capacitor mobile applications for Android and iOS**. It contains the app shell, six feature packages (`home`, `weather`, `events`, `workspace`, `settings`, `auth`), the per-profile IndexedDB store, a single sync engine shared by the page and the service worker, a weather client with a cache keyed by coordinates, and the Capacitor bridge (`CapacitorHttp`, `Filesystem`, `Share`, `@capgo/capacitor-passkey`).
2. **Developer portal** (`apps/developers`): a separate SPA on `api-egin.perricheno.com`, with its own session cookie and QR audience.
3. **API** (`apps/api`): Node 22+ and Express, with `node:sqlite` (`DatabaseSync` in WAL mode) on a persistent volume. It is organised as installable modules: auth and sessions, QR sign-in, sync and binary assets, cadastre adapter, news, 1C connector, external API and portal, and push.
4. **Gateway**: a permanent Nginx container (`egin-mobile-web-1`) that serves hashed static assets (`/assets/`, `/developer-assets/`) and proxies `/api` to the active **blue/green** slot.

Architectural style and the main decisions. Section 5 compares alternatives for each of these decisions.

Decision	Choice	Requirement driver
Client platform	Cross-platform Web & Mobile: Capacitor (Android & iOS) + PWA from one codebase (5.1)	NFR-CMP-1, NFR-USE-1, low entry barrier [1]
Data storage	IndexedDB on the device; SQLite WAL (`egin.sqlite`) on the server (5.2)	FR-SYNC-1, NFR-REL-3
Synchronisation	Monotonic server sequence (`seq`), per-record `version`, tombstones (`deleted=1`), payload-equality idempotency and explicit conflict resolution, without CRDT (5.3)	FR-SYNC-3…5, NFR-REL-1, [8], [9]

Decision	Choice	Requirement driver
Authentication	Passkeys (WebAuthn & `@capgo/capacitor-passkey` verified by `@simplewebauthn/server`) with a one-time recovery code; eGov pending (5.4)	FR-AUTH-1…5, [10], [11]
Cadastre source	Server-side adapter over the public EGKN map (`proj4` UTM → WGS84, PII stripping); GeoJSON and geolocation as fallbacks (5.5)	FR-FLD-1…5, NFR-SEC-7
Weather source	Open-Meteo by plot coordinates (5.6)	FR-WTH-1…3
Component interaction	REST/JSON over HTTPS through one gateway; there is no message broker	Simplicity of a single-node deployment, NFR-REL-2

7.2 System context diagram



System context Open full-size image · Editable source

Description. EGIN is used by five kinds of people: the farmer, the field worker, the accountant who keeps 1C, external developers and the operator. It depends on seven external systems. All outbound connections go through the **API**, except two: the weather forecast, which the client requests directly from Open-Meteo, and standard OpenStreetMap basemap tiles. This keeps credentials, rate limits and personal-data filtering on the server (NFR-SEC-4, NFR-SEC-5, NFR-SEC-7).

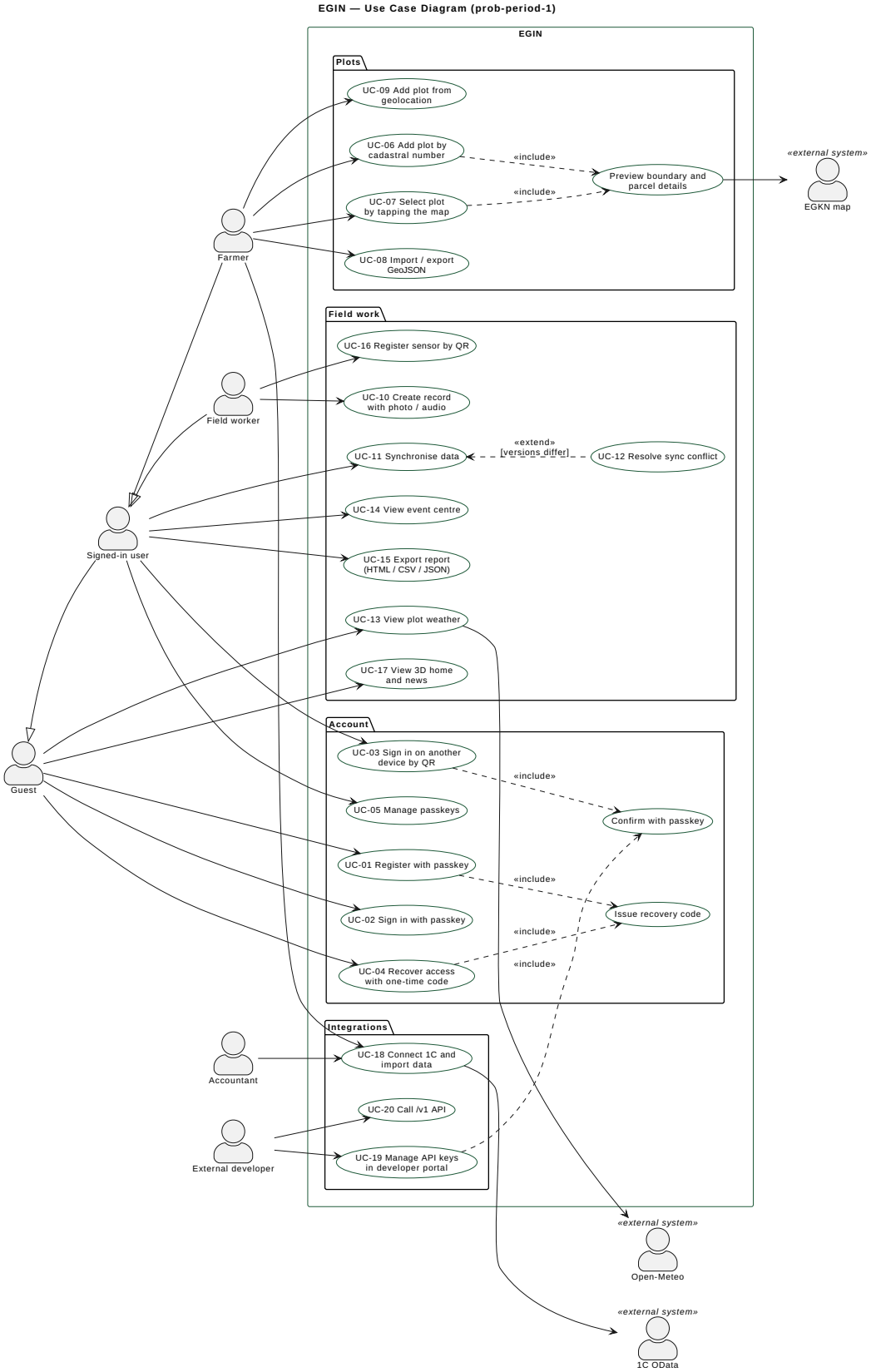
The dashed boxes, **eGov / Smart Bridge LandCoordinates** and **field sensor telemetry**, are deliberately outside the current boundary. They correspond to the *Won't* requirements FR-INT-3 and FR-SNS-3, and the UI shows them as "В разработке".

7. Architecture and Diagrams · Table 7.3

External system	Interface	Used for	Requirements	Failure behaviour
EGKN public map	HTTPS, public map endpoints, TLS verified with Sectigo intermediate CA	Cadastral search, tap-to-identify, tiles, parcel details	FR-FLD-1…3	Error message; GeoJSON and geolocation fallbacks (NFR-REL-4)
Open-Meteo	HTTPS REST	Forecast by coordinates	FR-WTH-1…3	24 h cached forecast labelled "stale"
OpenStreetMap, map.gov.kz aerial	HTTPS tiles (direct or proxied via `/api/cadastre/basemap/*`)	Basemap	FR-FLD-1	Map without basemap; contours stay visible
ElDala.kz	HTTPS HTML feed	News	FR-HOME-4	Last feed with `stale=true`
Customer 1C	OData over HTTPS/443, Basic Auth, read-only	Import of farm data	FR-INT-1…2	Previously imported history is kept and marked

External system	Interface	Used for	Requirements	Failure beha-viour
Web Push services	Web Push / VAPID (`fcm.googleapis.com`, `web.push.apple.com`, `up-dates.push.services.mozilla .com`)	Test notifications	FR-PUSH-1	Nothing is lost; push is optional
Device authenticator	WebAuthn / Capacitor Passkey bridge through the browser or OS	Passkey registration and assertion	FR-AUTH-1…4	Recovery code (FR-AUTH-2)

7.3 Use case diagram



Use case Open full-size image · Editable source

Description. The actors form a generalisation hierarchy: *Guest* → *Signed-in user* → *Farmer / Field worker*. A guest sees the 3D home screen, news and weather, and can register, sign in or recover access (FR-AUTH-6). The `<<include>>` relations show mandatory shared steps: issuing a recovery code, confirming with a passkey, and previewing a boundary before saving. `<<extend>>` shows that conflict resolution happens only when versions differ.

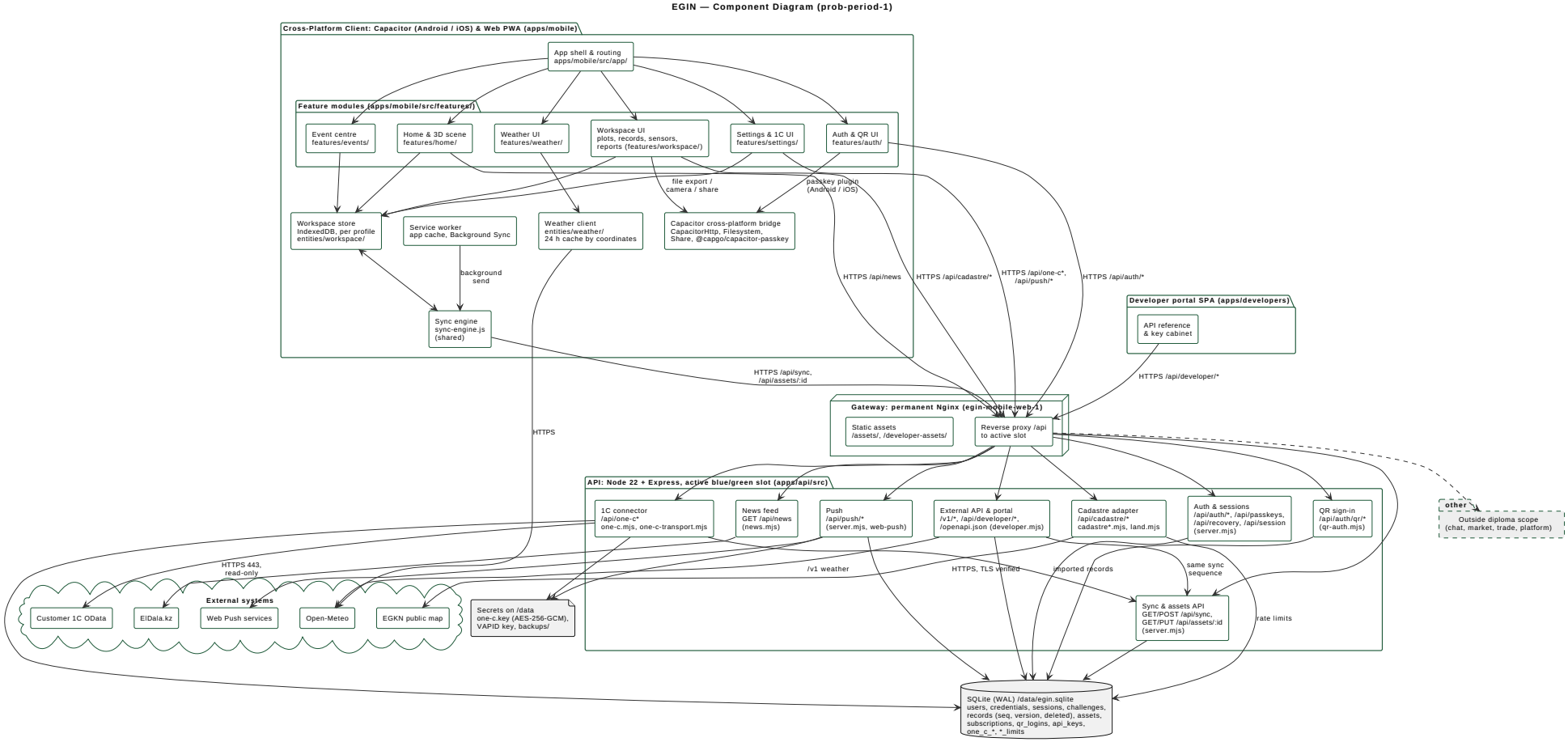
Use case → requirement → component traceability.

Use case	Requirements	Main components (7.4)
UC-01 Register with passkey	FR-AUTH-1, FR-AUTH-2	Auth & QR UI, Capacitor bridge, Auth & sessions
UC-02 Sign in with passkey	FR-AUTH-1	Auth & QR UI, Capacitor bridge, Auth & sessions
UC-03 Sign in on another device by QR	FR-AUTH-4	Auth & QR UI, QR sign-in, Auth & sessions
UC-04 Recover access	FR-AUTH-2	Auth & QR UI, Auth & sessions
UC-05 Manage passkeys	FR-AUTH-3	Settings & 1C UI, Auth & sessions
UC-06 Add plot by cadastral number	FR-FLD-1, FR-FLD-3, FR-FLD-7	Workspace UI, Cadastre adapter, Workspace store
UC-07 Select plot on the map	FR-FLD-2, FR-FLD-3	Workspace UI, Cadastre adapter
UC-08 Import / export GeoJSON	FR-FLD-4	Workspace UI, Workspace store
UC-09 Add plot from geolocation	FR-FLD-5	Workspace UI, Workspace store
UC-10 Create record with photo / audio	FR-REC-1, FR-SYNC-1	Workspace UI, Capacitor bridge, Workspace store
UC-11 Synchronise data	FR-SYNC-2, 4, 5, 6, 7, 8	Sync engine, Service worker, Sync & assets API
UC-12 Resolve sync conflict	FR-SYNC-3	Event centre / Settings UI, Sync engine, Sync & assets API
UC-13 View plot weather	FR-WTH-1…3	Weather UI, Weather client
UC-14 View event centre	FR-EVT-1	Event centre UI, Workspace store
UC-15 Export report	FR-RPT-1	Workspace UI, Capacitor bridge, Workspace store

Use case	Requirements	Main components (7.4)
UC-16 Register sensor by QR	FR-SNS-1, FR-SNS-2	Workspace UI, Workspace store
UC-17 View 3D home and news	FR-HOME-1…4	Home & 3D scene UI, News feed
UC-18 Connect 1C and import data	FR-INT-1, FR-INT-2	Settings & 1C UI, 1C connector
UC-19 Manage API keys	FR-API-2	Developer portal, External API & portal
UC-20 Call the /v1 API	FR-API-1, FR-API-3	External API & portal, Sync & assets API

The only functional requirement without a dedicated use case bubble is FR-PUSH-1 (Could); it is triggered from `Settings & 1C UI` (`POST /api/push/subscribe`, `POST /api/push/test`) and is omitted from the diagram to keep it readable.

7.4 Component diagram



Components Open full-size image · Editable source

Description of components. Each component maps to an exact path and route set verified in the repository.

7. Architecture and Diagrams · Table 7.5

Component	Repository path	Responsibility	Exact HTTP / internal interfaces	Requirements
App shell & routing	`apps/mobile/src/app/`	Hash routing, bottom navigation, lazy screens with error recovery	—	NFR-USE-1, NFR-USE-2
Home & 3D scene UI	`apps/mobile/src/features/home/`	3D crop scene (pauses off-screen), plot summary, ElDala news cards	`GET /api/news`	FR-HOME-1…4, NFR-PERF-3
Weather UI	`apps/mobile/src/features/weather/`	Hourly and 10-day forecast cards and charts	Uses Weather client	FR-WTH-1…3
Event centre UI	`apps/mobile/src/features/events/`	Unified feed of weather state, boundary imports, sync conflicts, 1C runs and sensors	Reads Workspace store	FR-EVT-1
Workspace UI	`apps/mobile/src/features/workspace/`	Plots (`FieldsPage`, `LocationPicker`, `FieldPreview`), records with media, QR sensor registry, HTML/CSV/JSON reports	`GET /api/cadastre/*`, reads/writes Workspace store	FR-FLD-1…7, FR-REC-1, FR-SNS-1…2, FR-RPT-1
Settings & 1C UI	`apps/mobile/src/features/settings/`	Categories, passkey management, sync queue & conflict resolver, 1C connection, Web Push controls	`/api/one-c*`, `/api/passkeys*`, `/api/push/*`	FR-AUTH-3, FR-SYNC-3, FR-INT-1…2, FR-PUSH-1
Auth & QR UI	`apps/mobile/src/features/auth/`	Passkey registration/sign-in, recovery code flow, cross-device QR scanner & confirmation	`/api/auth/*`, `/api/auth/qr/*`	FR-AUTH-1…5
Workspace store	`apps/mobile/src/entities/workspace/`	Per-profile IndexedDB: records (`entry`, `field`, `sensor`, `profile`), queue, revisions, binary assets	Local API to UI & Sync engine	FR-SYNC-1, FR-SYNC-8
Sync engine	`apps/mobile/src/entities/workspace/sync-engine.js`	Cursor pull, batch push, binary asset upload/download, version comparison, tombstones, conflicts. **One implementation** shared by the page and the service worker	`GET /api/sync?cursor=`, `POST /api/sync`, `GET/PUT /api/assets/:id`	FR-SYNC-2…7, NFR-MNT-2

Component	Repository path	Responsibility	Exact HTTP / internal interfaces	Requirements
Service worker	`apps/mobile/scripts/` (build)	Offline app shell, Background Sync, update prompt; sensitive API responses are not cached	Invokes `sync-engine.js`	NFR-PERF-2, NFR-CMP-2
Weather client	`apps/mobile/src/entities/weather/`	Open-Meteo requests, normalisation, 24 h cache by coordinates	HTTPS to `api.open-meteo.com`	FR-WTH-1…3
Capacitor cross-platform bridge	`apps/mobile/scripts/prepare-native.mjs`, `native-trust.mjs`	Cross-platform Android & iOS packaging of local `dist`; bridges `CapacitorHttp`, `Filesystem`, `Share`, `@capgo/capacitor-passkey`, camera, microphone and geolocation	Capacitor JS-to-OS plugin bridge	FR-AUTH-1, FR-RPT-1, NFR-CMP-1, NFR-MNT-3
Developer portal	`apps/developers/`	API reference rendered from `catalog.json`, key cabinet	`/api/developer/*`, `/openapi.json`	FR-API-2, FR-API-3
Gateway	`deploy/nginx.conf`, `deploy/manage.py`	Serves `/assets/` and `/developer-assets/`, proxies `/api` to active blue/green slot, runs health checks and rollback	HTTP `127.0.0.1:4934` / `:4938`	NFR-REL-2
Auth & sessions	`apps/api/src/server.mjs` (`@simplewebauthn/server`)	Passkey registration and assertion (Web & Capacitor origins), recovery code rotation, 30-day sessions	`GET /api/session`, `POST /api/auth/register/{options,verify}`, `POST /api/auth/login/{options,verify}`, `POST /api/auth/recover`, `POST /api/logout`, `GET /api/passkeys`, `DELETE /api/passkeys/:id`, `POST /api/recovery`	FR-AUTH-1…3, NFR-SEC-1
QR sign-in	`apps/api/src/qr-auth.mjs`	Two-minute QR requests bound to browser cookie; single-use session issue	`POST /api/auth/qr`, `GET/POST /api/auth/qr/:id`, `POST /api/auth/qr/:id/reject`, `POST /api/developer/qr`	FR-AUTH-4
Sync & assets API	`apps/api/src/server.mjs`	Owner checks, `BEGIN IMMEDIATE` version comparison, monotonic `sequence`, tombstones, payload-equality idempotency, binary assets, quotas (10,000 records, 128 MB per profile, 8 MB per asset)	`GET /api/sync`, `POST /api/sync`, `PUT /api/assets/:id`, `GET /api/assets/:id`	FR-SYNC-3…5, NFR-SEC-2, NFR-CAP-1

Component	Repository path	Responsibility	Exact HTTP / internal interfaces	Requirements
Cadastre adapter	`cadastre.mjs`, `cadastre-tiles.mjs`, `cadastre-raster.mjs`, `cadastre-details.mjs`, `land.mjs`	Search and tap-identify, UTM → WGS84 with `proj4`, tile reprojection, detail snapshot without personal data, rate limits	`GET /api/cadastre/{search,identify,labels,regions,district}`, `GET /api/cadastre/tiles/:z/:x/:y.png`, `GET /api/cadastre/basemap/:z/:x/:y.png`	FR-FLD-1…3, NFR-PERF-4, NFR-SEC-7
News feed	`apps/api/src/news.mjs`	ElDala parsing, 15-minute file cache (`/data/news-cache.json`), stale fallback	`GET /api/news`	FR-HOME-4
1C connector	`one-c.mjs`, `one-c-transport.mjs`	Encrypted credentials (AES-256-GCM with `/data/one-c.key`), OData XML `$metadata`, paged read-only import, SSRF protection	`/api/one-c*`	FR-INT-1…2, NFR-SEC-3, NFR-SEC-5
External API & portal	`developer.mjs`, `catalog.json`	`/v1` CRUD through the same sync sequence, SHA-256 hashed scoped keys, 60 req/min rate limit, `If-Match`, OpenAPI	`/v1/*`, `/api/developer/*`, `GET /openapi.json`	FR-API-1…3, NFR-SEC-4
Push	`apps/api/src/server.mjs` (`web-push`)	VAPID public key, device subscriptions (≤ 20 per user, allowlisted push domains), test delivery	`GET /api/push/key`, `POST /api/push/{subscribe,unsubscribe,test}`	FR-PUSH-1
SQLite (WAL)	`/data/egin.sqlite`	Persistent relational & document store shared by active/standby slots of one environment	`node:sqlite` (`DatabaseSync`)	NFR-REL-3, NFR-REL-5

Important internal dependency. Both the `/v1` API (`developer.mjs`) and the 1C import (`one-c.mjs`) write through the **same `records` and `sequence` tables** as `/api/sync`. Records created or updated by external systems therefore receive monotonic `seq` cursors and `version` increments identically to mobile client records (FR-INT-2, FR-API-1).

Outside the diploma scope. The repository also contains chat, market, trade and business-platform modules (`apps/api/src/{chat,market,trade,platform}`). They are shown as a single dashed package on the diagram and are outside the scope defined in Section 1.6 and Section 6.

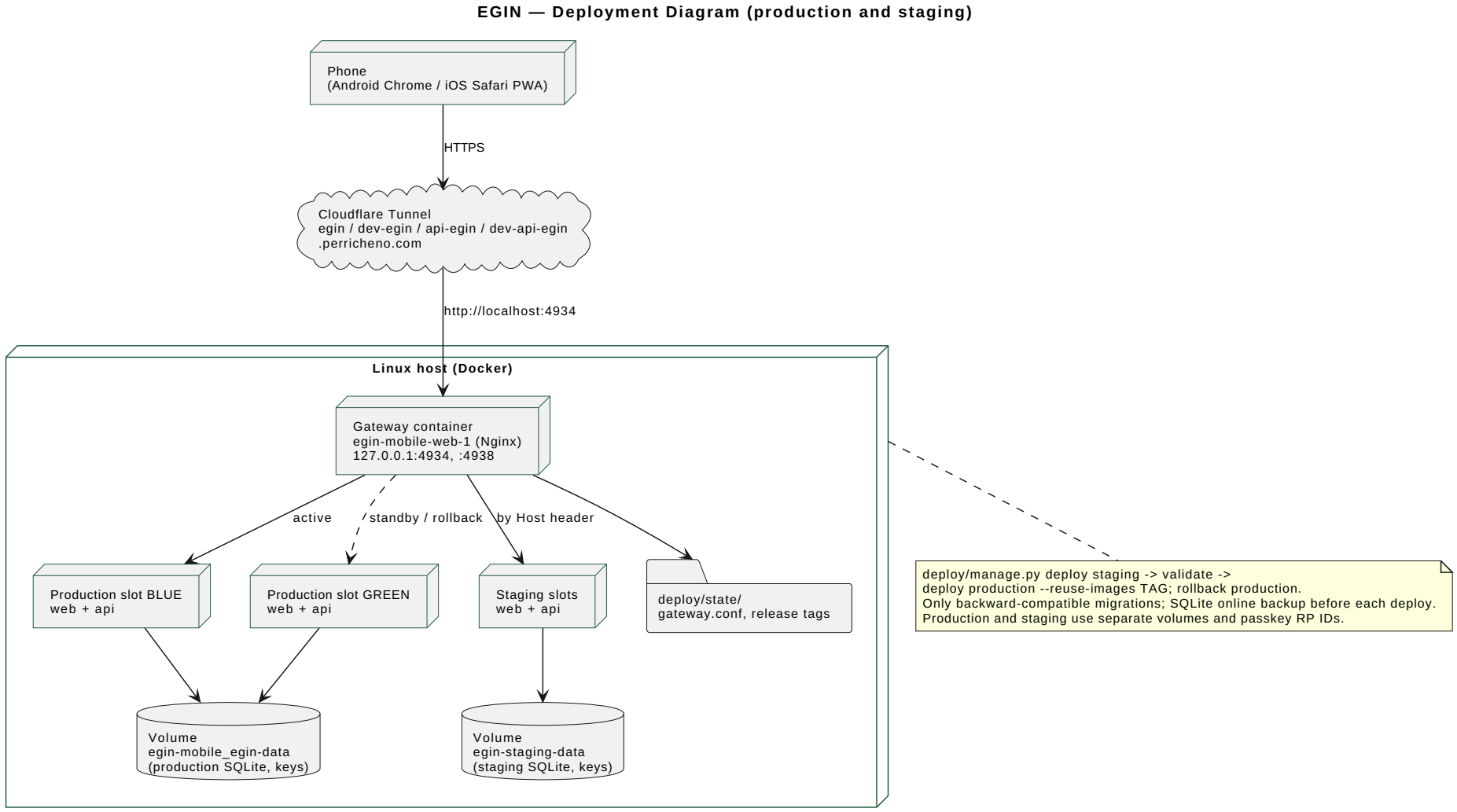
7.4.1 Physical & logical data model (verified against `apps/api/src/*.mjs`)

All core tables reside in `/data/egin.sqlite` (SQLite WAL mode). Application entities (`field`, `entry`, `sensor`, `profile`) are stored in a unified versioned `records` table on the server and mirrored in per-profile IndexedDB stores on the client.

SQLite table / file	Primary key & main columns	Purpose & constraints
`users`	`id TEXT PK`, `name TEXT NOT NULL`, `recovery TEXT`, `created INTEGER`	User profile and SHA-256 hash of the current one-time recovery code
`credentials`	`id TEXT PK`, `user_id FK`, `public_key BLOB`, `counter INTEGER`, `transports TEXT`, `name TEXT`, `created INTEGER`	Registered WebAuthn / Capacitor passkeys (`ON DELETE CASCADE`)
`challenges`	`id TEXT PK`, `type TEXT`, `challenge TEXT`, `user_id TEXT`, `name TEXT`, `expires INTEGER`	Short-lived WebAuthn registration and authentication challenges
`sessions`	`id TEXT PK`, `user_id FK`, `expires INTEGER`	30-day main-app sessions (`HttpOnly`, `Secure`, `SameSite=Lax`)
`qr_logins`	`id TEXT PK`, browser binding hash, `user_id`, `expires`, status	2-minute cross-device QR login ceremonies (`qr-auth.mjs`)
`records`	`PRIMARY KEY (user_id, id)`, `kind TEXT ('entry' | 'field' | 'sensor' | 'profile')`, `data TEXT` (JSON), `deleted INTEGER` (0/1 tombstone), `version INTEGER`, `seq INTEGER`	Unified versioned store for plots (including `boundary` and `cadastre.details`), journal entries, sensor registry and profile settings; capped at 10,000 rows per `user_id`
`sequence`	`n INTEGER` (single-row counter)	Monotonic change cursor (`UPDATE sequence SET n=n+1 RETURNING n`) used by `GET /api/sync?cursor=`
`assets`	`PRIMARY KEY (user_id, id)`, `mime TEXT`, `data BLOB`	Binary photos (`image/jpeg|png|webp`) and audio (`audio/webm|ogg|mp4|mpeg|wav`); ≤ 8 MB per row, ≤ 128 MB total per `user_id`
`subscriptions`	`PRIMARY KEY (user_id, endpoint)`, `data TEXT`	Web Push subscriptions (≤ 20 per user; allowlisted FCM / Apple / Mozilla HTTPS endpoints)
`api_keys`, `api_limits`, `api_usage`, `developer_sessions`	Key id, `open_key`, SHA-256 `secret_hash`, scopes, expiry, per-minute counters, sanitised audit log	External `/v1` API authentication, 60 req/min rate limiting, 12-hour portal sessions (`developer.mjs`)
`one_c_connections`, `one_c_sources`, `one_c_runs`, `one_c_revisions`, `one_c_previews`, `one_c_limits`	Owner-scoped OData URL, encrypted password ciphertext, deterministic source mappings, lease locks	Read-only 1C OData import, idempotent re-sync, 12 req/min rate limit (`one-c.mjs`)

SQLite table / file	Primary key & main columns	Purpose & constraints
`cadastre_limits`, `cadastre_tile_limits`, `cadastre_view_limits`	Per-user / per-window counters	Rate limiting across blue/green slots: 24/min search, 60/min metadata, 240/min tiles
`/data/one-c.key`, VAPID key, `/data/backups/`	Files on persistent volume (`0600`)	AES-256-GCM master key for 1C passwords, Web Push VAPID key, online SQLite backups

7.5 Deployment diagram (supplementary)



Deployment Open full-size image · Editable source

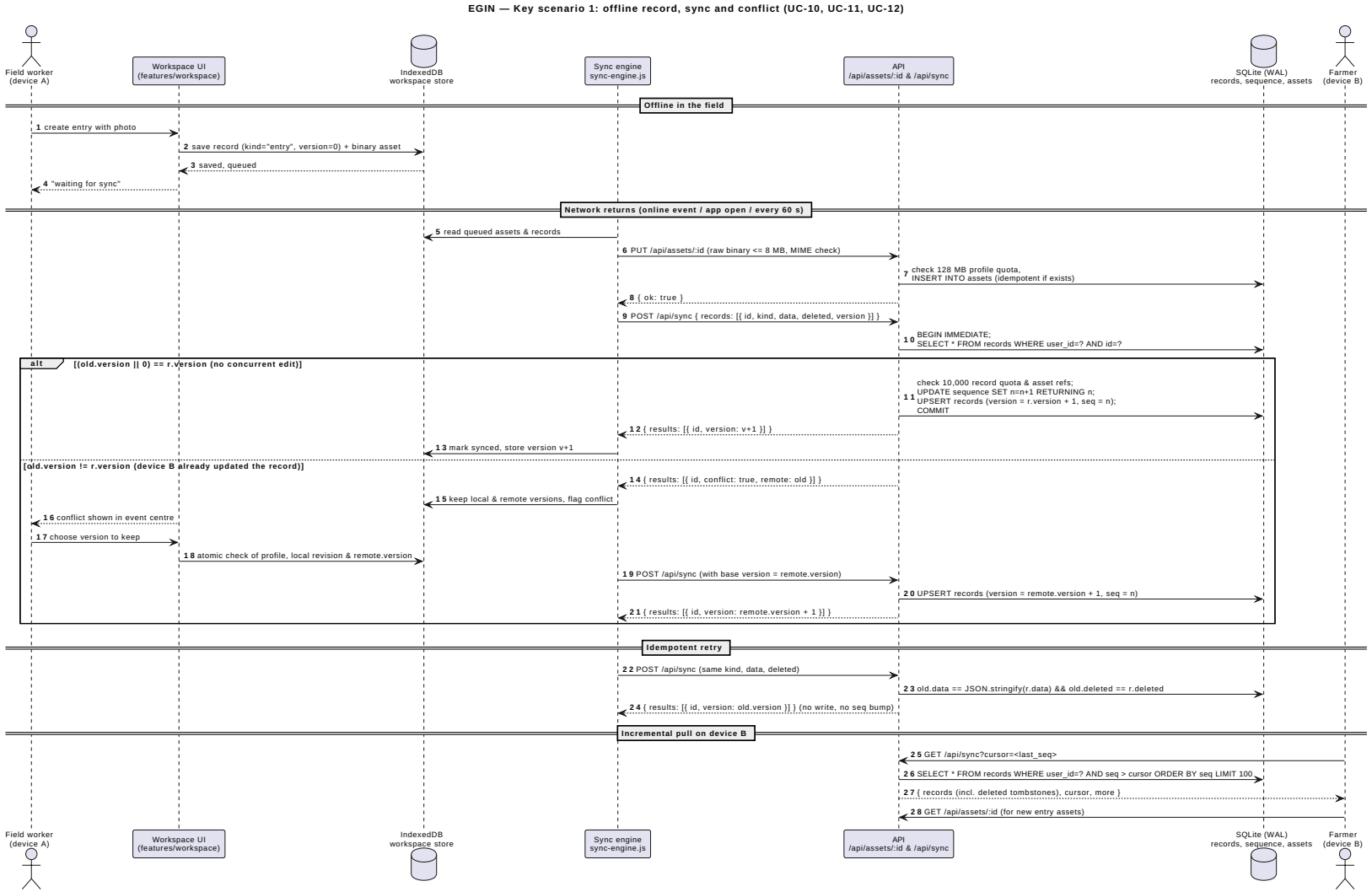
Requests from the phone pass through Cloudflare Tunnel to a single **permanent gateway** (`egin-mobile-web-1`). The gateway proxies to the active production slot (blue or green). The other slot stays available for an immediate rollback. Staging runs on the same host with **separate data volumes (`egin-staging-data`) and passkey RP IDs** (NFR-SEC-6).

Release workflow:

1. `python3 deploy/manage.py deploy staging`
2. Validation on `dev-egin.perricheno.com` and `dev-api-egin.perricheno.com`
3. `python3 deploy/manage.py deploy production --reuse-images TAG`, which promotes the exact container images verified on staging

Before each deploy, an online SQLite backup and a copy of `one-c.key` are written to `/data/backups/` (NFR-REL-3). Database migrations must be backward-compatible, because both production slots share one volume (NFR-REL-5).

7.6 Key scenario 1: offline record, sync and conflict (supplementary)

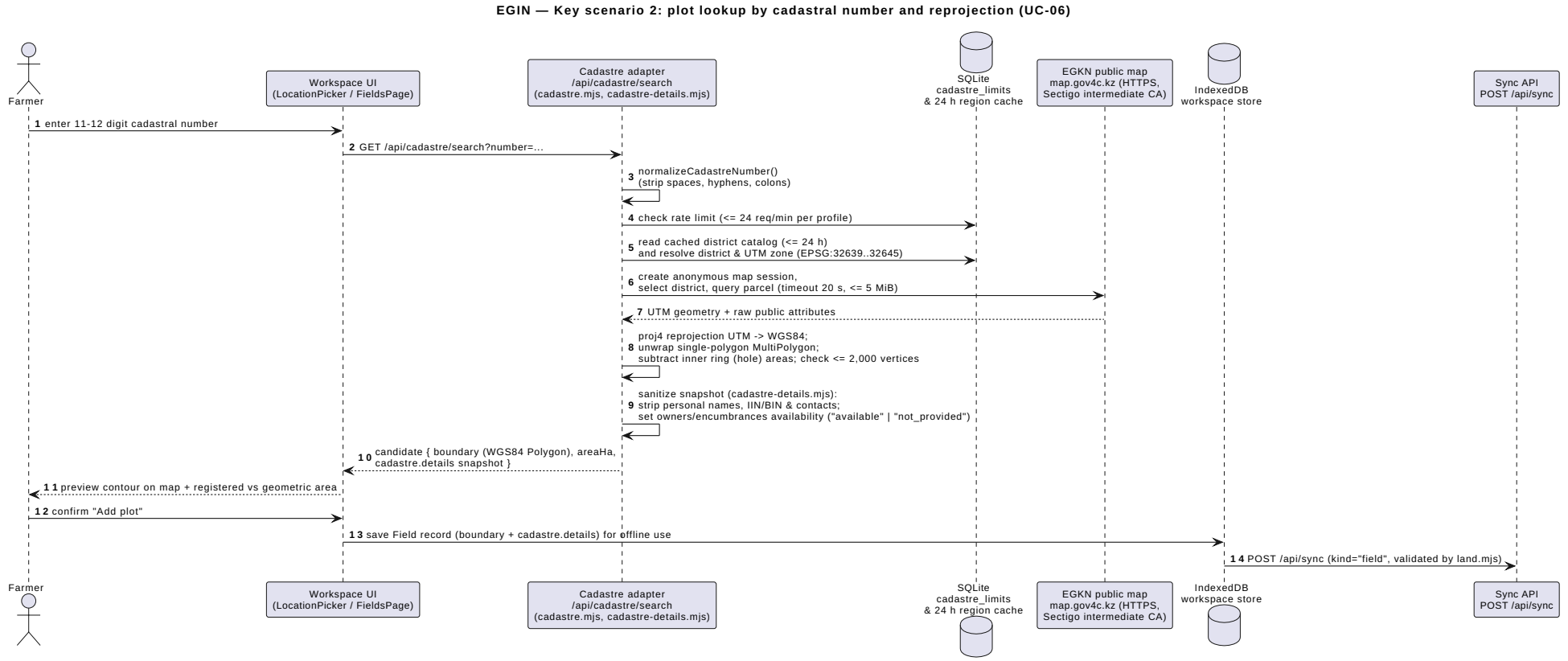


Sequence 1 Open full-size image · Editable source

This sequence diagram traces UC-10, UC-11 and UC-12 across the exact endpoints and SQLite statements in `server.mjs`:

1. **Offline write:** `Workspace UI` saves the record (`kind="entry"`, `version=0`) and its photo in `IndexedDB`.
2. **Binary upload first:** when connectivity returns, `sync-engine.js` uploads each binary attachment via `PUT /api/assets/:id` (checked for MIME type, ≤ 8 MB size and the 128 MB profile quota; idempotent if the asset ID already exists).
3. **Transactional batch sync:** `POST /api/sync` runs inside `BEGIN IMMEDIATE`, verifies that all referenced asset IDs exist in `assets`, checks the 10,000-record quota, compares `(old?.version || 0)` with `r.version`, increments the global `sequence` cursor (`seq`), and upserts `records` with `version = r.version + 1`.
4. **Explicit conflict:** if `old.version !== r.version`, the server does not overwrite the row; it returns `{ id, conflict: true, remote: old }`. After the user picks a version, the client resends the record with `version = remote.version`.
5. **Idempotent retry:** if a retried request sends a record whose `kind`, `data` and `deleted` flag already match the stored row, the server returns the existing `version` without writing or advancing `seq`.
6. **Incremental pull:** Device B calls `GET /api/sync?cursor=<last_seq>` (`SELECT * FROM records WHERE user_id=? AND seq>? ORDER BY seq LIMIT 100`), receiving updated records and tombstones (`deleted=true`), and then fetches any new attachments via `GET /api/assets/:id`.

7.7 Key scenario 2: plot lookup by cadastral number and reprojection (supplementary)



Sequence 2 Open full-size image · Editable source

This sequence diagram traces UC-06 across `LocationPicker.tsx`, `cadastre.mjs`, `cadastre-details.mjs` and `land.mjs` (FR-FLD-1, FR-FLD-3, FR-FLD-7, NFR-PERF-4, NFR-SEC-7):

1. The farmer enters an 11–12-digit cadastral number; `GET /api/cadastre/search?number=...` normalises separators and checks the 24 req/min profile limit in `cadastre_limits`.
2. Using the 24-hour cached district catalog, the adapter determines the district and its local UTM zone (`EPSG:32639`–`32645`) and queries `map.gov4c.kz` over verified HTTPS (using the bundled Sectigo intermediate CA, ≤ 20 s timeout, ≤ 5 MiB response).
3. The adapter reprojects the geometry from UTM to WGS84 with `proj4`, normalises single-polygon `MultiPolygon` geometries, subtracts inner hole areas, and verifies the ≤ 2,000-vertex limit.
4. `cadastre-details.mjs` builds a sanitised public snapshot (`Field.cadastre.details`), stripping all personal names, individual IINs, contacts and operator metadata, and distinguishing `"available"` from `"not_provided"`.
5. The UI previews the contour and displays both geometric and registered areas before saving. Once confirmed, the plot is written to `IndexedDB` for offline use and synchronised via `POST /api/sync` (`kind="field"`).

7.8 Consistency check: requirements, components, data and scenarios

Check	Result
Every **Must** FR is realised by at least one use case	**Yes** — mapped in the 7.3 traceability table (UC-01…UC-20)
Every use case is served by specific UI, cross-platform bridge and server components	**Yes** — decomposed into 6 feature packages (`home`, `weather`, `events`, `workspace`, `settings`, `auth`), the Capacitor bridge, and 8 API modules in 7.4
Every component maps to an exact repository path and verified HTTP routes	**Yes** — all routes (`/api/auth/*`, `/api/auth/qr/*`, `/api/sync`, `/api/assets/:id`, `/api/cadastre/*`, `/api/one-c*`, `/api/push/*`, `/v1/*`) verified against `apps/api/src/*.mjs`
Data model matches physical SQLite tables and record kinds	**Yes** — all tables in 7.4.1 (`users`, `credentials`, `sessions`, `challenges`, `qr_logins`, `records`, `sequence`, `assets`, `subscriptions`, `api_keys`, `one_c_*`, `*_limits`) match `CREATE TABLE` definitions in `apps/api/src/`
Key engineering scenarios are traced end-to-end	**Yes** — Sequence 1 (7.6) covers offline writes, binary assets, `seq`/`version` sync, conflicts and idempotency; Sequence 2 (7.7) covers cadastral search, UTM→WGS84 reprojection and PII filtering

Check	Result
Every external system in 7.2 appears as a dependency in 7.4	**Yes** — EGKN, Open-Meteo, OSM/basemap, ElDala, 1C OData, Web Push and WebAuthn / Capacitor Passkey authenticator
Every *Won't* item is outside the system boundary	**Yes** — eGov, LandCoordinates and sensor telemetry are dashed in 7.2 and excluded from 7.3 and 7.4

8. UX and Prototype

Editable section source · PART-8-UX-PROTOTYPE.md

8.1 Purpose and evidence boundary

The EGIN prototype demonstrates how a farmer can move from a mobile home screen to a plot, its forecast, available events, sensor registration and data export. A separate developer portal demonstrates how an external system can discover the API and obtain scoped access. These interfaces provide concrete interaction evidence for the criterion: *“Mockups of key user scenarios; for projects without a user interface, examples of APIs, commands, or input and output data.”*

This section is based on the source at commit `8cc0ca3f73052b030dcddb990b8a5460be9ddcdf`, inspected on **23 October 2026**, and on the actual interface captures in Section 8.5. The source audit and screenshot capture are different kinds of evidence:

Evidence level	Meaning in this section	Limitation
Observed interface	A real locally served web bundle rendered the screen, and the capture script checked the stated elements.	API responses and local records were fixtures; this is not a live-service acceptance test.
Implemented in source	Components, routes and associated API or test code exist in the inspected repository.	Presence of code does not establish that every interaction passes on the deployed service.
Partial	Some supporting functions exist, but a stated end-user scenario or external integration is incomplete.	The missing step must be disclosed during the defence.
Pending / outside this stage	The interface explicitly marks the function as unavailable, or its implementation is outside the approved scope.	It must not be demonstrated as a working integration.

Build provenance. Both web applications were freshly built from commit `b0b0f46` in an isolated temporary directory, and the nine screenshots were recaptured at **2026-10-23T17:12:27.705Z**. Its `apps` tree and the historical `8cc0ca3` review tree are identical (`3c5eda79a3be6e2c00533ce5da288c5446987f92`). The earlier missing optional Rolldown binding was resolved in the isolated dependency copy; shared dependencies were not modified. TypeScript checking and Vite builds passed for both web apps, and the mobile service worker was generated. The capture metadata, complete artifact fingerprints and web-build execution record identify the source, environment and output. This establishes source-to-screenshot traceability while preserving the fixture limitations below; broader verification results remain separate in Part 9.

8.2 Interaction principles and navigation

The main application follows a phone-first layout with a prominent crop illustration, dark green controls, a light background and bottom navigation. The five main tabs are **Home, Weather, Chat, Market and Settings**. Plot management and the event centre are reached from the home screen or Settings. The

active bottom-navigation item changes its icon and text colour without a filled tab background. Safe-area spacing, semantic controls, a skip link and reduced-motion handling are present in the source; this inspection is not a completed accessibility audit.

The interface currently uses Russian. English names in this report describe the actual Russian controls; the screenshots retain the original UI text. Hash routes, such as `/#/fields`, are used by the application. The canonical route inventory is in App.tsx; settings subsections are selected in SettingsPage.tsx.

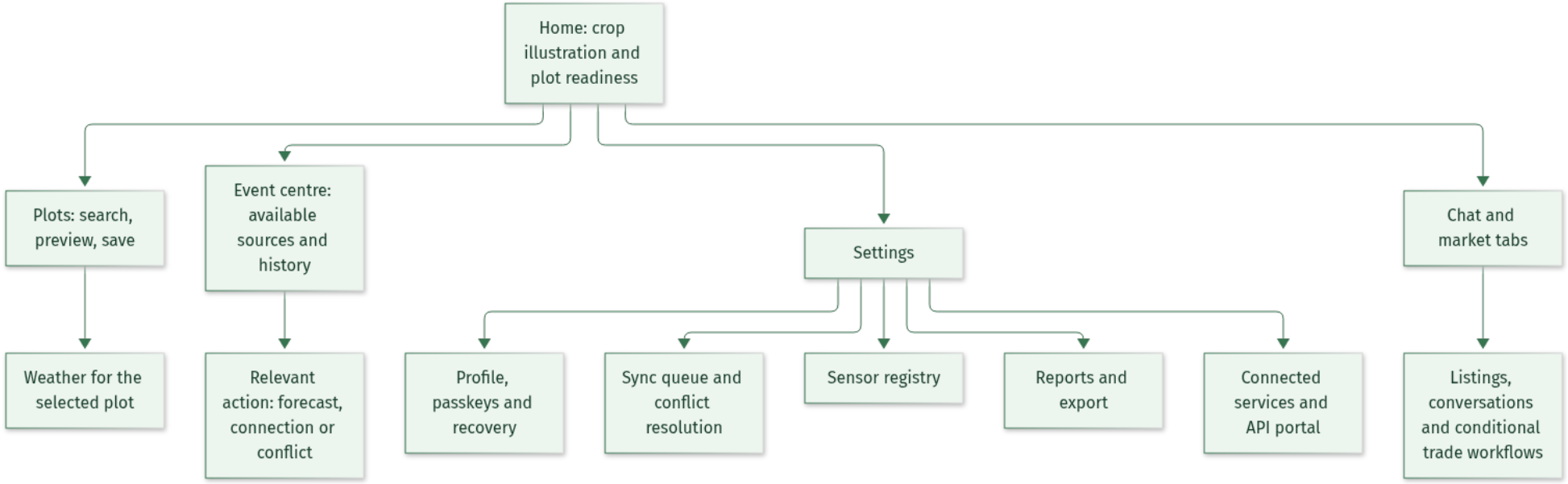


Figure 8.N — Prototype navigation and principal user flows. Open full-size image · Editable source

The visualisation is deliberately separated from agronomic measurement. HomePage.tsx and monitoring.ts label the crop as an illustration while monitoring data is unavailable. A visual weather preview does not constitute a crop prediction. Local and staging hosts also expose an experimental 2D/3D switch; the capture script selected 3D. This laboratory control should not be represented as a production-wide feature.

8.3 Screen inventory and implementation status

Requirement IDs refer to Part 6. “Observed” applies only to the screen state captured in this review. Existing automated scenarios are linked as evidence of available coverage, not as a claim that those suites were all executed successfully during this audit.

Screen / entry point	Main interaction and output	Current evidence and limits	Requirements / source
Home, `/#/`	Inspect the crop illustration, change its viewpoint, read plot readiness and open the next action.	**Observed interface; implemented in source.** No measured plant state or digital twin is demonstrated.	FR-HOME-1–4; HomePage, FieldReadiness
Plots, `/#/fields`	Search by cadastral number, select on the map or import GeoJSON; inspect the boundary before confirming; inspect/export a saved plot.	**Observed saved demo plot; acquisition implemented in source.** A synthetic boundary is clearly labelled; live EGKN search was not executed in the capture. Local guest storage is available; server synchronisation requires sign-in.	FR-FLD-1–7; FieldsPage, ParcelPassport
Weather, `/#/weather`	Read current, hourly and ten-day forecasts, change the displayed metric and inspect details.	**Observed fixture forecast; implemented in source.** Forecast values are model output, not on-site sensor readings. The screenshot contains the repository fixture's historical forecast date.	FR-WTH-1–3; WeatherPage
Events, `/#/events`; legacy `/#/journal`	Filter available events by source, plot, importance and text; follow an action; read preserved history and attachments.	**Observed interface; partial relative to manual record creation.** Both routes open the event centre. There is no current manual record composer.	FR-EVT-1; FR-REC-1 remains partial; EventsPage
Sign-in, `/#/auth`; security, `/#/settings/security`	Register/sign in with a passkey, use recovery, manage additional keys, approve another device's QR login.	**Observed sign-in screen; web authentication flows implemented in source.** Native iOS passkey configuration remains pending (FR-AUTH-1 is partial). No real or virtual passkey ceremony was performed by the screenshot script. eGov remains unavailable.	FR-AUTH-1–5; AuthPage, ProfilePage

Screen / entry point	Main interaction and output	Current evidence and limits	Requirements / source
Synchronisation, `/#/settings/sync`	Inspect pending records and conflicts, trigger synchronisation and choose a conflict resolution.	**Implemented in source; not captured end to end.** Local writes and server synchronisation are separate states. Direct fixture insertion of a history entry does not demonstrate a user-facing entry editor.	FR-SYNC-1–8; SyncPage, workspace browser scenario
Sensors, `/#/settings/sensors`	Register a sensor by QR/manual ID and link it to a plot.	**Observed registry state; registration implemented in source.** The UI states that receiving readings is still in development.	FR-SNS-1–3; SensorsPage
Reports, `/#/settings/reports`	Select plot, period, format and optional preserved history/media; generate an export and download/share it.	**Observed controls; generation implemented in source.** Formats include HTML, plot CSV, history CSV and JSON. No export download was executed by the capture script.	FR-RPT-1; ReportsPage, report-export
Connected services, `/#/settings/services`	Configure read-only 1C import, map fields, preview changes and review exchange history; prepare other connection requests.	**1C workflow implemented in source; partner integrations partial.** AgroSignal and AgroStream pages collect preparation details; their data exchange is not operational.	FR-INT-1–3; OneCPage, ServicesPage
Chat, `/#/chat`	Open conversations/communities and exchange supported messages and attachments.	**Implemented in source; authenticated interaction not captured.** This is an actual feature, not the former development placeholder.	Extension not yet assigned a Part 6 requirement ID; ChatPage, chat browser scenario
Market, `/#/market`	Search/filter listings, view details, save favourites and create a listing after sign-in.	**Observed empty catalogue; other workflows implemented in source.** The empty screenshot is a chosen fixture, not evidence of the number of real users or listings.	Extension not yet assigned a Part 6 requirement ID; MarketPage, PublicationWizard
Trade and organisation cabinet, `/#/market?view=trade` and `?view=business`	Manage harvest lots, offers, deals, documents and organisation/team details.	**Implemented in source, with conditional availability.** Organisation roles and feature flags control access; private demo data must be distinguished from real transactions. These workflows were not executed for the captures.	Extension not yet assigned a Part 6 requirement ID; TradePage, BusinessPanel

Screen / entry point	Main interaction and output	Current evidence and limits	Requirements / source
Developer portal, separate web application	Browse API methods, request/response examples, scope requirements and key management.	**Observed public documentation; key lifecycle implemented in source.** Separate QR audience/session and main-app passkey approval apply. No secrets were issued during this review.	FR-API-1–3; portal App, Cabinet

8.4 Key user flows

8.4.1 Establish a plot and inspect its forecast

Actor: farmer. **Goal:** reach plot-specific information without manually redrawing an existing cadastral boundary.

1. Open Home and follow the plot-readiness action to Plots.
2. Choose cadastral search, map selection or GeoJSON import. A point without boundaries is a fallback where appropriate.
3. Review the candidate boundary and available public details. An unsuccessful search must produce an actionable error; it must not silently create an empty plot.
4. Confirm the plot and select it as active. The local record becomes the context for the home summary, forecast and linked sensor registry.
5. Open Weather and inspect the selected plot name, source timestamp and relevant hourly/daily values.

Visible completion: the selected plot is present, its geometry or absence of boundaries is explicit, and the forecast names that plot. A cadastral snapshot does not prove ownership. Cached or unavailable data must not be presented as a fresh measurement. The corresponding source scenarios are map-picker.cjs, parcel-passport.cjs and home-weather.cjs.

8.4.2 Preserve local work and resolve a synchronisation conflict

Actor: farmer or field worker. **Goal:** keep locally supported changes available and reconcile differences between devices.

1. Create or edit a supported local object, such as a plot or sensor registry entry.
2. Inspect the synchronisation screen and distinguish local/pending state from confirmed server synchronisation.
3. Sign in and reconnect to synchronise the profile. If the server has a concurrent version, inspect the conflict rather than assuming an overwrite occurred.
4. Choose the appropriate version and synchronise again.

Visible completion: the pending state or conflict is resolved and the selected data is retained. A two-device demonstration requires a disposable authenticated profile and a separate test run. The existing workspace scenario seeds history entries directly into IndexedDB to test preservation and conflicts. This is useful storage/synchronisation evidence, but it does not prove that a user can create or edit such entries through the present UI. The screenshot capture does not test service-worker offline reload.

8.4.3 Review events and export available information

Actor: farmer. **Goal:** understand available plot information and prepare a shareable file.

1. Open the event centre from the home readiness card or Settings.
2. Filter by plot, source or importance, or search the event text.
3. Open the relevant forecast, connection or conflict screen, or inspect preserved history/materials.
4. Open Reports, select a plot and format, and optionally include available history and media.
5. Generate the report and download or share the result through supported browser controls.

Visible completion: the report represents the selected local data and period, and a file is produced. The report period applies to history; plot passports contain the latest saved details. Missing sources are disclosed. The event feed is not a continuous server archive of every weather change, and absence of events does not establish normal field conditions.

8.4.4 Register a sensor without implying telemetry

Actor: farmer. **Goal:** associate a physical device identifier with a plot for future integration.

1. Open Settings → Sensors → Add sensor.
2. Scan a supported EGIN sensor QR code, load its image, or enter the identifier manually.
3. Review the identifier, name, type and associated plot, then save.
4. Confirm that the item appears in the registry with the pending API-connection status.

Visible completion: a registry entry exists. This flow does not show live moisture, connectivity, agronomic alerts or an automatic crop phase. The distinction is visible in Figure 8.5 and enforced by the explicit `FR-SNS-2` requirement.

8.4.5 Sign in and grant external API access

Actors: account owner and external developer. **Goal:** provide controlled access to the owner's EGIN data.

1. Sign in to the main application with a passkey, retaining the recovery code securely when it is issued.
2. Open the separate developer portal and start its QR sign-in flow.
3. Confirm the correct destination and matching code in the main application, and approve with a fresh passkey assertion.
4. Select the required scopes and expiry when issuing a key pair.
5. Copy the secret at issuance, use it from the integrating server, and rotate/revoke it when required.

Visible completion: the developer session can manage scoped keys and an authorised request returns the owner's permitted data. Portal and main-app sessions are separate. Plaintext secrets are displayed only at issuance/rotation and must be excluded from screenshots. eGov is not involved in this process. The full flow requires a dedicated acceptance run; the present evidence shows the sign-in screen and public API reference only.

8.4.6 Explore communication and marketplace extensions

Actor: authenticated farmer or organisation member. **Goal:** communicate and find or publish agricultural offers.

The actual Chat tab leads to conversations and communities. The actual Market tab supports catalogue search, filters, listing detail, favourites and a publication wizard. Organisation trade screens provide additional lot, offer and deal interactions when the relevant feature flags and permissions are enabled. These extensions should be demonstrated separately from the core plot/weather scenario, with synthetic accounts and listings. Their source presence is verified; a successful complete commercial workflow or adoption by real farms is not established here. Dedicated requirement IDs and outcome measures remain to be reconciled with Parts 6 and 9 before these extensions are used to claim the diploma's aim has been achieved.

8.5 Captured interface evidence

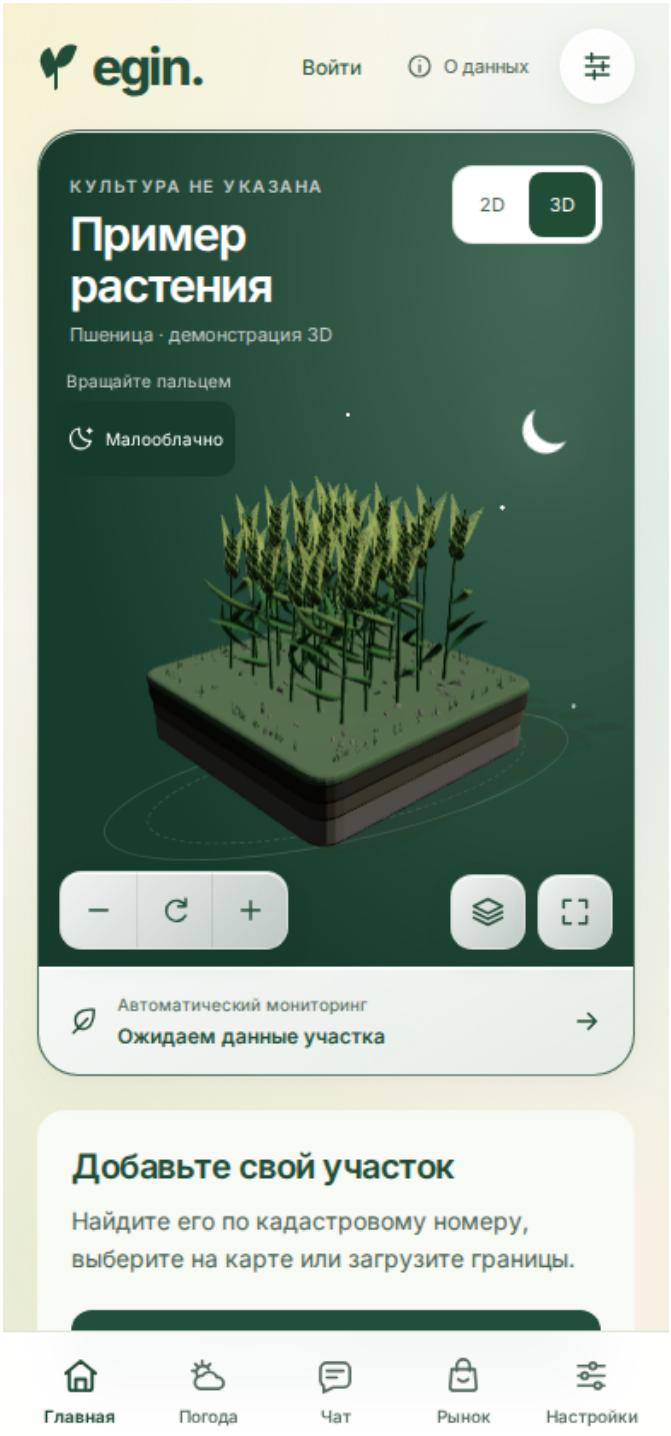
Nine screenshots were captured on **23 October 2026** in Chromium **153.0.8010.12**, using a **390 × 844 CSS px** mobile viewport and **1280 × 900 CSS px** for the developer portal. Device scale factor was 1; reduced motion was requested. These are screenshots of the application's actual rendered interface, not generated mockups.

All API responses were intercepted locally. Weather and news came from the repository's existing fixtures. A clearly labelled synthetic plot, sensor and historical entry were inserted only into the temporary browser profile's IndexedDB. The map basemap was intentionally unavailable, which exposes the real fallback message while the local boundary remains visible. No production data, real farmer record, passkey credential or API secret was used. The Russian text inside the images is the original interface, while captions and analysis are in English.

Figure	Screenshot	Demonstrated interaction/state	Evidence limitation
8.1	Home and 3D crop illustration	First-use home, model controls and the invitation to add a plot.	Weather/news fixtures; local laboratory switch visible; no measured crop state.
8.2	Saved plot and boundary	Synthetic contour, demo label, plot details and GeoJSON export control.	Full-page capture; basemap disabled; not an EGKN response or land-rights record.
8.3	Plot-specific weather screen	Plot name, forecast timestamp, hourly graph and daily-tab entry.	Historical fixture data dated 8 October; not a forecast retrieved on the capture date.

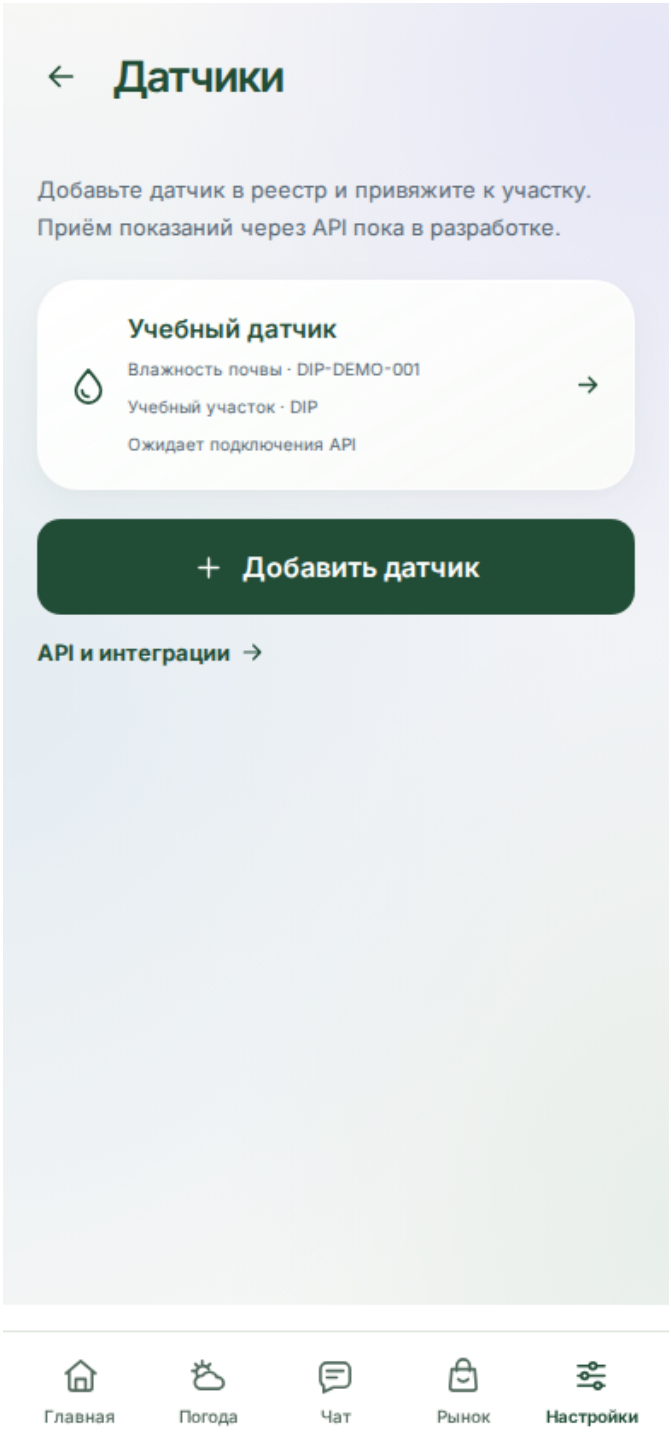
Figure	Screenshot	Demonstrated interaction/state	Evidence limitation
8.4	Event centre	Source/plot filters, demo-plot event, sensor registration and preserved history.	Full-page capture; generated from synthetic local records.
8.5	Sensor registry	Plot-linked sensor and explicit pending API-connection state.	Registration record only; no telemetry or field measurement.
8.6	Report controls	Plot/format/period selection and optional history/media.	Full-page capture; generation/download was not executed by this script.
8.7	Sign-in and pending eGov option	Passkey, recovery and EGIN QR entry points; unavailable eGov.	Full-page capture; no successful authentication claim.
8.8	Marketplace catalogue	Search, categories, filters, publication entry and empty-state guidance.	Deliberately empty fixture catalogue; no live inventory or trade transaction.
8.9	Developer API reference	Separate interface for method discovery and API usage.	Public documentation only; key issuance and live requests not executed.

Figure 8.1 — actual mobile home screen, isolated fixture environment.



Actual EGIN home screen with a 3D crop illustration and plot onboarding; local fixture capture Open full-size image

Figure 8.5 — actual sensor-registry screen with pending telemetry status.



Actual EGIN registry screen with a synthetic sensor and an explicit pending API-connection status Open full-size image

The capture script checked real screen rendering, 3D canvas readiness, 24 hourly forecast entries, the absence of a manual event composer, the pending sensor-connection message and the empty catalogue state. The final capture completed with **zero browser page errors**. Reproduction instructions and machine-readable metadata describe the environment. This small capture check is not a replacement for authenticated browser tests, offline/PWA checks, live integration validation or a user study.

Additional interface figures from the same controlled capture. Original files and their provenance remain linked. Tall screens are split into consecutive readable panels.

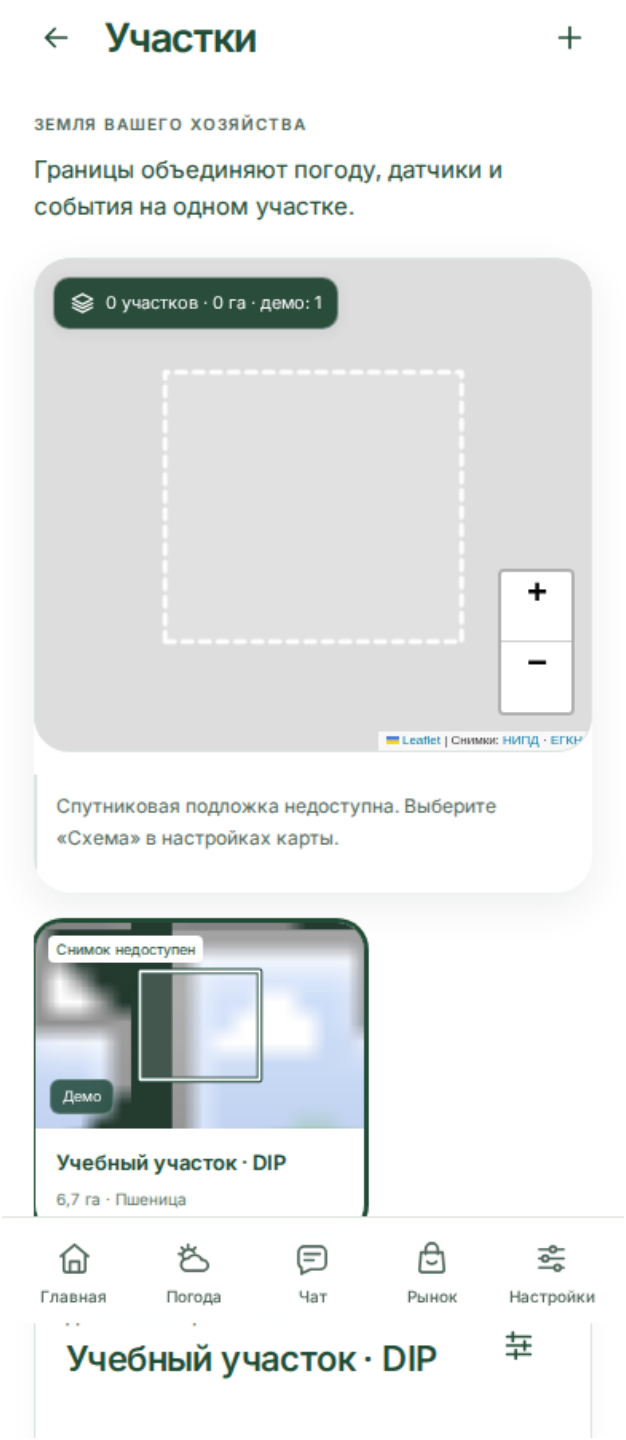


Figure 8.2, panel 1/2 — Saved synthetic plot, boundary and plot controls; basemap unavailable. Open the complete original screenshot

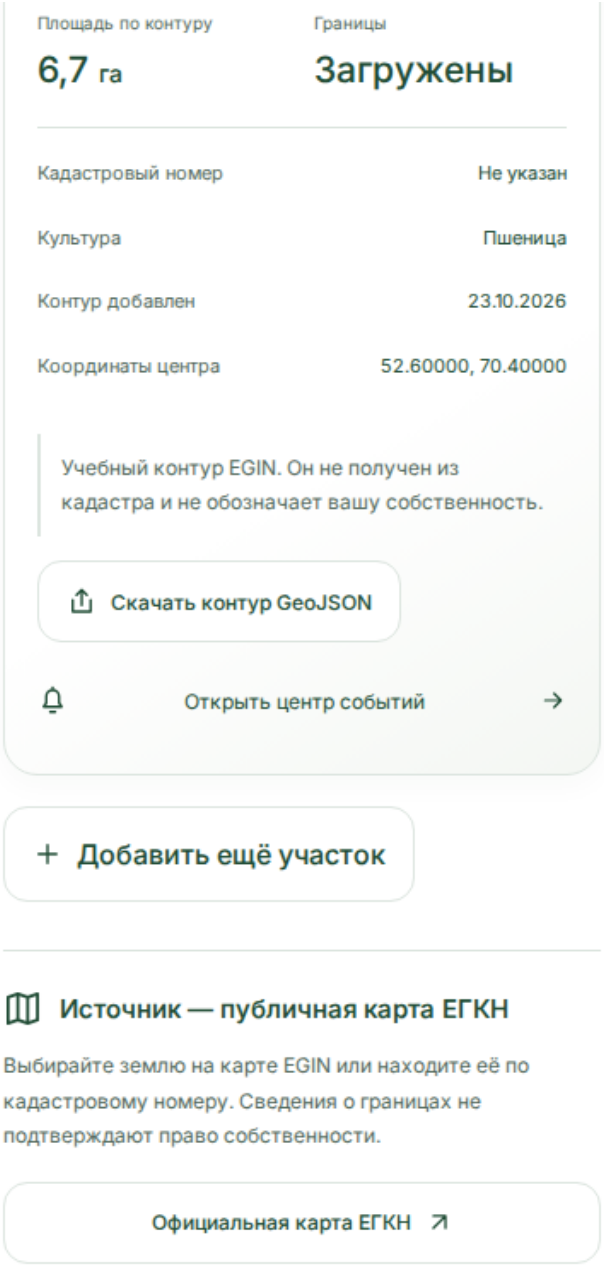


Figure 8.2, panel 2/2 — Saved synthetic plot, boundary and plot controls; basemap unavailable. Open the complete original screenshot

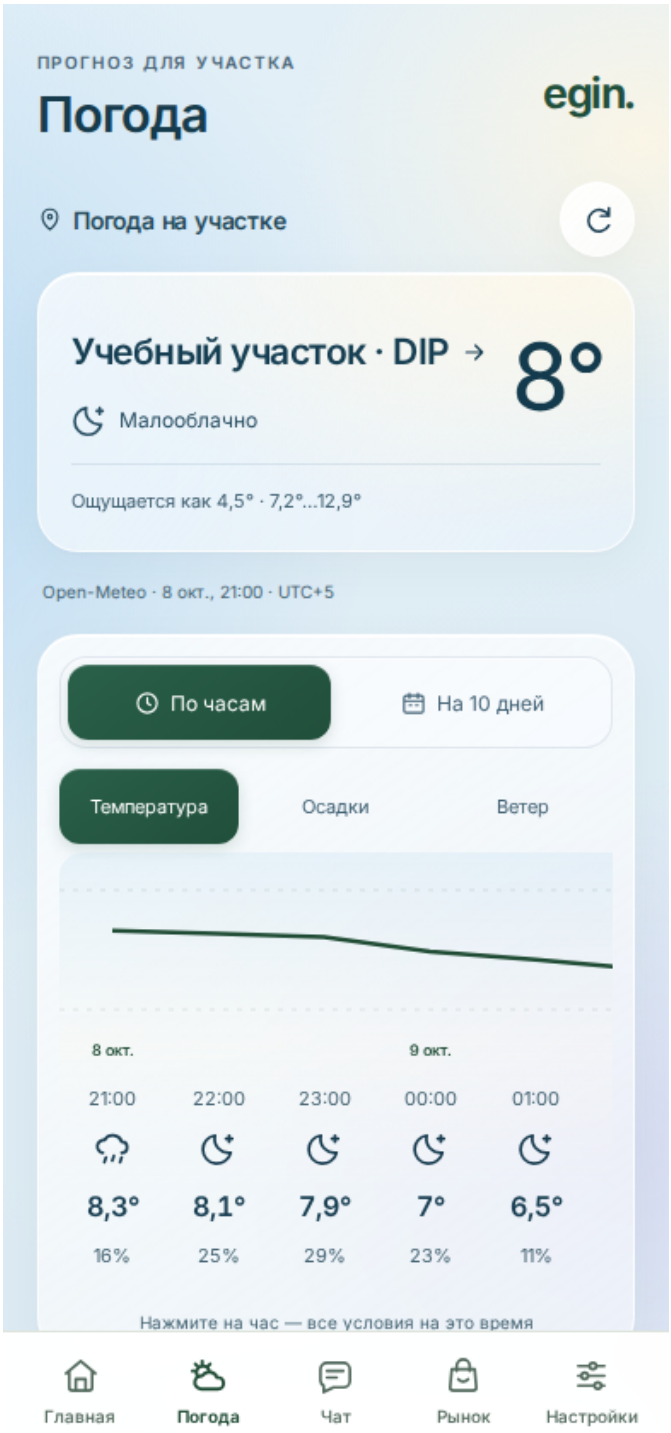


Figure 8.3 — Plot-specific forecast using the repository weather fixture. Open full-size image

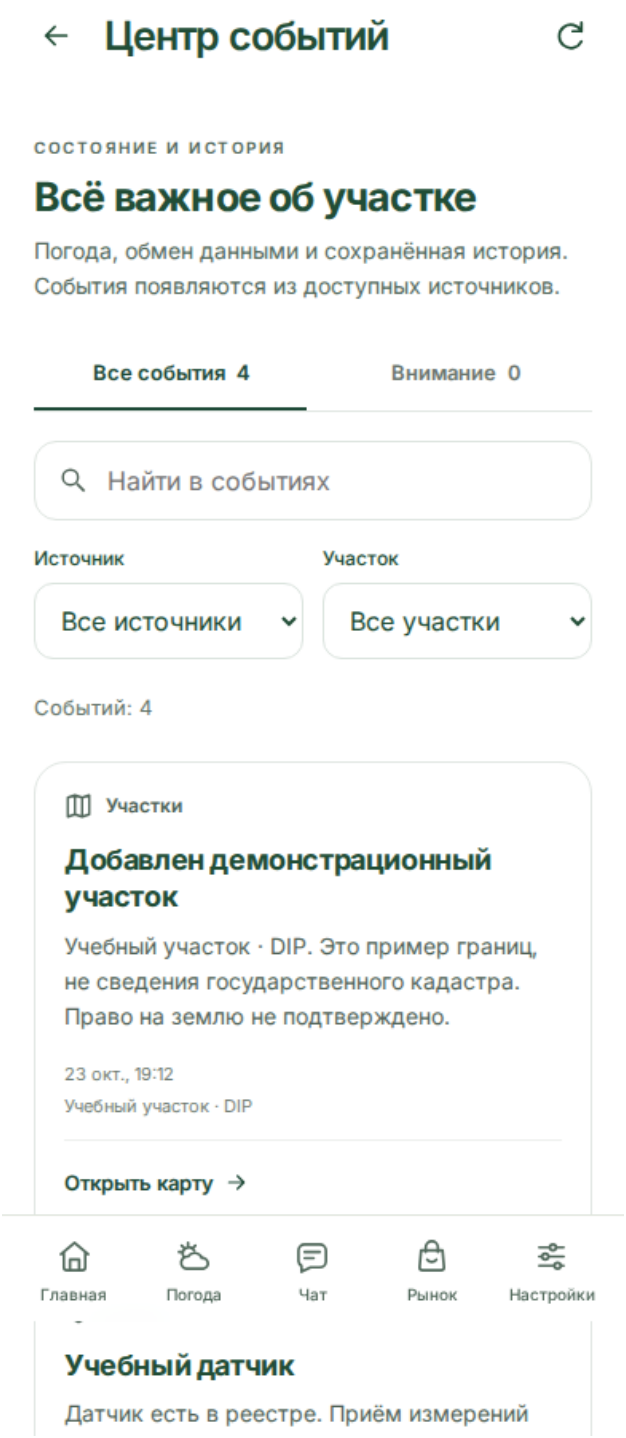


Figure 8.4, panel 1/3 — Event centre and preserved synthetic history. Open the complete original screenshot

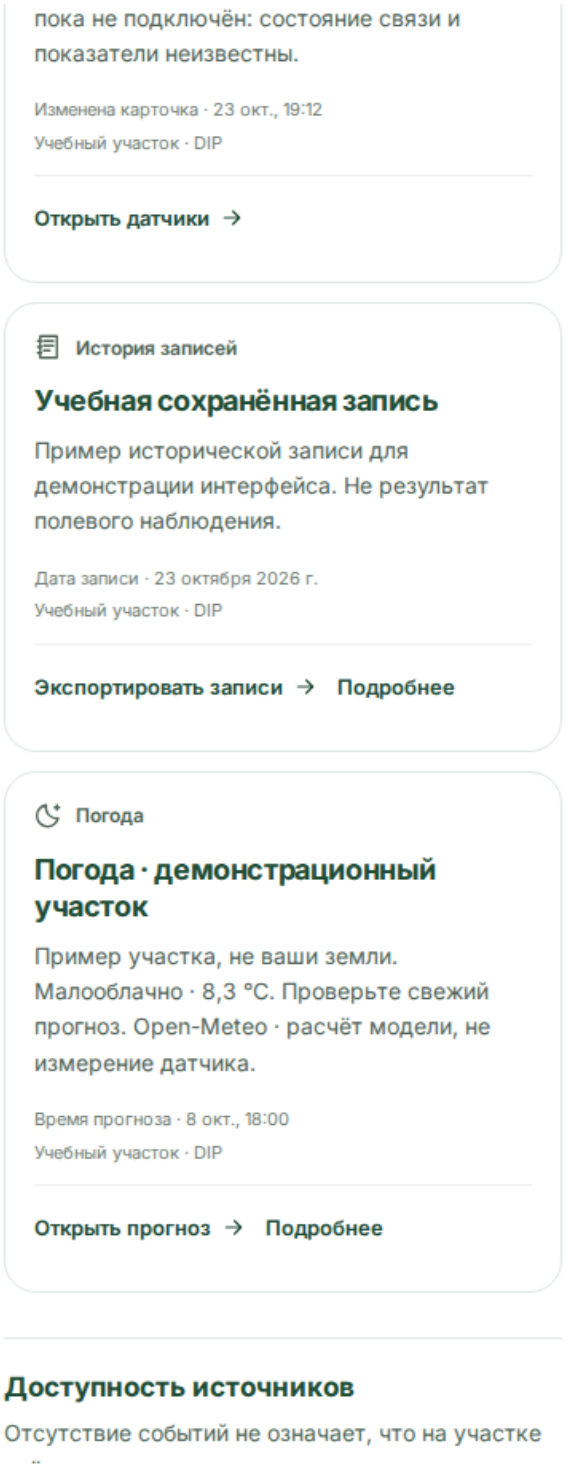


Figure 8.4, panel 2/3 — Event centre and preserved synthetic history. Open the complete original screenshot

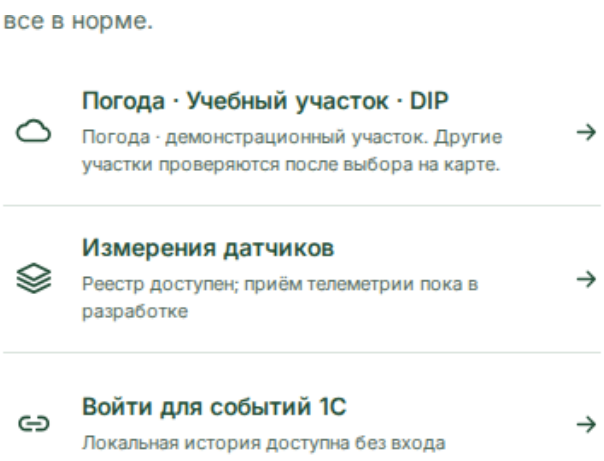


Figure 8.4, panel 3/3 — Event centre and preserved synthetic history. Open the complete original screenshot

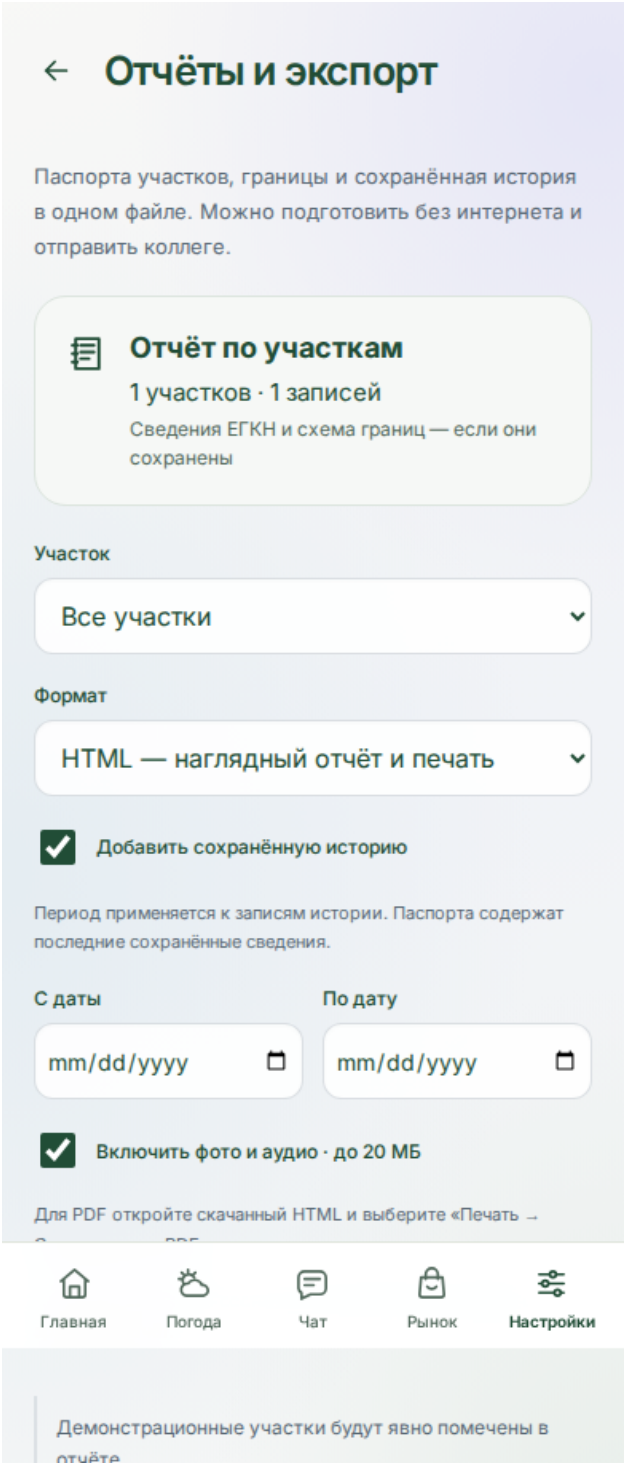


Figure 8.6, panel 1/2 — Report-export controls; generation was not exercised by this capture. Open the complete original screenshot

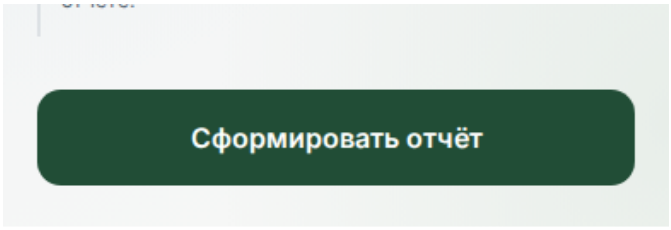


Figure 8.6, panel 2/2 — Report-export controls; generation was not exercised by this capture. Open the complete original screenshot

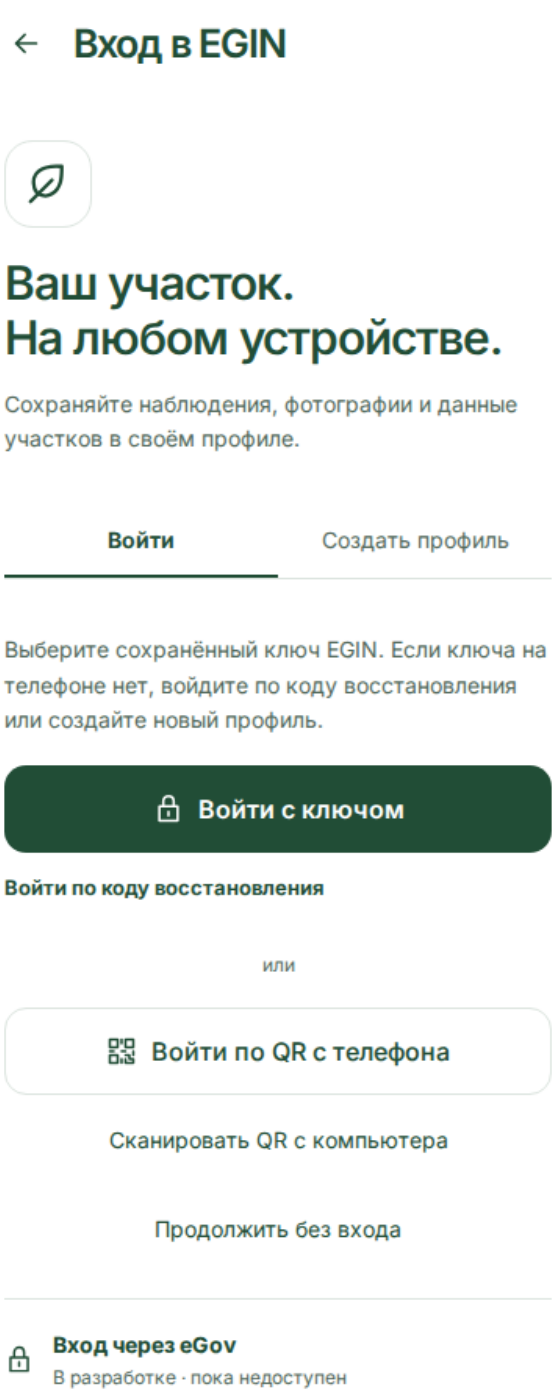


Figure 8.7 — Sign-in interface; no authentication ceremony was executed. Open full-size image

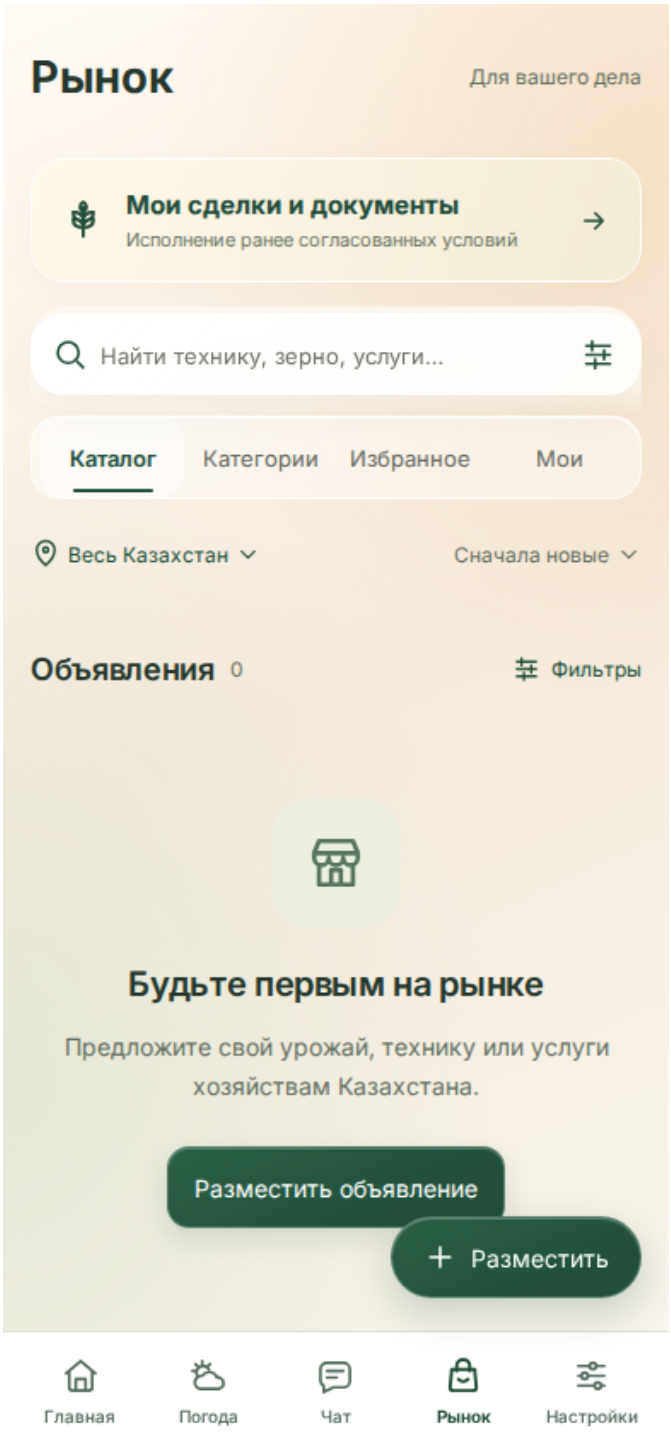


Figure 8.8 — Marketplace with an intentionally empty fixture catalogue. Open full-size image

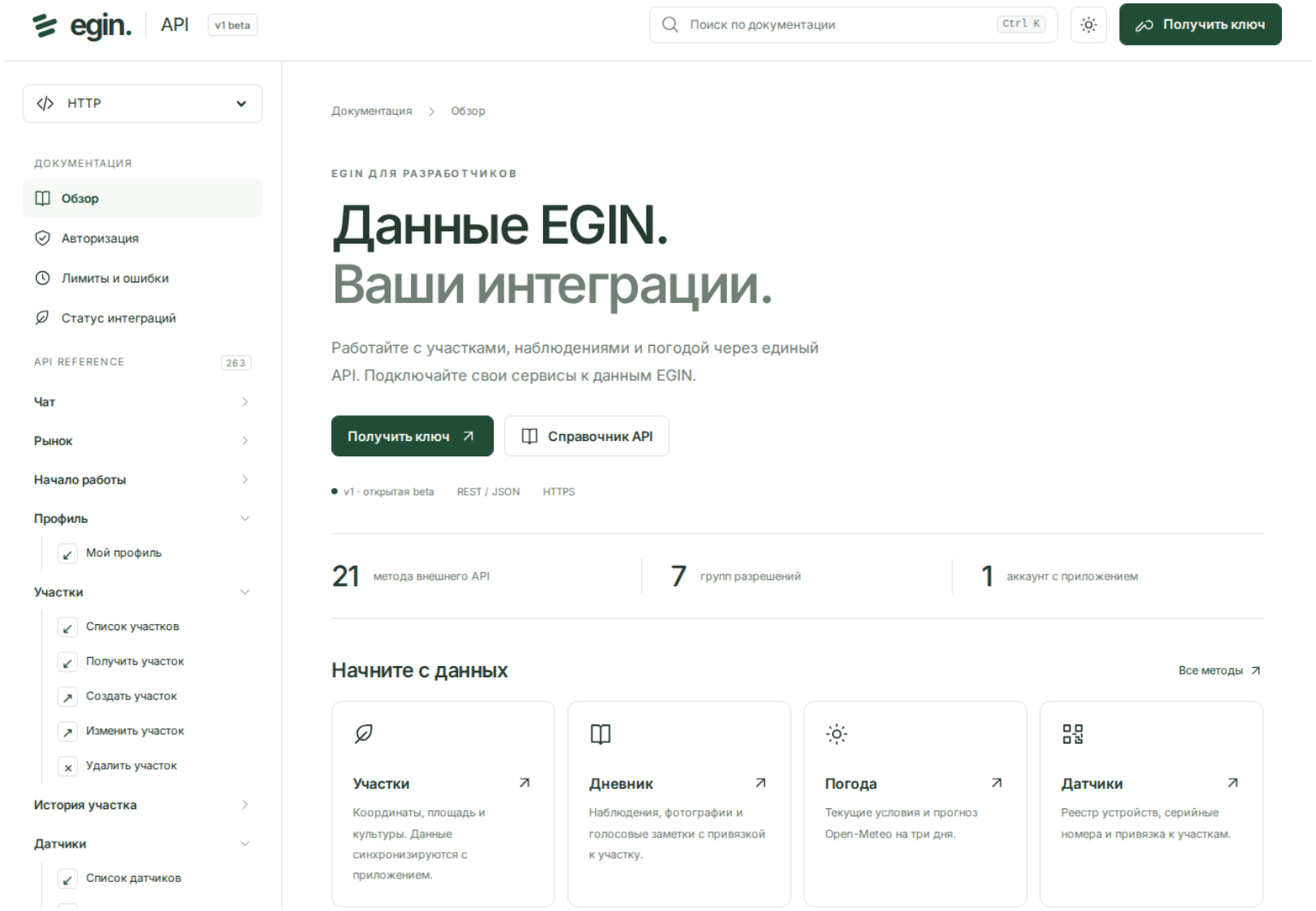


Figure 8.9 — Public developer API reference; no keys were issued. Open full-size image

8.6 External-system interaction example

The developer portal supplements the visual prototype with inspectable request/response contracts generated from catalog.json. For example, the `fields-list` operation uses `GET /v1/fields`, requires the `fields:read` scope and returns a paginated collection of the profile's plots. The following is an **illustrative request**, not a request executed during this review:

```
GET /v1/fields?limit=20 HTTP/1.1
Host: dev-api-egin.perricheno.com
X-EGIN-Key: <issued-open-key>
Authorization: Bearer <issued-secret-key>
```

The documented successful response envelope contains `success`, `result`, `next_cursor` and `request_id`; plot records include a version. Invalid or expired credentials, insufficient scopes and stale update versions are separate rejection cases. The exact methods, parameters and examples are browsable in the portal. For a defence demonstration, use a disposable staging account, show a permitted read followed by a rejected request with insufficient scope, and redact secrets from every capture. This demonstration remains to be executed under the conditions in Part 9.

8.7 Remaining work for the UX criterion

Item	Current position	Required next evidence
Key screen and navigation coverage	Actual UI screenshots, route inventory and source-backed flows are provided in this section.	Include the relevant figures in the defence slides and walk through one coherent scenario.
Screenshot-to-commit traceability	Completed for these captures: fresh web builds from `b0b0f46`, identical application tree to `8cc0ca3`, recorded commands and artifact hashes.	Preserve the build manifest with the submission; rebuild and recapture if the application source changes.
Complete core scenario	Plot/weather/event/report screens exist; the capture shows selected states only.	Record a disposable-account walkthrough: sign-in, plot acquisition, forecast, supported offline edit, synchronisation and export; retain the execution log.
Manual history-entry workflow	Preserved history and attachments remain readable; storage/API support exists.	Resolve the scope of FR-REC-1 and FR-SYNC-1 with the supervisor: implement/restore a composer if required, or explicitly document the reduced UI scope. Do not present seeded records as user-entered records.
Cross-device and accessibility evidence	Responsive styles and browser scenarios exist.	Run the relevant current-commit scenarios and carry out real-device, keyboard, contrast and touch-target checks; retain results.

Item	Current position	Required next evidence
User validation	No observed usability study is attached to this section.	Run the proposed tasks with representative users and record completion, errors and time as planned in Part 9. Interest in the idea is not usability-test evidence.
Extended chat/market/trade scope	Current source includes these functions beyond the earlier core specification.	Approve their requirement mapping, choose representative permitted workflows and record results separately.
External integrations and native packages	eGov, sensor telemetry and partner exchanges remain pending; native delivery is outside this review.	Keep these limits visible. Build Android/iOS packages only after the owner's explicit release request; browser/PWA evidence must not be described as an APK/IPA test.

The provided material makes the intended interaction visible and traceable. Successful operation of every end-to-end scenario, measurable usability improvement and deployed/native compatibility remain separate claims requiring the additional evidence specified above.

9. Verification and Evaluation Plan

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9.1 Purpose, scope and evidence conventions

Verification will establish whether the implemented software satisfies its stated requirements. Evaluation will examine whether the available workflows help agricultural users complete defined tasks more effectively than a comparable manual process. Automated checks alone cannot demonstrate agricultural benefit, usability or stakeholder acceptance.

The evaluation covers the website and PWA: parcel management, cadastral boundaries, forecast presentation, the event centre, retained records and reports, local storage and synchronisation, authentication, the 1C connector and the developer API. Requirements retain their identifiers from Part 6. The architecture combines a React/TypeScript client and IndexedDB with a Node.js/Express API and SQLite. Chat, marketplace, trade and business-platform modules additionally depend on PostgreSQL; semantic marketplace search also depends on an inference service. These additional modules have regression tests but remain outside the diploma core described in Part 7.

Sensor registration is evaluated as a registry, not as live telemetry. The 3D scene is evaluated as an illustrative interface, not as a validated crop model. eGov authentication, contractual cadastral integrations and unsupported third-party integrations remain outside the implemented scope. Native Android/iOS packaging and device certification are not part of this documentation exercise and require a separate explicit release request.

The following evidence levels are used throughout this section:

Evidence level	Meaning
Executed — passed	A named command or observation completed successfully, with its conditions and result recorded.
Executed — failed / blocked	A command was attempted but did not complete successfully; the recorded reason limits what can be concluded.
Inspected	Source code, test definitions or configuration exist and were examined; execution is not implied.
Planned	A measurement or acceptance exercise remains to be conducted. No result is asserted.

This audit uses source commit `8cc0ca3f73052b030dcddb990b8a5460be9ddcdf`, inspected on 23 October 2026. The repository had no published tags at that historical inspection. The later `prob-period-1` tag packages the report and supporting evidence; the historical unit/API and CI results remain attributed to their inspected source and recorded environment. The later web build and UI recapture are identified separately in Section 9.2.1. An implementation label or a test filename in an earlier part is not, by itself, proof that its acceptance criteria have passed.

Current scope correction. The `/journal` route opens the event centre. The browser workspace test explicitly checks that the manual “New record” button is absent and seeds retained records through a test helper. Consequently, retained-entry display, storage and export can be verified, but manual diary creation through the current user interface must not be presented as an available scenario. `FR-REC-1` and the record-editing interpretation of `FR-SYNC-1` require reconciliation with this current product behaviour before final acceptance.

9.2 Verification already performed

9.2.1 Local execution

The historical unit/API run used existing dependencies and Node.js 22.23.3 with pnpm 10.32.1. No dependency installation, production access, deployment or native build was performed during that run. The API tests that could run used temporary stores and fixture servers. Database and inference service URLs were not configured. The later web-build/capture run is recorded separately below and used an isolated dependency copy with missing platform packages supplied.

Check	Observed result	Interpretation
`pnpm typecheck`	Passed for both `@egin/mobile` and `@egin/developers`; exit code 0.	The current TypeScript source passes static checking. This does not establish browser correctness.
Mobile portion of `pnpm test`	**73 passed, 0 failed, 0 skipped.**	Executed unit coverage includes geometry, weather, event derivation, report escaping, sync-batch limits, QR parsing and chat presentation helpers.
API portion of `pnpm test`	**38 passed, 20 failed, 1 skipped** in 59 reported entries; overall command exit code 1.	All 20 failures occurred while importing `sharp`: its Linux x64 native runtime was unavailable in the existing dependency installation. Assertions in those test files did not execute.
Available API adapter tests	Passed cadastral privacy/geometry/transport tests, news parsing/cache tests, 1C transport tests and two marketplace-search helper tests.	These fixture-based tests establish bounded behaviour of the tested adapters, not current availability of live providers.

Check	Observed result	Interpretation
Real ONNX inference case	Explicitly skipped because `MARKET_TEST_INFERENCE_URL` was absent.	No semantic-search quality result was obtained.
PostgreSQL and media-dependent cases	Not exercised; several containing files failed to import before their conditional skips could be evaluated.	Real database concurrency, media processing and additional-module integration are not locally verified by this run.
Fresh web builds, subsequent run	**Passed:** TypeScript checks and Vite builds for mobile and developer apps; mobile service-worker generation. Node.js `24.21.0`; source `b0b0f46`.	Its `apps` tree matches `8cc0ca3`. Missing Linux glibc build packages were supplied only in an isolated dependency copy. This resolves the earlier web-build blocker, not the historical `sharp` API-test failure.
UI fixture recapture, subsequent run	**9 screenshots**, zero reported page errors; Chromium `153.0.8010.12`; **2026-10-23T17:12:27.705Z**. See capture metadata.	Limited rendering checks completed against the fresh `b0b0f46` web builds. API responses were intercepted, and guest records were synthetic. This does not validate live authentication, synchronisation, export, service-worker behaviour or the backend.

The condensed command transcript, exact failing file list and limitations are preserved in local verification evidence. The API count combines individual tests with failed file-level entries; it must not be converted into a requirement-coverage percentage. The dependency failure is an environment problem observed during this run, but the overall test command remains **failed** until a reproducible rerun succeeds.

The subsequent web-build and capture execution record records source extraction, runtime, commands and outcomes. Artifact fingerprints cover all 72 generated files. The optional build packages were `@rolldown/binding-linux-x64-gnu` 1.2.13 and `lightningcss-linux-x64-gnu` 1.33.0. No backend suite, native package or production deployment was executed in this later run. Its successful build and screenshot checks do not change the earlier API or CI outcomes.

9.2.2 Continuous-integration evidence

The following GitHub Actions results were inspected for the same source commit at 15:55 UTC on 23 October 2026. These are remote execution records, distinct from the local results above.

Workflow	Observed status	Evidence and limitation
Mobile checks	Success	Run 39977774247. The `verify` job passed; `android` was skipped. The workflow defines typechecking, unit/API checks, web builds and browser scenarios. Retain its artifacts with the defence evidence.
Chat checks	Failure	Run 39977774228. The failing step was “Real PostgreSQL, realtime, permissions and media validation”. Investigate it before making an additional-module readiness claim.
Trade, billing and video notes	Failure	Run 39977774136. The failing step was “Cameras, offline outbox, admin controls and trade cards”. The workflow does not constitute completed additional-module acceptance.
Market checks	Cancelled	Run 39977774255. The job was cancelled during installation of Chromium/WebKit dependencies. Cancellation is not a successful result or proof of an application defect.

The inspected branch response reported `protected: false` for `main`. Therefore, the protected-branch acceptance criterion in `NFR-MNT-1` is not demonstrated. A successful Mobile workflow does not establish that all workflows are green or that merges are protected.

Browser capture evidence in Part 8 must be interpreted according to its own fixture and build-provenance notes. The capture used Node.js 24.21.0 and local static servers; it is separate from the Node.js 22 unit/API run above. Screenshots and limited screen assertions are not substitutes for the complete browser suite, live authentication or a usability study.

9.3 Functional and non-functional testing plan

Each test case below traces to Part 6. “Expected result” is an acceptance target, not a claimed observation. Test paths are relative to the repository root; mobile browser scripts are in `apps/mobile/tests/`, and API tests are in `apps/api/tests/`.

9. Verification and Evaluation Plan · Table 9.4

Test Case	Objective	Method	Expected Result	Metric	Status
TC-01 — Authentication and recovery (FR-AUTH-1…4, NFR-SEC-1)	Prevent invalid authentication and account lockout.	Execute api.test.mjs, qr-auth.test.mjs and auth.cjs with an isolated account and virtual authenticator; add physical-device passkey checks.	Invalid challenge/origin/signature and reused recovery codes are rejected; last-passkey deletion is prevented; valid recovery rotates the code.	All enumerated positive and negative cases pass; zero unintended sessions.	API files blocked locally by sharp; browser scripts inspected; physical-device acceptance planned. Native iOS passkey configuration remains pending, as qualified in Part 6.
TC-02 — Parcel acquisition and privacy (FR-FLD-1…3, NFR-SEC-7, NFR-PERF-4)	Return only an exact, bounded, non-personal parcel result.	Execute cadastral adapter tests with valid, invalid, ambiguous, timeout and oversized fixtures; then map-picker.cjs on a controlled build.	Correct projected boundary; no broad search or silent ambiguity; no personal names or individual identifiers; saving requires confirmation.	Zero privacy leaks or false exact matches in fixtures; configured timeout/size limits enforced.	Adapter tests executed and passed; browser acceptance not rerun locally.
TC-03 — GeoJSON and retained boundaries (FR-FLD-4…6)	Preserve valid geometry and reject malformed imports.	land.test.ts, land.test.mjs; import/export round trip, denied geolocation and offline reload in browser.	Holes and geometry survive the round trip; invalid/oversized geometry is rejected; saved contours remain readable offline.	100% valid-fixture round trips; zero accepted invalid fixtures; zero lost stored contours.	Mobile geometry tests passed; API file blocked; browser/device acceptance planned.
TC-04 — Offline synchronisation (FR-SYNC-1…8, NFR-REL-1)	Prevent silent loss, duplication and cross-profile leakage.	sync-batches.test.ts, api.test.mjs, workspace.cjs, sync-ui.cjs; two contexts, offline writes, reconnect, repeated payloads, deletion and conflict fixtures.	Every queued operation is acknowledged or remains explicitly pending/in conflict; profiles remain isolated; deleted records do not reappear.	Zero silent losses, duplicates or cross-owner records; convergence after explicit conflict resolution.	Batch tests passed; API blocked; full browser lifecycle planned. Retained-entry seeding does not prove a manual diary editor exists.
TC-05 — Weather and external-source failure (FR-WTH-1…3, FR-HOME-4, NFR-REL-4)	Preserve coordinate binding and honest freshness labels.	Weather unit tests, news.test.mjs, then home-weather.cjs and news.cjs; fresh/missing/stale/offline fixtures.	Correct selected-plot coordinates; missing values remain absent; cached forecasts/news are labelled and usable without fabricated warnings.	Zero cross-plot forecast substitutions or fabricated readings; all defined fallback cases pass.	Unit and news adapter tests passed; browser/provider validation planned.

Test Case	Objective	Method	Expected Result	Metric	Status
TC-06 — Events, reports and retained records (FR-EVT-1, FR-RPT-1, retained-data part of FR-REC-1)	Produce traceable event summaries and safe local exports.	events.test.ts, reports.test.ts, then events.cjs, reports.cjs; seeded retained history and malicious text/formula fixtures.	Filters preserve relevant events; HTML/CSV/JSON match the selected dataset; HTML is escaped and CSV formulas neutralised.	Zero unsafe exports or unexplained omissions; expected field/record counts match all three formats.	Unit tests passed; download/media/browser acceptance planned.
TC-07 — Honest product boundaries (FR-AUTH-5, FR-SNS-1…2, FR-HOME-3)	Avoid presenting planned integrations or illustrative data as live services.	QR/parser and field-readiness tests; readiness.cjs; inspect eGov, sensor and 3D labels.	eGov remains “В разработке”; EGIN QR is identified correctly; a registered sensor does not imply telemetry; 3D assumptions are visible.	Zero misleading states in the agreed fixture matrix.	Relevant unit tests passed; remaining UI checks inspected/planned.
TC-08 — Read-only 1C import (FR-INT-1…2, NFR-SEC-3…5)	Preserve credentials, owner boundaries and existing edits.	one-c-transport.test.mjs, one-c.test.mjs, one-c-flow.test.mjs; local OData fixtures for pagination, repeat import, disconnect and conflicts.	Only allowed read requests; encrypted secrets; no duplicate imports, silent overwrites or credentials in responses.	Zero forbidden outbound requests, duplicate source records or exposed secrets; repeat import is idempotent.	Transport tests passed; full connector/flow files blocked locally; client-sandbox validation planned.
TC-09 — Developer portal and API (FR-API-1…3)	Enforce API contracts, scopes and the separate portal audience.	developer.test.mjs, including installed-route inventory; developers.cjs; expired/revoked/rotated keys, missing scopes and stale If-Match.	Required 401/403/412 responses; secret shown only on issue/rotation; no cross-audience session use; catalog matches installed routes.	Zero missing/extraneous catalog routes or unauthorised successful requests.	API file blocked locally; suite inspected and must be rerun.
TC-10 — Responsive layout and accessibility (NFR-USE-1…3, FR-HOME-1)	Make primary workflows usable on phones.	Chromium/WebKit viewport checks at 320/360/390/430/480 CSS px, physical safe-area checks, automated accessibility audit and manual keyboard/focus review.	No horizontal overflow or obscured controls; prominent 3D home view; text/icon active navigation; readable contrast and touch targets.	0 overflow cases; contrast and target-size thresholds in 9.4; documented unresolved findings.	Test scripts inspected; complete accessibility and device measurements planned.

Test Case	Objective	Method	Expected Result	Metric	Status
TC-11 — Performance and renderer lifecycle (NFR-PERF-1…3)	Measure load responsiveness and background work.	Fixed Lighthouse mobile profile; production web bundle; cold/warm runs; scene-performance.cjs and service-worker reload tests.	Load targets met; offline shell available after caching; hidden/off-screen scene stops requesting frames.	LCP ≤ 4.0 s, score ≥ 70; 0 requested idle frames over the specified 5 s window.	Planned; local test-command duration is not a performance result.
TC-12 — Deployment, restore and environment isolation (NFR-REL-2…3,5, NFR-SEC-6)	Establish recoverability without changing production during evaluation.	Controlled staging rehearsal of blue/green switching and rollback; restore test copy; inspect RP origins, volumes, assets and compatible schemas.	Gateway and both asset namespaces remain available; previous version operates with migrated data; restore is consistent; origins/stores remain separate.	No failed health probes during the measured interval; rollback ≤ 60 s; SQLite integrity check ok; restore contents match expected records.	Configuration inspected; measured rehearsal and recovery evidence planned.
TC-13 — Quotas, privacy and secret handling (NFR-CAP-1, NFR-SEC-2…4,7)	Reject excess data and unauthorised access without partial writes.	Boundary cases at/below/above quotas, cross-owner API calls, rate-limit cases, repository secret scan and log review.	Controlled 4xx responses; no partial writes, leaked secrets or retained personal cadastral fields.	Zero unauthorised reads/writes or secret findings; every agreed boundary case passes.	Cadastral privacy/rate cases passed; remaining quota/security checks planned or blocked with their API suite.
TC-14 — PWA compatibility and update safety (NFR-CMP-1…2, NFR-MNT-2…3)	Preserve local data across supported browser and PWA updates.	app-update.cjs, offline/reopen tests, Chromium and WebKit, physical Android browser/iPhone PWA; inspect shared sync engine and native job guard.	Updated shell preserves data/queue; unsupported background sync falls back on reopening; native jobs remain manually gated.	Zero lost queued records; recorded browser/device matrix; native job guard present.	Scripts/configuration inspected; physical-device and update measurements planned. No native packages built.
TC-15 — Additional-module regression (outside diploma core)	Prevent chat/market/business changes from being misrepresented as core acceptance.	Dedicated Chat, Market and Business workflows with isolated PostgreSQL, media tools and inference fixtures.	Relevant workflow assertions pass and failed cases have tracked owners.	Per-workflow passed/failed/skipped counts and artifact references.	Local dependency block and remote failures recorded in 9.2; not accepted by this document.

9.4 Proposed outcome metrics and acceptance thresholds

The targets below are **prospective**. They define how outcomes will be judged and are not measured achievements. Targets already specified in Part 6 are retained; supplementary thresholds are explicitly proposed for the evaluation protocol.

Metric	Definition and measurement	Target / decision rule	Requirement / interpretation
Task completion	Correct, unassisted completions divided by all assigned tasks; report numerator and denominator and also results per task.	Proposed target: ≥ 90% across the planned task set, with every failure described.	Practical usefulness; small-sample descriptive evidence, not population validation.
Parcel-addition time	Seconds from opening “Plots” to a correctly saved parcel; final geometry and selected identifier must match the reference.	Median ≤ 60 s and lower than the matched manual method.	`NFR-USE-4`; successful-only time is reported alongside failures/timeouts.
Workflow time improvement	For each participant and task: `(baseline time − EGIN time) / baseline time × 100%`; report paired values and median.	Proposed exploratory target: median improvement ≥ 20%, without a reduction in correctness.	A decision aid for iteration, not a promised gain or an already verified interview finding.
Critical transcription errors	Wrong parcel identifier, coordinates, area label, forecast location or required report content against a prepared answer key.	Proposed target: 0 critical errors; report all lesser omissions separately.	Correctness must not be traded for speed.
Offline data preservation	For a controlled batch, compare queued operation IDs and final content with server acknowledgements, explicit conflicts and pending errors.	100% accounted for; 0 silent losses or duplicates across at least 20 interruption/reconnect cycles.	`NFR-REL-1`; includes unsuccessful cycles, not only successful demonstrations.
Reconnection recovery time	Time from network restoration to acknowledgement of a 50-record fixture batch without attachments or intended conflicts.	Proposed target: p95 ≤ 10 s on the specified controlled connection; failures remain counted.	Supplementary diagnostic metric; not a promise for weak networks or media uploads.
First-load performance	Lighthouse mobile measurements of the web production bundle under the frozen profile; retain every run.	Median LCP ≤ 4.0 s and Performance score ≥ 70; report all five run values and spread.	`NFR-PERF-1`: Slow 4G and 4× CPU throttling.

Metric	Definition and measurement	Target / decision rule	Requirement / interpretation
Layout, contrast and touch size	Measure all primary screens at specified widths; inspect representative physical safe areas; calculate contrast; inspect target bounding boxes.	0 horizontal-overflow cases; normal text contrast ≥ 4.5:1; interactive targets ≥ 44×44 CSS px unless a documented exception is accepted.	`NFR-USE-1…3`; automated auditing does not replace manual review.
Deployment recovery	Timed staging rehearsal with health probes once per second throughout the switch; compare before/after data and asset availability.	No observed failed probes; rollback ≤ 60 s; valid restored database.	`NFR-REL-2…3`; 1 s sampling cannot prove absence of every shorter interruption.
Perceived task ease	After each task, a consistent 1–7 ease question, with optional comments and no interviewer coaching.	Proposed exploratory target: median ≥ 5/7.	Qualitative explanation of observed difficulty; no questionnaire result currently exists.

Agricultural yield, profit, sensor accuracy, adoption and long-term retention are not evaluation claims at this stage. Demonstrating them would require separate longitudinal datasets and an appropriate study design.

9.5 Measurement conditions and reproducibility

1. **Freeze the candidate.** Record the Git SHA, lockfile, build command, browser/OS versions, device model, environment and fixture version. Rebuild both web apps from that SHA. A previously generated `dist` directory without verified source provenance must not be treated as a reproducible release artifact.
2. **Separate the test environments.** Use temporary data for automated tests and a dedicated staging dataset for acceptance sessions. Never use production personal records or real payment/partner operations as fixtures. Preserve separate production/staging volumes and passkey RP origins. Use HTTPS for actual passkey and PWA device checks.
3. **Use deterministic source data.** Prepare valid and invalid cadastral numbers, one polygon with a hole, one malformed import, two plot coordinate sets, a missing-data weather response, fresh and stale forecasts, retained history and safe/unsafe export strings. Record expected answers before the session. Test controlled failures with fixtures; run live-provider smoke checks separately and date their results.
4. **Fix the browser matrix.** Run automation in Chromium and WebKit; record their exact versions. Repeat essential flows on a physical Android browser and an iPhone Safari/installed PWA. Use 320, 360, 390, 430 and 480 CSS px for layout checks. An emulated browser viewport is not evidence of a native build or every physical device.

5. **Separate connection conditions.** Measure usability under one stable connection and one declared offline/reconnect scenario. For the proposed sync-latency metric use 10 Mbps downstream, 5 Mbps upstream and 100 ms added latency, a 50-record batch of approximately 1 KiB per record and no attachments. Capture the exact shaping tool/settings. Do not mix these timings with the Lighthouse Slow 4G/4× CPU experiment.
6. **Control cache and state.** Distinguish cold loads from warm/offline reloads. Clear browser state for cold measurements; preserve the specified account and cached dataset for offline trials. Use a built service worker for PWA/update tests; a development server is insufficient evidence for those claims.
7. **Retain repetitions and exceptions.** Collect five performance runs and at least 20 sync interruption cycles. Record each failure, timeout, retry and missing observation. Reset the dataset between repetitions and do not silently remove slow runs.
8. **Capture auditable evidence.** Store timestamped command logs, CI links, screenshots, task sheets, anonymised timing records and defect IDs. Separate synthetic fixtures, self-reported stakeholder feedback and directly observed measurements. Restrict interview recordings and personal data according to participants' consent.

The current local `sharp` import failure must be resolved in a reproducible test environment before rerunning the blocked API checks. The environment repair and a subsequent successful run must have their own evidence; this document does not assume that repair has occurred.

9.6 Baseline comparison and user study

The proposed baseline is a **manual workflow using separate tools**: a public cadastral map or supplied parcel source, a weather page and a spreadsheet or document template. This is a candidate comparator derived from the problem definition, not a claim that every interviewed farmer uses exactly this combination. Confirm its realism with participants and document the exact tools, versions and starting materials before evaluation.

Task	Baseline workflow	EGIN workflow	Common success criterion
B1 — Identify and save a parcel	Find the supplied cadastral identifier; transfer its boundary/location and labelled area into the supplied record template. For the specific manual-drawing hypothesis, digitise the same reference boundary using an agreed map tool.	Search by cadastral number, inspect the boundary and confirm addition.	Correct parcel and consistent boundary/location; registered and calculated areas are not confused.
B2 — Obtain plot-specific weather	Use the same coordinates with the chosen weather source and record the requested forecast facts and timestamp.	Select the plot and read the corresponding forecast.	Correct coordinates, forecast period, units and freshness interpretation.

Task	Baseline workflow	EGIN workflow	Common success criterion
B3 — Find an event requiring action	Locate the relevant item among supplied separate weather/history/sync notes and identify the next action.	Filter the event centre and follow its action link.	Correct event, source, timestamp and next action; no historical item mistaken for a new alert.
B4 — Prepare a parcel report	Populate an agreed template from the supplied boundary, metadata and retained-history dataset.	Select the parcel/period and export the equivalent report.	Required information is complete and correct; both methods use the same reference dataset.

Two comparisons must remain distinct: the practical cadastral-number workflow against a realistic scattered-tool process, and the narrower manual-boundary-drawing hypothesis in `NFR-USE-4`. A participant must not be forced to draw a boundary when their normal baseline already provides an equivalent downloadable contour; report that alternative separately.

Recruit a target of **five potential users**, including farmers and agronomists, with a minimum of three completed sessions for the limited `NFR-USE-4` pilot. This is a proposed sample, not the number already tested. Previous discussions reported in Part 2 do not automatically count as usability sessions. Record participants' relevant role, device familiarity and consent using anonymised identifiers.

Use a within-participant design: every participant tries both methods. Counterbalance the order as evenly as possible between baseline-first and EGIN-first, and use different but comparable parcel datasets to reduce memorisation. Provide equal training and practice time. Use the same device and connection for both methods where practical; if real workflows require different equipment, record that difference and avoid attributing all timing differences to the software.

Start timing at the same task instruction and stop only after the common success criterion is met. Predeclare a three-minute limit for B1–B3 and a five-minute limit for B4; record timeouts as unsuccessful attempts. Report completion rates over all attempts, successful-task times separately, and a sensitivity view that includes timeout limits. Record assistance and critical errors rather than quietly converting coached completions into successes. Report paired differences, medians and ranges. A small convenience sample supports prototype refinement and descriptive evidence, not a general statistical claim about all farmers in Kazakhstan.

The study does not include manual diary creation, live sensor monitoring or eGov login as user tasks because those are not available user flows in the audited scope. Unsupported tasks cannot be used to claim a product benefit.

9.7 Acceptance decision and remaining evidence

Achievement of the project aim will be assessed through a combined evidence chain: a confirmed user need from Part 2, a supported interaction in Part 8, an explicit requirement in Part 6, a reproducible verification case and a measured user outcome. No single successful build or screenshot is sufficient.

Before the final defence, the team must:

1. Reconcile requirement status with current routing and retained-record behaviour; preserve stable IDs and record any accepted scope change.
2. Repair the isolated test environment and rerun the blocked API and browser checks, including route inventory, authentication, sync conflicts, API scopes and 1C import.
3. Review failed/incomplete CI workflows and retain the final results; separately decide and document whether to enable required checks and branch protection.
4. Use the traceable web candidate for physical-device, accessibility, performance and staging recovery measurements; rebuild and record a new source revision if implementation changes are needed.
5. Conduct the counterbalanced user sessions, preserve consented evidence and calculate the predeclared outcome metrics.
6. Maintain an acceptance register linking requirement ID → test case → evidence → result → defect or scope decision. A failing or unmeasured Must criterion stays open; any revision requires a documented scope decision and updated acceptance criterion.

The present evidence supports the existence of a substantial implementation and several successful local/remote checks. It does not yet establish that all Must requirements, operational recovery targets or usability hypotheses have been validated.

10. Project Management and Risks

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This section distinguishes **repository evidence**, **documented team responsibilities**, and **proposed management activities**. A source file or test suite demonstrates that an artefact exists; it does not, by itself, demonstrate acceptance by users or a successful test run. The evaluation evidence and outstanding measurements are specified in Part 9.

10.1 Repository and current project state

The project repository is Adilzhan17/egin. The inspection baseline for this report is commit `8cc0ca3`, before the addition of Parts 2, 8, 9 and 10. This identifies the implementation examined. The later `prob-period-1` tag identifies the report-submission package. The table below preserves the historical repository observations from 23 October 2026; it is not a fresh inspection of the tagged submission.

Item	Observed evidence	Interpretation / limitation
Default development line	Local `main`, `origin/main` and the authenticated GitHub branch response resolve to `8cc0ca3f73052b030dcddb990b8a5460be9ddcdf`; `origin/HEAD` points to `origin/main`.	The report can refer to a reproducible source baseline. Record the final submission SHA separately.
History	174 commits are reachable from the inspected `main`. Representative changes are linked below.	Commit history demonstrates development activity, not hours worked or equal contribution.
Branches	The remote reference query returns one branch, `main`. The working copy also contains a local archive branch and a tracking reference for an upstream repository.	Local archival/tracking references are not additional active GitHub project branches.
Tags	No tags were returned by the remote reference query or authenticated GitHub inspection at the historical baseline.	The subsequent `prob-period-1` tag packages the submission; implementation inspection and executed test evidence retain their recorded commit references.
Application structure	`apps/mobile`, `apps/developers`, `apps/api` and `deploy` contain the user application, developer portal, server and deployment tooling.	The current implementation includes the original land-management core and later chat, marketplace and business/trade extensions.

Item	Observed evidence	Interpretation / limitation
Automated checks	`checks.yml`, `chat.yml`, `market.yml` and `business.yml` define automated checks.	At `8cc0ca3`, the authenticated GitHub snapshot at **2026-10-23T15:55:27Z** reported Mobile checks: success (`verify` succeeded; Android packaging was skipped), Chat checks: failure at “Real PostgreSQL, realtime, permissions and media validation”, Trade/billing/video: failure at “Cameras, offline outbox, admin controls and trade cards”, and Market: cancelled during Playwright browser installation. Step locations identify where the runs stopped; they do not establish the underlying cause or successful completion of skipped scenarios.
Branch protection	The authenticated GitHub response reports `protected: false` for `main`.	Mandatory passing checks before merge are a proposed control, not an enforced property demonstrated by the current repository.
Task-management records	`EGIN3-IMPLEMENTATION.md`, `EGIN3-REPORT.md` and feature documentation record implementation work and remaining boundaries.	These are useful engineering records, but do not establish that a YouTrack project or a historical issue workflow exists.

The history has been rewritten to correct commit co-author attribution. Accordingly, links and evidence should use the **current commit identifiers**. An old screenshot or link using a superseded identifier needs to be re-associated with the current revision. Git author names and account aliases are not sufficient to independently establish each student's contribution; responsibility should also be supported by task records, reviews and an explanation of the relevant implementation.

10.2 Allocation of responsibilities

The allocation below is recorded in Part 1. It is a **documented allocation**, not an independently verified attribution of every commit.

Team member	Documented responsibilities	Representative implementation evidence
Rustam Seitzhan — Full-Stack Developer	Offline synchronisation engine; WebAuthn and Capacitor passkeys; cross-device QR sign-in; security; CI/CD and blue/green deployment.	`sync-engine.js`, `server.mjs`, `qr-auth.mjs`, `manage.py`, automated workflows and their tests.
Adilzhan Amangeldi — Full-Stack Developer	EGKN cadastral adapter and coordinate reprojection; 3D crop scene and weather module; 1C OData connector; external API and developer portal.	`cadastre.mjs`, `features/home`, `features/weather`, `one-c.mjs`, `developer.mjs`, `apps/developers`.

The proposal identifies **Shuteyeva Gulnur Serikovna** as the supervisor. This does not imply that the supervisor implements features or administers the repository.

Ownership of user-research records, task-management records, final evaluation and the newer chat/marketplace/business modules is not allocated by the proposal. These responsibilities require explicit agreement between the two documented team members. They remain **unassigned in this report** rather than being inferred from Git aliases. For cross-cutting changes, the proposed process is to name one responsible implementer and have the other team member review the change.

10.3 Work register and task-management approach

GitHub history, code, workflow runs and the engineering documents are the project's primary available management evidence. No YouTrack project URL, exported board or integration configuration was found in the inspected tracked documentation and configuration. The commit prefix `DIP-1` is visible in Git history, but a prefix alone does not prove that a corresponding YouTrack issue exists. The project owner considers a separate YouTrack project optional. Consequently, this report proposes traceability through GitHub and the work register below; if the assessor requires the named YouTrack artefact in the rubric literally, its acceptance or substitution should be clarified before the defense.

The following is a **proposed local work register**. Identifiers beginning with `PM-` are report-local references, not existing YouTrack issue keys. “Implemented” means that code and related artefacts are present; it does not imply that all evaluation gates have passed. Historical implementation records should retain their actual evidence and must not be backdated when imported into a tracker.

Local ID	Work item	State and evidence	Responsible role / next ownership decision
PM-01	Offline workspace, authentication and cross-device sign-in	Implemented: `0688142` and `a19cfe3`; workspace, API and QR test suites exist. Evaluation status is recorded separately in Part 9.	Rustam, according to the proposal.
PM-02	Cadastral plot lookup and offline parcel snapshots	Implemented: `0b64472` and `79468ce`; cadastral modules and tests are present. Live upstream availability remains a dependency.	Adilzhan, according to the proposal.
PM-03	Weather, 3D home, 1C import and developer API	Implemented artefacts: home/weather features, `4f7f88a` for 1C and `7614495` for the portal. Sensor telemetry and eGov are excluded.	Adilzhan, according to the proposal.
PM-04	Website deployment and operational safeguards	Implemented tooling: `deploy/manage.py`, compatibility tests and environment-specific data configuration. A current backup/restore and rollback demonstration is still required for the evaluation evidence.	Rustam, according to the proposal.

Local ID	Work item	State and evidence	Responsible role / next ownership decision
PM-05	Chat, marketplace and business/trade extensions	Code and implementation records are present: `8b916fd`, `fe25f2d`, `9338b8a`, `865e4e6`. The defense scope and ownership of these extensions require confirmation.	Unassigned in the proposal.
PM-06	Prepare assessment documentation	Parts 1 and 3–7 were added in `8cc0ca3`; Parts 2, 8, 9 and 10 were completed in `b0b0f46`. The combined report and updated capture evidence are packaged under `prob-period-1`. Outstanding user-research and acceptance evidence remains explicitly identified in the relevant parts.	Documentation owner to be agreed.
PM-07	Substantiate the user-research evidence	Pending: attach anonymised records of 3–5 substantive discussions or a real client agreement; distinguish reported meetings from verified requirements as specified in Part 2.	Research coordinator to be agreed; participants must approve any attributed material.
PM-08	Consolidate task traceability	Planned: assign owners and link the work register or actual GitHub issues/PRs to requirements, changes and verification records. A YouTrack mirror is optional, subject to the assessor's interpretation of the rubric.	Management-record owner to be agreed.
PM-09	Investigate failed or incomplete CI runs and record reproducible verification	Open: inspect the failed Chat and Trade/billing/video steps and complete the cancelled Market run, then execute any relevant Part 9 checks with stated dependencies. Save results and distinguish defects, setup failures, cancellations and skipped tests.	Rustam for core security/deployment and Adilzhan for integrations; owners for extension failures and overall test coordination to be agreed.
PM-10	Conduct outcome and physical-device evaluation	Planned: usability/baseline comparison, mobile performance, real-device permissions and recovery checks, using the conditions in Part 9.	Evaluation owner and participant appointments to be agreed.

Local ID	Work item	State and evidence	Responsible role / next ownership decision
PM-11	Freeze and rehearse the defense package	Planned: select the final commit, repair stale claims/ links, collect screenshots and measurement records, and rehearse the agreed scenarios.	Both team members; final sign-off process to be agreed with the supervisor.
PM-12	Release native packages or enable external commercial integrations	Deferred: requires a separate scope decision, explicit native-release instruction where applicable, provider access and validation.	Relevant implementation owner after the scope is approved.

If a board is adopted in GitHub or YouTrack, the proposed states are **Backlog → Ready → In Progress → Review → Verification → Done**, with a separate **Blocked** state and a recorded reason. Each issue or work-register entry should include its requirement IDs from Part 6, objective, acceptance criteria, assignee, dependency, risk reference where relevant, implementation link and verification evidence. Link any actual issue identifiers to the local `PM-` references; do not present local IDs as proof of an existing external board.

10.4 Milestones and acceptance gates

No approved calendar schedule or dated milestone commitments were supplied. The milestones below are therefore **ordered acceptance gates**, not invented deadlines. Dates should be added only after the team confirms the academic submission date, available participants and testing resources.

Milestone	Current position	Exit evidence	Dependency
M1 — Define and substantiate the problem	Proposal and literature analysis exist; user discussions are reported by the team.	Reviewed proposal; anonymised discussion records; evidence-to-requirement mapping; assumptions explicitly labelled.	Access to original notes and participant permission; PM-07.
M2 — Establish requirements and design	Parts 6 and 7 exist; the implementation has subsequently expanded.	Requirements and diagrams reconciled with the selected defense scope and actual current behavior; acceptance criteria remain measurable.	M1 and scope review of later extensions.
M3 — Deliver the functional website prototype	Source modules and deployment tooling are present.	Reproducible web build and demonstration of the Part 8 scenarios at a recorded revision, with screenshots and visible limitations.	M2; a test environment and suitable test accounts/data.

Milestone	Current position	Exit evidence	Dependency
M4 — Demonstrate functional and security verification	Automated suites and historical reports exist; current CI includes failures requiring investigation.	Current run logs, environment description, requirement-to-test mapping, and reviewed failures/skips; no unsupported “all tests passed” claim.	M3; isolated SQLite/PostgreSQL and required media tools; PM-09.
M5 — Evaluate the project aim	Measurement plan prepared; completed outcome measurements are not assumed.	Baseline comparison, usability results, performance data, real-device checks and operational recovery evidence.	M4; representative participants and devices; PM-10.
M6 — Complete management and defense evidence	In preparation.	GitHub/work-register evidence, named owners, task-to-commit-to-test links, risk review, final source SHA and rehearsed presentation. A YouTrack artefact is added only if required by the assessor.	PM-08; M1–M5 evidence or explicitly disclosed unresolved items.
M7 — Optional native/commercial release	Outside the current documentation task.	Explicit owner release request, signing/provider configuration and separate release acceptance.	Required accounts/agreements and validation; no automatic native build on push or PR.

10.5 Risk assessment

Probability and impact are **qualitative planning judgements**, not measured probabilities. *High impact* denotes possible data exposure/loss, failure of a core user scenario or failure to substantiate an assessment criterion; *medium impact* denotes limited functional degradation or rework. Risk owners should be confirmed when the corresponding work items are assigned.

Risk	Probability	Impact	Mitigation Strategy
R1 — Reported interviews cannot be substantiated or are interpreted as validated feature demand.	Medium	High	Preserve original notes and consent; use anonymised participant IDs; distinguish direct evidence, team reports and hypotheses. Obtain 3–5 substantive discussion records without inventing quotations or retrospective dates.
R2 — Task ownership or the assessor's expected management artefact is unclear.	Medium	Medium	Assign owners and demonstrate the GitHub/work-register task-to-commit-to-test chain. Clarify whether the named YouTrack artefact is required; provide it only if needed, without inventing historical issues.

Risk	Probability	Impact	Mitigation Strategy
R3 — Public EGKN, map tiles, weather or news services change or become unavailable.	Medium	High for cadastral lookup; Medium for secondary feeds	Keep bounded requests, controlled errors and tested fixtures; demonstrate GeoJSON/geolocation fallbacks and labelled cached data. Review upstream terms and access before broader operation.
R4 — Offline retries, concurrent edits or interrupted updates lose or duplicate user data.	Medium	High	Test idempotent retries, record versions, tombstones, conflict resolution, user isolation and service-worker updates. Retain exports/backups and demonstrate recovery with disposable test records.
R5 — Passkey, portal-session, API-scope or organisation-membership errors expose private data or lock out users.	Medium	High	Maintain negative ownership/scope/origin tests, QR audience separation and recovery-code tests; store only hashed API secrets; review permission changes and avoid production credentials in evidence.
R6 — Deployment or schema changes make rollback unsafe or damage retained data/assets.	Medium	High	Deploy to staging first, validate and promote the same images. Keep the permanent gateway, environment-specific volumes, `deploy/state/`, both asset namespaces and SQLite/PostgreSQL backups. Require backward-compatible migrations and retain the existing privacy/demo compatibility guards.
R7 — Browser automation is mistaken for physical Android/iPhone validation.	Medium	High	Record browser/OS versions and test actual devices for permissions, passkeys, media and installation. Keep native packaging separate and trigger it only after the explicit release instruction.
R8 — Expanding chat, marketplace and business scope prevents completion of the core evaluation.	High	High	Agree the defense scope, prioritise Part 6 Must requirements and the evidence gates, and defer optional extensions/integrations. Track new scope as a separate issue with an explicit acceptance criterion.

Risk	Probability	Impact	Mitigation Strategy
R9 — Documentation, commit links and test results refer to different revisions after history changes.	High until reconciled	Medium	Use current SHAs, record the final submission revision, validate links, remove unsupported tag/status claims and attach run metadata. Retain historical reports as historical evidence.
R10 — Provider access, commercial settings, signing accounts or test participants are unavailable.	Medium	High for affected milestones	Record blockers and owners early; use explicit fixtures and test accounts for technical verification; distinguish these from live-provider or user validation. Do not claim eGov, banking, EDS or sensor integration without the required access.
R11 — Synthetic plots, private trade demos or a 3D example are mistaken for measured agricultural outcomes.	Medium	High	Preserve visible demo/source labels and isolation; explain that 3D is illustrative until real analytics exists. Show no invented sensor readings, growth phase, commercial transactions or interview outcomes.
R12 — Insufficient coordination between two developers leaves cross-cutting work unreviewed.	Medium	Medium	Confirm one owner per issue, use peer review and short risk/blocker reviews, and record handover notes. Schedule work against confirmed availability rather than assumed dates.
R13 — Failing CI remains unexplained and `main` permits changes without mandatory passing checks.	High while unresolved	High	Investigate the actual failed run logs, track corrective work and re-run affected suites. Agree appropriate protection or review controls before the final baseline; do not equate a successful Mobile job with success of all workflows.

10.6 Technical and organisational dependencies

Dependency	Purpose and effect if unavailable	Management action
Node.js, pnpm and web tooling	The repository pins pnpm in `package.json`; CI uses Node 22. A mismatched environment can invalidate reproduction.	Record actual versions and use the lockfile; build both web applications.

Dependency	Purpose and effect if unavailable	Management action
SQLite, PostgreSQL and private file storage	SQLite supports identity/workspace data; later marketplace, chat, platform and trade modules use PostgreSQL. Media and documents also depend on stored files.	Provide isolated test databases and document backup/restore coverage for all required stores; do not describe the complete current application as SQLite-only.
Docker, Nginx, DNS/TLS and deployment state	Required for the documented website deployment and origin separation. Loss of persistent volumes or state affects recovery.	Preserve the permanent gateway and deploy state; validate main-app and portal origins in each environment.
Media-processing and browser-test dependencies	Some integration scenarios need PostgreSQL, ffmpeg/qpdf and installed Playwright browsers. A default test run can skip dependency-bound suites.	Record setup, execute the applicable suites explicitly and disclose every skipped test.
External data providers and customer 1C	Cadastral/weather/news features and live OData import depend on provider behavior and valid customer access.	Separate fixture coverage from live validation; use read-only 1C credentials and document fallback scenarios.
Representative users and permission to use evidence	Needed to substantiate relevance and conduct outcome measurements.	Agree participation, anonymise records and preserve consent; public support for promotion is not an official partnership or contract.
GitHub review access and, if required, a YouTrack project	Needed to demonstrate traceability and review private repository evidence.	Arrange authorised access to the GitHub evidence. Confirm whether a separate tracker is necessary; avoid publishing credentials or participant data to make the assessment easier.
Supervisor feedback and confirmed submission schedule	Needed to settle scope, milestone dates and presentation requirements.	Confirm the scope and dates before committing to a calendar plan; record the resulting decisions.
Physical devices and optional release credentials	Needed for device acceptance; Android/iOS packaging additionally depends on release authorisation and signing configuration.	Reserve devices for website/PWA checks. Keep native release work deferred until explicitly requested; iOS trust/signing readiness requires separate verification.

10.7 Traceability and change-control workflow

The proposed traceability chain is:

Evidence in Part 2 → requirement in Part 6 → work item / actual GitHub issue → reviewed code change/current Git SHA → test or measurement record in Part 9 → milestone acceptance.

1. Record the user need or engineering problem, identify its source and distinguish verified facts from assumptions.
2. Record a work item or create an actual GitHub issue with a named owner, requirement IDs, scope, acceptance criteria and dependencies. If YouTrack is used, link its real issue key without replacing or fabricating historical evidence.

3. Implement the agreed change in a working branch and request peer review. Include the actual work-item/issue reference in the commit or PR; historical commits may be linked from a new issue without claiming an earlier tracker history.
4. Run checks appropriate to the change and attach their actual results. Inspect failed/skipped checks and agree required branch-protection controls; `main` was unprotected at this review. A workflow file alone is not proof that merging is blocked on failure.
5. For a website release, use `python3 deploy/manage.py deploy staging`, validate both web applications, then promote with `python3 deploy/manage.py deploy production --reuse-images TAG`. Production rollback must preserve data and respect compatibility guards. Native packages are not built during routine website changes.
6. Move the issue to Done only when its acceptance evidence is attached. Update the risk register and milestone state; keep incomplete provider/device/user validation visible as separate open work.

This is a recommended process for the remaining work. The repository inspection does not establish that every past change followed it.

10.8 Evidence to show at the defense

Evidence	Availability at this review	What to present / prepare
Repository, implementation and current history	Available.	Open the project tree, selected current commits and the final submission SHA; explain the role of each application and the current scope.
Responsibility allocation	Documented in Part 1.	Present the two named roles and obtain agreement on the currently unassigned research, tracker and evaluation responsibilities. Demonstrate individual understanding of relevant code.
Automated test and deployment configuration	Available; results must be tied to actual runs.	Show workflow files, the selected CI run and relevant local/integration logs, including failures and skips. Demonstrate backup/restore separately from the existence of backup code.
Task and implementation notes	Available in engineering documentation.	Show representative completed artefacts and explain the difference between implementation status and evaluation acceptance.
Actual YouTrack project and board	Not evidenced; optional under the owner's stated approach.	If required by the assessor, provide a real project URL, members/assignees and linked issues. Otherwise demonstrate the agreed GitHub/work-register substitute and disclose the absence of YouTrack.
Milestones and risks	Proposed in this section.	Confirm owners and dates, review dependencies and show how at least one unresolved risk leads to a concrete task.

Evidence	Availability at this review	What to present / prepare
User-research and outcome evidence	Required additions are specified in Parts 2 and 9.	Attach consented/anonymised discussion records, measurement data and baseline results; identify outstanding work candidly.

The project has substantial implementation and version-control evidence. Remaining management work consists of confirming ownership, linking milestone acceptance to actual verification results and resolving the failed CI runs. A separate YouTrack project is an optional addition unless the assessor requires the named tool explicitly.

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Stakeholder Background (Section 2)

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